

Arithmetic Local–Global Dichotomies for ProveIt’s Three-Variable Keller Map

Split integral Hasse failures, a dense parameter family,
and exact p -adic fiber laws

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Abstract

We study the arithmetic image of the explicit polynomial map with constant Jacobian determinant -2 developed in ProveIt’s Jacobian-conjecture project. The complex geometry and finite-field fiber laws are prior results ([write] of Gao and of the `royvanrijn/jacobian-research` notes respectively; neither is part of ProveIt); our focus is the distinction between rational, integral, and locally integral inverse images. We construct an explicit family of targets having exactly three rational preimages, none integral, but at least one integral preimage over $\mathbb{Z}_p$ for every prime p . More generally, a two-parameter split-root family admits an exact greatest-common-divisor criterion for local solubility. Its locally soluble parameters have density $27/(4\pi^2)$ and its targets are Zariski dense in the plane with third coordinate 2. The failure persists over every prescribed ring of S -integers and under integral stable equivalence.

At the exceptional prime 2, we prove that integral fiber cardinality is determined modulo 8, and has the Haar probability law $(21, 7, 3, 1)/32$ for cardinalities 0, 1, 2, 3. Consequently the 2-adic image has measure $11/32$. A general inverse-ball lemma turns a small finite certificate into a theorem at every precision. In contrast, the rational image satisfies a Hasse principle over every number field, by a cubic Galois argument. We also give a quantitative density-zero bound for rationally soluble integral targets and a precise formalization roadmap. All new claims have explicit proofs; computational identities and finite tables are supplied as replayable certificates, not represented as Lean proofs.

Status. The split integral family, its parameter density, and the dyadic refinement are candidate new contributions relative to the sources reviewed here. Global priority has not been established. This is not a new disproof of the Jacobian conjecture, nor a solution of the plane Jacobian problem.

[write] Batch 36, manuscript 03, placed in 1a1396d4d: a report of ProveIt’s research-report collection that continues the formal project `Algebra/JacobianConjecture`. AI-assisted, unrefereed, not formalized; labels carry the prefix `alg:`. See Section 1.3 and Appendix C. [Added 30 September 2026, batch 58:] Part II (Section 14, labels `alg:dh:`, not formalized) answers Research question 12.4 and, for completely split fibers in square boxes, Research question 12.3. [Added 30 September 2026, batch 64:] Part III (Section 28, labels `alg:ns:`, not formalized) answers Research questions 12.2 and, for square boxes in all sectors, 12.3.

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1 The research question and its exact scope

1.1 From a formal polynomial witness to an arithmetic problem

ProveIt contains independent Lean and Rocq/Coq developments for a three-variable polynomial map announced by Levent Alpöge, including its constant Jacobian determinant, explicit collisions, symmetry, and stable changes of variables [1, 2]. We use the following compact presentation, over $\mathbb{Z}$:

$$\begin{aligned} u &= 1 + xy, & h &= u^2z + y^2(1 + 3u), \\ F(x, y, z) &= (P, Q, R) = (uh, y + 3xh, x(5 - 3u - x^2z)). \end{aligned} \tag{1.1}$$

Thus

$$\begin{aligned} P &= (1 + xy)^3z + y^2(1 + xy)(4 + 3xy), \\ Q &= y + 3x(1 + xy)^2z + 3xy^2(4 + 3xy), \\ R &= 2x - 3x^2y - x^3z. \end{aligned}$$

The formal identity $\det JF = -2$ makes F an *étale* morphism over $\mathbb{Z}[1/2]$: there the Jacobian determinant is a unit. Over $\mathbb{Z}$ it is not étale at 2. This distinction will matter in the local counting theorem, but it will *not* explain away the global integral obstruction.

A collision establishes noninjectivity. It does not answer which integral targets occur as values, how much congruences detect, or whether rational and integral local–global principles agree. We therefore pose the concrete question

$$\mathbf{b} \in \mathbb{Z}^3, \quad \mathbf{b} \in F(\mathbb{Z}_p^3) \text{ for every prime } p \stackrel{?}{\implies} \mathbf{b} \in F(\mathbb{Z}^3). \tag{1.2}$$

We answer it negatively by explicit complete fibers. We then classify a large family of such failures and determine the local image at the unique bad prime.

1.2 What is inherited, and what is contributed here

The choice of this problem followed a review of the actual repository statements rather than an assumption that every unexplained phenomenon was open. In particular, Gao’s Theorems 3.3–3.4 already give the generic geometric degree and complete complex fiber stratification of this map [3]. The archived note *Exact finite-field value distribution* in `royvanrijn/jacobian-research` already gives the cubic inverse model and exact finite-field histograms, including characteristic 3 [4]. We reproduce short proofs because the arithmetic arguments depend on their precise boundary conditions, not because these are being claimed as new theorems.

There is also prior work on a *different* integral Jacobian-one map with a locally integral fiber of geometric degree five having *no rational point* [5]. Our examples instead have three rational points, all of which fail global integrality. Consequently they neither supersede that stronger rational obstruction nor conflict with its minimum-degree discussion. The two constructions isolate different failures of local–global reasoning.

Statement	Role in this article	Evidence boundary
Map F , determinant, collisions	Repository input; rederived	Prior formal developments; exact identity here
Complex fiber sizes 3, 1, 0	Prior geometry	Gao; self-contained root proof here
Odd finite-field histograms	Prior arithmetic input	Archived note; proof and replay here
Split integral Hasse family and density	Candidate new refinement	Explicit formulas and elementary proofs
S -integer and stable-equivalence persistence	Consequences of the family	Exact algebraic arguments
Dyadic law (21, 7, 3, 1)/32	Candidate new refinement	Inverse-ball theorem plus finite certificate
Rational Hasse principle; thin image bound	Applications, not priority claims	Cubic algebra, Chebotarev, elementary sieve

Our source review is dated September 24, 2026, with ProveIt pinned to e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9 and the second research repository pinned to caf5685db1099cb90e608556d4b8d825734f29cb. The review did not locate the same split-plane or dyadic statements in the inspected material. That is a bounded literature finding, not a proof of worldwide novelty. Nor do we claim that the question in (1.2) was a previously published named open problem. It is a natural unresolved arithmetic question arising from the repo, and the proofs below resolve it for this map.

1.3 Attribution and the formal project

[write] This subsection was added when the manuscript was written into ProveIt. It makes the attribution of the inherited material precise.

- *The map* was announced by Alpöge. ProveIt’s Jacobian project formalizes it, in Lean 4 as `counterexample` (`Algebra/JacobianConjecture/Lean/JacobianConjecture/Counterexample.lean`, lines 95–101, namespace `LeanProofs.JacobianCounterexample`) and in Rocq/Coq as `counterexample` (`Algebra/JacobianConjecture/Coq/Counterexample.v`, line 110), with exactly the coordinates P, Q, R displayed below.
- *The determinant and the collisions* are kernel-checked there: `jacobianDet_counterexample` ($\det JF = -2$ as a polynomial identity over every commutative ring), the integral collision `collision` with `collision0 = (-1, 1, 5)`, `collision1 = (0, -2, -16)` and `collisionValue = (0, -2, 0)`, the rational triple collision `counterexample_triple_collision` over $(-\frac{1}{4}, 0, 0)$, the mirror symmetry `counterexample_equivariant`, the torus action `counterexample_scaling` $F(x/s, sy, s^2z) = (s^2P, sQ, R/s)$, the one-parameter collision family `collisionFamily`, and the refutation `jacobianConjectureInDimensionThree_false`; the Coq development proves the corresponding statements (`jacobian_det_is_minus_two`, `counterexample_scaling`, `jacobian_conjecture_dimension_three_is_false`).
- *The complex fiber stratification* (fibers of size 3, 1, 0; [corollary 2.4](#)) is Gao’s [3, Theorems 3.3–3.4]. It is *not* in ProveIt. At the write it was checked on arXiv that Gao’s Theorem 3.3 concerns this very map (component degrees 7, 6, 4, determinant -2 , three points in the generic fiber) and that his polynomial $c_3 = 27v_1^2v_3^2 - 18v_1v_2v_3 + 16v_1 + v_3^3v_3 - v_2^2$ is $-\Delta/4$ in the notation of (2.9).

- *The cubic inverse model and the finite-field histograms* ([theorem 7.1](#) and the count [\(7.4\)](#)) are from the archived note [\[4\]](#) of [royvanrijn/jacobian-research](#), which already uses the variable $t = y + 1/x$ and the cubic [\(2.4\)](#). They are *not* in ProveIt. At the write it was checked that the pinned note states the formulas [\(7.1\)](#) and [\(7.2\)](#).
- *Nothing arithmetic about the fibers* is in ProveIt: the project has no material on fiber stratification, finite fields, p -adic or dyadic fibers, or Hasse principles, at the pin and at the placement commit. The theorems of [sections 3 to 6](#) and [8 to 10](#) are this manuscript’s (with the priority caveats stated above), and *none of them is formalized*. The stable representatives used in [proposition 5.3](#) are the project’s: its degree-six one is kernel-checked (`simplerCounterexample`), the lower-degree ones have SymPy certificates only.

1.4 Notation and the headline results

We write $\mathbf{b} = (A, B, C)$ for a target. The letter p denotes a rational prime, while q is a finite field’s cardinality. Haar measure on $\mathbb{Z}_p^d$ is normalized to give the whole space measure one. For a finite set S of primes, $\mathbb{Z}[S^{-1}]$ denotes the rationals whose denominators use only primes in S .

The main explicit example is especially small:

$$F^{-1}(0, -4, 2)(\overline{\mathbb{Q}}) = \left\{ \left(-\frac{1}{2}, 2, 36 \right), \left(\frac{1}{6}, -4, -288 \right), \left(\frac{1}{3}, -4, 0 \right) \right\}. \quad (1.3)$$

Every prime sees an integral point in this fiber, but no point is integral over $\mathbb{Z}$. The complete proof, including exhaustiveness, is in [section 3](#). The larger split-plane family and density theorem follow in [section 4](#). The dyadic law is proved in [section 9](#):

$$\Pr_{\mathbf{b} \in \mathbb{Z}_2^3} (\#F^{-1}(\mathbf{b})(\mathbb{Z}_2) = j) = \frac{1}{32} (21, 7, 3, 1)_j, \quad j = 0, 1, 2, 3.$$

Both results depend on distinguishing actual source points from roots of an inverse polynomial which have escaped to infinity.

2 A complete simple-root model

2.1 An elementary coordinate change

The following calculation supplies all the inverse formulas needed later. It is valid over any field of characteristic different from 2. On the source chart $x \neq 0$, introduce

$$w = \frac{1}{x}, \quad t = y + \frac{1}{x}, \quad C = R(x, y, z). \quad (2.1)$$

Then $u = xt$, and solving the definition of R gives

$$x = \frac{1}{w}, \quad y = t - w, \quad z = 5w^2 - 3tw - Cw^3. \quad (2.2)$$

Direct substitution into [\(1.1\)](#) now yields the much smaller formulas

$$A = wt + t^2 - Ct^3, \quad B = 2w + 4t - 3Ct^2, \quad R = C. \quad (2.3)$$

For example, substitution first gives $u = t/w$ and $h = w^2 + tw - Ct^2w$, from which both first coordinates follow immediately. Eliminating w gives

$$g_{A,B,C}(T) = CT^3 - 2T^2 + BT - 2A, \quad g_{A,B,C}(t) = 0, \quad (2.4)$$

and the derivative is

$$d(t) := g'_{A,B,C}(t) = 3Ct^2 - 4t + B = 2w = \frac{2}{x}. \quad (2.5)$$

In particular, every finite source point on this chart determines a *simple* root. A repeated root cannot be reconstructed by dividing by its derivative.

Proposition 2.1 (A short determinant proof). *As an identity in $\mathbb{Z}[x, y, z]$, one has $\det JF = -2$.*

Proof. In the localization where x is invertible, the change $(x, y, z) \mapsto (w, t, C)$ has determinant $(-x^{-2}) \cdot 1 \cdot (-x^3) = x = 1/w$. From (2.3), the determinant of $(w, t, C) \mapsto (A, B, C)$ is

$$\det \begin{pmatrix} t & w + 2t - 3Ct^2 & -t^3 \\ 2 & 4 - 6Ct & -3t^2 \\ 0 & 0 & 1 \end{pmatrix} = -2w.$$

The chain rule gives -2 in the localization. Since $\mathbb{Z}[x, y, z]$ injects into that localization, it gives the same polynomial identity globally. This is a derivation of the identity, independent of the accompanying symbolic expansion check. $\square$

Theorem 2.2 (Complete field-valued inverse). *Let k be a field with $\text{char } k \neq 2$, and $(A, B, C) \in k^3$. Every simple root $t \in k$ of $g_{A,B,C}$ determines exactly one preimage with $x \neq 0$, namely*

$$w = \frac{g'_{A,B,C}(t)}{2}, \quad (x, y, z) = \left(\frac{1}{w}, t - w, 5w^2 - 3tw - Cw^3 \right). \quad (2.6)$$

These are all the preimages with $x \neq 0$. There is a preimage with $x = 0$ exactly when $C = 0$, and then it is uniquely

$$(0, B, A - 4B^2). \quad (2.7)$$

Equivalently, the complete fiber is in natural bijection with the simple k -rational projective roots of

$$G_{A,B,C}(T, S) = CT^3 - 2T^2S + BTS^2 - 2AS^3. \quad (2.8)$$

Proof. The forward construction and simplicity follow from (2.1)–(2.5). Conversely, set $w = g'(t)/2 \neq 0$. The second expression in (2.3) is B . The first expression minus A equals $g(t)/2$ after this substitution, so it vanishes. This proves that (2.6) is a two-sided inverse on the chart. If $x = 0$, (1.1) reduces to $F(0, y, z) = (z + 4y^2, y, 0)$, proving (2.7). Finally, the projective point $[1 : 0]$ is a root of G exactly when $C = 0$. On the chart $T = 1$, its derivative with respect to S is -2 , so it is always simple. All other projective roots are the affine roots already classified. $\square$

Remark 2.3 (Why a root count can be wrong). A cubic may have three roots counted with multiplicity while the actual fiber has only one point, or none. Also, setting $C = 0$ lowers the degree of the affine inverse polynomial but *adds* the source point (2.7). Both issues must be handled before making any arithmetic inference. In particular, a resultant equation alone is not an exhaustive inverse description.

2.2 The known geometric stratification

The discriminant of the binary cubic, or of g with its cubic coefficient retained, is

$$\Delta = 4B^2 - 4CB^3 - 64A + 72CAB - 108C^2A^2. \quad (2.9)$$

Over an algebraically closed field of characteristic different from 2 and 3, [theorem 2.2](#) immediately gives the following result.

Corollary 2.4 (Previously known complex fiber law). *There are three preimages if $\Delta \neq 0$, one if $\Delta = 0$ but the binary cubic is not a cube, and none on the triple-root locus*

$$C \neq 0, \quad 3BC = 4, \quad 27AC^2 = 4. \quad (2.10)$$

There is no fiber of cardinality two over an algebraically closed field.

Proof. A separable degree-three divisor on $\mathbb{P}^1$ has three simple points. A degree-three divisor with a repeated point, but not supported at a single point, has one double point and one simple point. A cube has no simple point. Comparing $C(T - t_0)^3$ with g gives $t_0 = 2/(3C)$ and then (2.10). A triple root at infinity is impossible because the coefficient of T^2S is -2 . $\square$

This recovers Gao’s stratification [3, Theorem 3.4]; indeed his polynomial c_3 is $-\Delta/4$. We use a different inverse coordinate because the derivative identity (2.5) is especially convenient for denominators. The geometric law does not imply that every *integral* fiber has size 0, 1, or 3; integrality can discard just one of three rational points.

2.3 The projective chart at infinity

For completeness, the projective-root picture can be made into an explicit scheme description over $\mathbb{Z}[1/2]$. Let Y be the incidence subscheme $G = 0$ in $\mathbb{A}^3 \times \mathbb{P}^1$, and let Y^{simp} be the open set where the corresponding root is simple. The map from the source is

$$(x, y, z) \mapsto (F(x, y, z), [1 + xy : x]).$$

The projective pair is globally defined because $(1 + xy) - yx = 1$. On $S \neq 0$, [theorem 2.2](#) gives its inverse. On $T \neq 0$, put $s = S/T$; the incidence equation becomes

$$C = 2s - Bs^2 + 2As^3.$$

Set

$$k = 1 - Bs + 3As^2, \quad Y_0 = B - 3As.$$

Simplicity is precisely $k \neq 0$, because the derivative of $C - 2s + Bs^2 - 2As^3$ is $-2k$. An inverse chart is

$$x = \frac{s}{k}, \quad y = Y_0, \quad z = Ak^3 - k(k + 3)Y_0^2. \quad (2.11)$$

Indeed $1 + xy = 1/k$, $h = Ak$, and the three coordinates of F become $A, B, 2s - Bs^2 + 2As^3$. These identities prove that the two chart inverses glue and that the source is isomorphic to Y^{simp} . This is a coordinate interpretation, not a separate novelty claim. No scheme-theoretic machinery from this paragraph is needed in the integral arguments below.

3 An explicit infinite integral Hasse failure

3.1 The family and all its points

Theorem 3.1 (A split three-point obstruction). *For every integer $n \geq 2$, let*

$$\mathbf{b}_n = (0, -2n(n-1), 2). \quad (3.1)$$

Then

$$\mathbf{b}_n \in F(\mathbb{Z}_p^3) \text{ for every prime } p, \quad \mathbf{b}_n \in F(\mathbb{Q}^3) \subset F(\mathbb{R}^3), \quad \mathbf{b}_n \notin F(\mathbb{Z}^3).$$

The complete geometric fiber consists of three rational points. More precisely, with

i	t_i	D_i
0	0	$-n(n-1)$
1	n	$n(2n-1)$
2	$1-n$	$(n-1)(2n-1)$

(3.2)

those points are

$$v_i(n) = \left(D_i^{-1}, t_i - D_i, 5D_i^2 - 3t_i D_i - 2D_i^3 \right). \quad (3.3)$$

Proof. The inverse cubic factors exactly as

$$g_{\mathbf{b}_n}(T) = 2T^3 - 2T^2 - 2n(n-1)T = 2T(T-n)(T+n-1).$$

For $n \geq 2$, its roots $0, n, 1-n$ are distinct. Half its derivative at these roots is respectively the three values in (3.2). Since the third target coordinate is nonzero, there is no $x = 0$ point. The complete inverse theorem therefore gives exactly (3.3), even over $\overline{\mathbb{Q}}$.

All second and third coordinates of these points are integers. Their first coordinates are $1/D_i$, with $|D_i| > 1$, so none is an integer point. This proves global insolubility without a bounded search.

The three integers $n, n-1, 2n-1$ are pairwise coprime: subtracting adjacent integers proves the first assertion, and $2n - (2n-1) = 1$, $(2n-1) - 2(n-1) = 1$ prove the other two. A prime divides at most one of them. If it divides n , select v_2 ; if it divides $n-1$, select v_1 ; if it divides $2n-1$, select v_0 . The selected denominator is a p -adic unit. If it divides none, all three points are p -adically integral. Thus every prime has an integral preimage. The displayed points themselves are rational and real. $\square$

Example 3.2 (The smallest displayed parameter). At $n = 2$, the points are exactly those in (1.3). At the prime 2, use $(1/3, -4, 0)$. At the prime 3, use $(-1/2, 2, 36)$. At every other prime all three points are integral. A single rational point integral at every prime would be a $\mathbb{Z}$ -point, but here the locally selected point changes with the prime. The quantifiers

$$\forall p \exists v_p \quad \text{and} \quad \exists v \forall p$$

are genuinely different, despite the fiber having only three points.

3.2 A denominator criterion for every split finite fiber

The preceding mechanism is elementary and reusable.

Lemma 3.3 (Denominator covering). *Let $f \in \mathbb{Z}[X_1, \dots, X_d]^d$ and $\mathbf{b} \in \mathbb{Z}^d$. Suppose its complete geometric fiber is a finite nonempty set of rational points $v_1, \dots, v_r$. Let d_i be the least positive common denominator of the coordinates of v_i . Then*

$$\begin{aligned} \mathbf{b} \in f(\mathbb{Z}^d) &\iff d_i = 1 \text{ for some } i, \\ \mathbf{b} \in f(\mathbb{Z}_p^d) &\iff p \nmid d_i \text{ for some } i, \\ \mathbf{b} \in f(\mathbb{Z}_p^d) \text{ for every } p &\iff \gcd(d_1, \dots, d_r) = 1. \end{aligned}$$

Proof. A rational vector is integral over $\mathbb{Z}_p$ exactly when its reduced coordinate denominators are all prime to p . Because every geometric point is rational, extension to $\mathbb{Q}_p$ creates no additional points in the fiber. Hence local membership is exactly the stated disjunction over the existing finite set. A prime fails every disjunct precisely when it divides their greatest common divisor. The global assertion is the corresponding denominator-one condition. $\square$

The split hypothesis matters: a nonsplit factor can acquire new points over some completions. We will not silently apply the lemma to a cubic with an irreducible quadratic factor.

3.3 An actual solution of every congruence

Corollary 3.4. *For every $n \geq 2$ and every positive integer m , there is an integer vector a with $F(a) \equiv \mathbf{b}_n \pmod{m}$, although $F(a) = \mathbf{b}_n$ has no integer solution.*

Proof. For each prime power $p^e \mid m$, select a point $v_i(n)$ with $p \nmid D_i$. Reduce its integer coordinates normally and its first coordinate by $D_i^{-1} \pmod{p^e}$. This gives a source modulo p^e . Combine these source coordinates using the Chinese remainder theorem. Since F has integer coefficients, its value has the required reduction at every prime power. $\square$

The accompanying function `congruence_preimage(n,m)` performs exactly this construction. For example,

$$F(20587, 20252, 13536) \equiv (0, -4, 2) \pmod{21600}.$$

This is an exact certificate, not a numerical approximation to a nonexistent integer solution.

4 A dense two-parameter family and its exact density

4.1 Complete classification on the integral split-root locus

The one-parameter family is the specialization $(a, b, c) = (0, n, 1 - n)$ of a much larger construction.

Theorem 4.1 (Split-plane classification). *Let $a, b, c \in \mathbb{Z}$ be pairwise distinct with $a + b + c = 1$, and define*

$$\Phi(a, b) = (abc, 2(ab + ac + bc), 2), \quad c = 1 - a - b. \quad (4.1)$$

For $r \in \{a, b, c\}$ put

$$D_r = \prod_{s \in \{a, b, c\}, s \neq r} (r - s).$$

Then the complete geometric fiber of F over $\Phi(a, b)$ consists of

$$\left(D_r^{-1}, r - D_r, 5D_r^2 - 3rD_r - 2D_r^3 \right), \quad r = a, b, c. \quad (4.2)$$

It has no integral point. Writing

$$g = \gcd(a - b, a - c) = \gcd(a - b, 3a - 1), \quad (4.3)$$

we have

$$\Phi(a, b) \in F(\mathbb{Z}_p^3) \text{ for every prime } p \iff g = 1. \quad (4.4)$$

More precisely, a prime has no integral preimage exactly when it divides g .

Proof. The coefficient identities for three roots of sum 1 give

$$g_{\Phi(a,b)}(T) = 2(T - a)(T - b)(T - c).$$

Thus [theorem 2.2](#) gives (4.2), with no boundary point. If $|D_r| = 1$, both other roots must be integers at distance one from r . Because they are distinct, they must be $r - 1$ and $r + 1$. Their sum with r would be $3r = 1$, an impossibility for integral r . Hence all three first coordinates are nonintegral.

For the exact gcd, put $u = a - b$ and $v = a - c$. Up to signs the three D 's are

$$uv, \quad u(v - u), \quad v(v - u).$$

After writing $u = gu_0$, $v = gv_0$, the integers $u_0, v_0, v_0 - u_0$ are pairwise coprime. No prime can divide all three of their pairwise products. Consequently

$$\gcd(|D_a|, |D_b|, |D_c|) = g^2. \quad (4.5)$$

All other coordinates in (4.2) are integral. Apply [lemma 3.3](#). Finally, $(a - b) + (a - c) = 3a - 1$, proving the second expression for g . $\square$

Remark 4.2 (Exact local multiplicities in this family). If $p \nmid g$ and exactly two roots agree modulo p , only the point corresponding to the third root is integral over $\mathbb{Z}_p$. If all three root residues are distinct, all three points are integral. If $p \mid g$, no point is integral. In particular, for every locally soluble target of this family, the integral fiber has exactly one point at $p = 2$ and at $p = 3$. At 2 three distinct residues are impossible. At 3 three distinct residues would sum to 0, whereas the root sum is 1.

4.2 A positive density theorem in the parameter plane

Density here refers to ordered integer parameters (a, b) , not to all integer targets in $\mathbb{A}^3$. The distinction is essential.

Theorem 4.3 (Parameter density). *Let $M(H)$ count pairs $(a, b) \in [-H, H]^2 \cap \mathbb{Z}^2$ such that the three integers $a, b, 1 - a - b$ are distinct and $\gcd(a - b, 3a - 1) = 1$. As $H \rightarrow \infty$ through positive integers,*

$$M(H) = \frac{27}{4\pi^2} (2H + 1)^2 + O(H \log H). \quad (4.6)$$

Every counted pair gives a locally integral but globally nonintegral fiber of the fixed map F .

Proof. First ignore coincidences among the roots, and let $L = 2H + 1$. Since $3a - 1 \neq 0$, the gcd in question is positive and at most $3H + 1$. The Möbius identity for coprimality gives the count

$$\sum_{d \leq 3H+1} \mu(d) \#\{(a, b) \in [-H, H]^2 : d \mid a - b, d \mid 3a - 1\}.$$

The inner set is empty when $3 \mid d$. Otherwise 3 is invertible modulo d , and there is exactly one residue pair:

$$a \equiv b \equiv 3^{-1} \pmod{d}.$$

Its count is $L^2/d^2 + O(L/d + 1)$, uniformly in d . Summing the errors gives $O(H \log H)$, while extending the absolutely convergent main sum to infinity costs $O(H)$. Its value is

$$\sum_{3 \nmid d} \frac{\mu(d)}{d^2} = \prod_{p \neq 3} \left(1 - \frac{1}{p^2}\right) = \frac{1}{\zeta(2)(1 - 3^{-2})} = \frac{27}{4\pi^2}.$$

The three root-coincidence equations $a = b$, $2a + b = 1$, and $a + 2b = 1$ account for only $O(H)$ pairs. Removing them does not affect the displayed asymptotic. The last assertion is [theorem 4.1](#). $\square$

The approximate value of the density is 0.683918. Exact enumeration, used only as a check on the proof, gives the following data.

H	$M(H)$	$M(H)/(2H + 1)^2$
10	310	0.702948
25	1764	0.678201
50	7034	0.689540
100	27706	0.685775
250	171878	0.684770

Corollary 4.4 (Zariski density in a fixed target plane). *The set*

$$\{\Phi(a, b) : a, b, 1 - a - b \text{ distinct, } \gcd(a - b, 3a - 1) = 1\}$$

is Zariski dense in the plane $C = 2$. Every target in it is an integral Hasse failure for F in the sense of (1.2).

Proof. A nonzero polynomial in two variables has only $O(H)$ zeros in $[-H, H]^2 \cap \mathbb{Z}^2$: view it as a polynomial in one variable and separate the finitely many values where all coefficient polynomials vanish. By [theorem 4.3](#), the allowed parameter pairs have positive two-dimensional density, so are Zariski dense in $\mathbb{A}^2$.

The map Φ is dominant onto the target plane. Indeed, for any $(A, B) \in \overline{\mathbb{Q}}^2$, the monic cubic

$$T^3 - T^2 + \frac{B}{2}T - A$$

has three roots over $\overline{\mathbb{Q}}$, counted with multiplicity, whose sum is 1; those roots give a parameter pair mapping to $(A, B, 2)$. A polynomial relation vanishing on the allowed target set would therefore pull back to a relation vanishing on a dense parameter set, hence would vanish identically on the target plane. $\square$

Remark 4.5 (What the density does not say). The map from ordered distinct root triples to a fixed target has six permutations. Moreover, a parameter box has coefficient heights $|A| = O(H^3)$ and $|B| = O(H^2)$, not a uniform target box. Thus (4.6) is not a density formula in target space. It does prove that the integral obstruction is much more than an isolated collision or a single exceptional arithmetic progression.

5 S -integers, stable equivalence, and congruence closure

5.1 Inverting finitely many primes does not remove the obstruction

Theorem 5.1 (Persistence over every prescribed S). *For every finite set S of rational primes, infinitely many $n \geq 2$ satisfy*

$$\mathbf{b}_n \in F(\mathbb{Z}_p^3) \text{ for every } p, \quad \mathbf{b}_n \notin F(\mathbb{Z}[S^{-1}]^3).$$

In particular, this failure persists for the étale map over $\mathbb{Z}[1/2]$.

Proof. Choose distinct primes ℓ_1, ℓ_2 outside S . By the Chinese remainder theorem, there are infinitely many $n \geq 2$ satisfying $n \equiv 0 \pmod{\ell_1}$ and $n \equiv 1 \pmod{\ell_2}$. In (3.2), the denominators D_0, D_1 are divisible by ℓ_1 , and D_2 is divisible by ℓ_2 . Their numerators are one, so no cancellation is possible. None of the three complete rational preimages lies in $\mathbb{Z}[S^{-1}]^3$. Local integrality at every prime was proved for every n in theorem 3.1. $\square$

For example, $n = 6$ gives the target $(0, -60, 2)$ and denominators $-30, 66, 55$. None is a power of two up to sign, so this example has no $\mathbb{Z}[1/2]$ -point. The obstruction is therefore not a defect caused solely by bad reduction at 2.

Corollary 5.2 (The same dense obstruction over every fixed S). *For a fixed finite set S , let $M_S(H)$ count the parameter pairs in theorem 4.3 whose target has no preimage over $\mathbb{Z}[S^{-1}]$. Then*

$$M_S(H) = \frac{27}{4\pi^2}(2H+1)^2 + O_S(H \log H).$$

Their targets are Zariski dense in $C = 2$ and have integral preimages over $\mathbb{Z}_p$ for every prime, including the primes of S .

Proof. All second and third coordinates in (4.2) are integers. A preimage belongs to $\mathbb{Z}[S^{-1}]^3$ exactly when some D_r has no prime factor outside S . Then both nonzero integer differences $r - s$ and $r - t$ have all prime factors in S as well.

In the parameter box, all differences have absolute value at most $3H + 1$. The number of signed nonzero integers of this size supported on S is at most

$$2 \prod_{p \in S} \left(1 + \left\lfloor \frac{\log(3H+1)}{\log p} \right\rfloor \right) = O_S((\log H)^{|S|}).$$

Given two such differences $u = r - s$, $v = r - t$, the root sum forces $3r = 1 + u + v$ and hence determines the root triple, up to its finitely many orderings. Thus only $O_S((\log H)^{2|S|})$ parameter pairs in the whole box can give an S -integral point. Removing this set from the locally soluble pairs of theorem 4.3 preserves its stated asymptotic; for fixed S the polylogarithmic error is absorbed into $O_S(H \log H)$. Positive parameter density implies target Zariski density by the proof of corollary 4.4. The all-prime local assertion was not weakened when these pairs were removed. $\square$

5.2 Transport to the repository’s stabilized maps

Proposition 5.3 (Integral stable-equivalence invariance). *Let R be either $\mathbb{Z}$ or $\mathbb{Z}[S^{-1}]$, let $f \in R[X]^d$, and let*

$$h = \beta \circ (f \times \text{id}_{\mathbb{A}^m}) \circ \alpha,$$

where α, β are polynomial automorphisms over R with polynomial inverses over R . For a target $\mathbf{b} \in R^d$, put $\mathbf{b}' = \beta(\mathbf{b}, 0)$. Over every R -algebra T there is a bijection

$$h^{-1}(\mathbf{b}')(T) \simeq f^{-1}(\mathbf{b})(T).$$

Consequently integral local–global failures transport to h over R .

Proof. Apply β^{-1} to the target equation and then α to a source point. The equation becomes $(f \times \text{id})(v, u) = (\mathbf{b}, 0)$, which forces $u = 0$ and $f(v) = \mathbf{b}$. Each step is invertible over every T because all polynomial coefficients and inverse coefficients lie in R . $\square$

The source and target shears in ProveIt’s stable degree reductions are of this form [2]. Thus the arithmetic obstruction is not tied to the expanded degree-seven presentation. It transfers to each such integral stable representative. The proposition is unconditional; when applied to a particular repository reduction, the equality defining that reduction retains the proof status documented by the repository. We do not upgrade its symbolic certificates to kernel-checked proofs.

5.3 The exact profinite closure

Theorem 5.4 (Closure of the integral image). *For any polynomial map $f \in \mathbb{Z}[X_1, \dots, X_d]^r$,*

$$\overline{f(\mathbb{Z}^d)}^{\widehat{\mathbb{Z}}^r} = f(\widehat{\mathbb{Z}}^d) = \prod_p f(\mathbb{Z}_p^d). \quad (5.1)$$

For the map F , the difference between this closure and $F(\mathbb{Z}^3)$, intersected with $\mathbb{Z}^3$, contains every $\mathbf{b}_n$ and is Zariski dense in the target plane $C = 2$.

Proof. The profinite integers $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$ are compact, and polynomial evaluation is continuous. Hence $f(\widehat{\mathbb{Z}}^d)$ is closed. The diagonal image of $\mathbb{Z}^d$ is dense by the Chinese remainder theorem, so its image is dense in the compact image: every neighborhood of $f(x)$ pulls back to a neighborhood of x meeting $\mathbb{Z}^d$. This proves the first equality. Polynomial evaluation on the product is coordinatewise, and choices of preimages can be made separately at each prime, proving the second equality. The final assertions follow from [theorem 3.1](#) and [corollary 4.4](#). $\square$

Thus even *all* congruences, not merely all congruences below a fixed bound, fail to characterize the integer image. On a split fiber, [lemma 3.3](#) identifies exactly what they miss: local points can live on different rational components.

6 The contrasting rational Hasse principle

6.1 A cubic obstruction at infinitely many primes

We use one standard external theorem in this section: Chebotarev’s density theorem. In the form needed here, every conjugacy class in the Galois group of a finite Galois extension of number fields occurs as Frobenius at infinitely many unramified finite places. The factorization of a separable polynomial at a good place is given by the cycle lengths of that Frobenius action. See Milne [6, Theorem 8.31 and Example 8.36]. No effective version is required.

Lemma 6.1. *If an irreducible cubic over a number field K has a root in every finite completion K_v , a contradiction follows. In fact, it has no root at infinitely many finite places.*

Proof. Its Galois group acts transitively on three roots and is either the cyclic group of order three or the full symmetric group S_3 . In either case it contains a three-cycle. Chebotarev supplies infinitely many good places with that Frobenius cycle type. The reduction is an irreducible cubic there. After excluding the finitely many places where its monic normalization is not integral or its discriminant is not a unit, a local root would be integral and would reduce to a root of this irreducible cubic. This is impossible. $\square$

Theorem 6.2 (Rational local–global equality). *For every number field K and $\mathbf{b} \in K^3$,*

$$\mathbf{b} \in F(K^3) \iff \mathbf{b} \in F(K_v^3) \text{ for every finite place } v. \quad (6.1)$$

Archimedean places can be added without changing the equality.

Proof. The forward implication is immediate. Write $\mathbf{b} = (A, B, C)$ and assume local solubility at every finite place. If $C = 0$, the point $(0, B, A - 4B^2)$ lies in K^3 . Suppose $C \neq 0$. By [theorem 2.2](#), local solubility means that $g_{A,B,C}$ has a simple root in every K_v .

An irreducible g is impossible by [lemma 6.1](#). A reducible cubic has a root in K . If some K -root is simple, reconstruction finishes the proof. If a K -root has multiplicity two, the remaining linear factor is also defined over K and is simple. The only remaining possibility is a triple root, in which case no simple root exists even geometrically and every local fiber is empty. This contradicts the assumption. Thus a simple K -root always exists. $\square$

Remark 6.3 (Reconciliation with the integral counterexamples). The targets $\mathbf{b}_n$ do have rational preimages, so they satisfy [theorem 6.2](#). What fails is simultaneous integrality of one of those points. The earlier degree-five construction with no rational preimage [5] is of a different kind: a product of an irreducible cubic and its quadratic discriminant-field polynomial can have a root at every completion without a rational root. A single cubic cannot realize that covering mechanism. We claim neither a new general Hasse theorem for arbitrary polynomial maps nor a contradiction to the known degree-five examples.

6.2 A practical exact rational inverse algorithm

The proof gives a short, complete rational algorithm. On input $(A, B, C) \in \mathbb{Q}^3$, factor $g_{A,B,C}$ over $\mathbb{Q}$. Retain each rational root of multiplicity one, reconstruct it by (2.6), and add (2.7) when $C = 0$. To decide integer or S -integer membership, inspect the reduced denominators of this finite list. An irreducible quadratic factor can be ignored for the rational list, but not for a local integral-image problem.

The file `code/rational_inverse.py` implements this procedure using exact rational arithmetic and SymPy’s rational factorization. It verifies every reconstructed point by substitution. The mathematical completeness of the procedure is [theorem 2.2](#), rather than any heuristic search bound.

7 Known finite-field laws and their arithmetic use

7.1 Odd-characteristic fiber counts

For a finite field $k = \mathbb{F}_q$ of odd characteristic, let

$$N_j(q) = \#\{\mathbf{b} \in k^3 : \#F^{-1}(\mathbf{b})(k) = j\}.$$

The following formulas are prior results from [4]. We include their elementary proof to fix exactly which roots are counted.

Theorem 7.1 (Finite-field law, credited to the archived note). *If char $k > 3$, the only fiber sizes are 0, 1, 3, and*

$$N_0 = \frac{(q-1)(q^2+2)}{3}, \quad N_1 = \frac{q^3+q^2-2q+2}{2}, \quad N_3 = \frac{(q-1)(q^2+2)}{6}. \quad (7.1)$$

If char $k = 3$, the only fiber sizes are again 0, 1, 3, but

$$N_0 = \frac{q^2(q-1)}{3}, \quad N_1 = \frac{q^2(q+1)}{2}, \quad N_3 = \frac{q^2(q-1)}{6}. \quad (7.2)$$

Proof. When $C \neq 0$, a cubic has 0, 1, or 3 simple roots in its ground field: two distinct simple rational roots force the third root to be rational and simple. A repeated-root cubic has at most one simple root. When $C = 0$, the boundary point is always present; its affine quadratic has either zero or two simple rational roots. This gives the same list of possible fiber sizes.

For $C \neq 0$, a three-point fiber is determined by an unordered triple of distinct roots $\{r, s, t\}$ with nonzero sum. The sum determines $C = 2/(r + s + t)$, and the other elementary symmetric functions determine B, A . For $C = 0$, a three-point fiber is determined by an unordered pair of distinct roots of $-2T^2 + BT - 2A$. Therefore

$$N_3 = \binom{q}{2} + \binom{q}{3} - Z_q, \quad (7.3)$$

where Z_q counts unordered distinct zero-sum triples.

There are q^2 ordered zero-sum triples in total. In characteristic not three, each equality event $r = s$, $s = t$, or $t = r$ has q solutions; all their pairwise and triple intersections consist only of $(0, 0, 0)$. The union has $3q - 2$ elements. Hence $6Z_q = q^2 - 3q + 2 = (q-1)(q-2)$. In characteristic three, any equality in a zero-sum triple forces all three entries equal, and there are q such triples. Thus $6Z_q = q^2 - q$ instead. Substitution in (7.3) gives both values of N_3 .

Finally, counting targets and source points respectively gives $q^3 = N_0 + N_1 + N_3$ and $q^3 = N_1 + 3N_3$. Thus $N_0 = 2N_3$ and $N_1 = q^3 - 3N_3$, proving every formula. $\square$

7.2 A fixed-leading-coefficient refinement

We will need one more known count from [4], proved here independently by trace counting. Fix $C \neq 0$ and assume char $k > 3$. As (A, B) varies, g/C ranges over all monic cubics with the fixed coefficient $-2/C$ of T^2 . The irreducible ones number

$$\frac{q^2 - 1}{3}. \quad (7.4)$$

Indeed, the trace map $\mathbb{F}_{q^3} \rightarrow \mathbb{F}_q$ has q^2 elements of any prescribed trace. Exactly one of those lies in $\mathbb{F}_q$, since the trace of a ground-field element is three times that element. Every other element has degree three and belongs to a Frobenius orbit of size three. This proves (7.4). Across all $C \neq 0$, the number of target residue classes whose inverse binary cubic has *no* projective root is consequently

$$I(q) = \frac{(q-1)(q^2-1)}{3} = \frac{(q-1)^2(q+1)}{3}. \quad (7.5)$$

These classes form a useful congruence obstruction to rational solubility. They should not be confused with *all* omitted targets of F over k , because a repeated projective root is still a root but contributes no finite source point.

8 An inverse-ball theorem at every precision

8.1 The exact image of a sufficiently small ball

We now establish the analytic bridge which makes a finite dyadic calculation conclusive. The lemma is a quantitative version of the p -adic inverse-function argument, with its image lattice retained explicitly. We give a proof rather than relying on a convergence claim.

Lemma 8.1 (Exact inverse ball). *Let $f : \mathbb{Z}_p^d \rightarrow \mathbb{Z}_p^d$ be given by polynomials with coefficients in $\mathbb{Z}_p$, let $a \in \mathbb{Z}_p^d$, and set $J = Df(a)$. Suppose $v_p(\det J) = e < \infty$. For every integer $k > e$, the restriction of f to $a + p^k \mathbb{Z}_p^d$ is a bijection onto*

$$f(a) + p^k J \mathbb{Z}_p^d. \quad (8.1)$$

This image is a union of target residue classes modulo p^{k+e} . For every $m \geq k + e$, a target residue class modulo p^m has either zero or exactly p^e preimages modulo p^m within the source class $a \pmod{p^k}$. It has p^e precisely when its lifts lie in (8.1).

Proof. Taylor expansion, with integral polynomial coefficients, gives

$$f(a + p^k t) = f(a) + p^k J t + p^{2k} R(t),$$

where $R : \mathbb{Z}_p^d \rightarrow \mathbb{Z}_p^d$ is an integral polynomial map and is 1-Lipschitz for the maximum norm. The adjugate formula shows that $p^e J^{-1}$ is integral. Define

$$\Psi(t) = t + p^k J^{-1} R(t).$$

Its correction term is Lipschitz with constant at most $p^{-(k-e)} < 1$. For every $v \in \mathbb{Z}_p^d$, the fixed-point iteration

$$t \mapsto v - p^k J^{-1} R(t)$$

is a contraction of the complete space $\mathbb{Z}_p^d$ into itself. It has a unique fixed point, proving that Ψ is bijective. The same strict inequality gives $\|\Psi(t) - \Psi(t')\|_p = \|t - t'\|_p$, so Ψ induces a bijection modulo every p^r . Since

$$f(a + p^k t) = f(a) + p^k J \Psi(t),$$

we have both injectivity and the exact image (8.1).

Also $p^e \mathbb{Z}_p^d \subset J \mathbb{Z}_p^d$, so the image contains a full target ball of radius $p^{-(k+e)}$ around each of its points. It is therefore a union of residue classes modulo p^{k+e} .

For the finite count, the Smith normal form of J has diagonal entries with valuations $e_1, \dots, e_d \geq 0$ and $\sum e_i = e$. With $r = m - k \geq e$, the congruence $Jt \equiv v \pmod{p^r}$ has $p^{e_1 + \dots + e_d} = p^e$ solutions whenever it is soluble. The bijection induced by Ψ leaves that count unchanged. Finally, $p^m \mathbb{Z}_p^d \subset p^k J \mathbb{Z}_p^d$ for $m \geq k + e$, so solubility modulo p^m is equivalent to membership in the actual image lattice. This proves the last assertion without extrapolating from finite precision. $\square$

Corollary 8.2 (Uniform determinant valuation). *Suppose $v_p(\det Df(a)) = e$ for every $a \in \mathbb{Z}_p^d$, and fix $k > e$. Then the actual integral fiber cardinality $J_f(\mathbf{b}) = \#f^{-1}(\mathbf{b})(\mathbb{Z}_p)$ is finite and determined by $\mathbf{b} \pmod{p^{k+e}}$. For every $m \geq k + e$,*

$$\#\{a \pmod{p^m} : f(a) \equiv \mathbf{b} \pmod{p^m}\} = p^e J_f(\mathbf{b}). \quad (8.2)$$

In particular, both the image and its fiber-cardinality strata are clopen.

Proof. Partition the source into its p^{kd} residue balls modulo p^k . Each ball contributes either one actual preimage or none, and its image membership depends only on the target class modulo p^{k+e} by lemma 8.1. Sum the point counts and the finite congruence counts over these disjoint source balls. $\square$

8.2 The good-prime consequences

For F , every odd prime has $e = 0$, so we can take $k = 1$. Each source point modulo p lifts uniquely above every target lift with that residue. Thus, exactly and not just asymptotically,

$$\Pr_{\mathbf{b} \in \mathbb{Z}_p^3} (\#F^{-1}(\mathbf{b})(\mathbb{Z}_p) = j) = \frac{N_j(p)}{p^3}, \quad p \text{ odd}, j = 0, 1, 3. \quad (8.3)$$

This is the direct p -adic consequence of the previously known finite-field formulas. In particular,

$$\rho_p := \text{vol}(F(\mathbb{Z}_p^3)) = \begin{cases} 7/9, & p = 3, \\ \frac{2}{3} + \frac{1}{3p} - \frac{2}{3p^2} + \frac{2}{3p^3}, & p > 3. \end{cases} \quad (8.4)$$

The same lifting proof works over the ring of integers of an unramified extension of $\mathbb{Q}_p$, with p replaced in the finite-field counts by its residue cardinality. We do not use that extension below.

9 The exceptional prime: the exact dyadic law

9.1 From 64 source balls to 512 target classes

At $p = 2$, the determinant has constant valuation one. Taking $e = 1$, $k = 2$ in [lemma 8.1](#) partitions the source into 64 balls $a + 4\mathbb{Z}_2^3$. Their images are

$$F(a) + 4J_F(a)\mathbb{Z}_2^3. \quad (9.1)$$

Because $v_2(\det J_F(a)) = 1$, the matrix $J_F(a) \bmod 2$ has rank two. Thus each image in (9.1) consists of exactly four classes modulo 8, specifically

$$\{F(a) + 4J_F(a)h \pmod{8} : h \in \{0, 1\}^3\}. \quad (9.2)$$

Every target in such an image has exactly one preimage in that source ball. Therefore the desired integral fiber cardinality is obtained by counting how many of the 64 four-element sets contain the target class.

Theorem 9.1 (Exact dyadic integral law). *For the fixed map F in (1.1), the number $j(\mathbf{b}) = \#F^{-1}(\mathbf{b})(\mathbb{Z}_2)$ is determined by $\mathbf{b} \bmod 8$. Its Haar distribution is*

$$\begin{array}{c|cccc} j & 0 & 1 & 2 & 3 \\ \hline \Pr(j(\mathbf{b}) = j) & 21/32 & 7/32 & 3/32 & 1/32. \end{array} \quad (9.3)$$

In particular,

$$\text{vol}(F(\mathbb{Z}_2^3)) = \frac{11}{32}. \quad (9.4)$$

For every $m \geq 3$, the possible fiber sizes of $F : (\mathbb{Z}/2^m\mathbb{Z})^3 \rightarrow (\mathbb{Z}/2^m\mathbb{Z})^3$ are 0, 2, 4, 6, with counts

$$2^{3m} \left(\frac{21}{32}, \frac{7}{32}, \frac{3}{32}, \frac{1}{32} \right) \quad (9.5)$$

respectively. Modulo 8 is necessary for image membership: modulo 4 does not suffice.

Proof. The reduction to (9.2) proves that a finite table at modulus 8 settles the infinite-precision assertion. The exact table is below. Each row fixes the parity of (A, B, C) and thus contains 64 target classes modulo 8.

$(A, B, C) \bmod 2$	$j = 0$	$j = 1$	$j = 2$	$j = 3$
(0, 0, 0)	0	48	0	16
(0, 0, 1)	32	32	0	0
(0, 1, 0)	48	0	16	0
(0, 1, 1)	64	0	0	0
(1, 0, 0)	32	32	0	0
(1, 0, 1)	64	0	0	0
(1, 1, 0)	48	0	16	0
(1, 1, 1)	48	0	16	0
Total	336	112	48	16

Here is a complete finite enumeration defining and checking the total row using only integer arithmetic. The derivative is not needed in this version: [corollary 8.2](#) says that each actual preimage contributes exactly two source residues modulo 8.

```

from collections import Counter
from itertools import product

def F(x, y, z):
    u = 1 + x*y
    h = u*u*z + y*y*(1 + 3*u)
    return u*h, y + 3*x*h, x*(5 - 3*u - x*x*z)

count = Counter(tuple(v % 8 for v in F(*a))
                 for a in product(range(8), repeat=3))
assert all(c % 2 == 0 for c in count.values())
actual = {b: count[b] // 2
          for b in product(range(8), repeat=3)}
assert Counter(actual.values()) == {0:336, 1:112, 2:48, 3:16}

```

The accompanying certificate independently constructs all 64 sets in (9.2) using the product-rule Jacobian and verifies equality *for every target residue*, not merely agreement of the total histogram. It also reproduces each parity row. This finite integer enumeration is the computational part of the proof; the all-precision part is [lemma 8.1](#).

Divide the total row by 512 to obtain (9.3) and sum its nonzero columns to obtain (9.4). Equation (8.2) multiplies actual cardinalities by two at every modulus 2^m , $m \geq 3$, proving (9.5). Finally, the target classes (0, 1, 0) and (0, 1, 4) agree modulo 4; the former has two integral 2-adic preimages and the latter none, as certified by the same pointwise table. Hence image membership cannot be determined modulo 4. $\square$

9.2 A two-point integral fiber and the importance of the domain

The target (0, 1, 0) has three rational preimages:

$$(0, 1, -4), \quad (2, -1/2, 5/4), \quad (-2, 1, 2).$$

Only the first and third are in $\mathbb{Z}_2^3$, so its dyadic integral fiber has two points. This is compatible with the three-point rational fiber and the known geometric law. In fact, it also has exactly two ordinary integer preimages.

Remark 9.2 ([write] An intake observation: the project’s collision fiber). This remark is *not* a claim of the manuscript. It was observed when the manuscript was taken into ProveIt and checked there by exact

computation (SymPy 1.14.0: evaluation, and a lex Gröbner basis whose x -polynomial is $x(x-1)(x+1)$). Apply [theorem 2.2](#) to the project’s collision value $(0, -2, 0)$ (`collisionValue`, [section 1.3](#)). Since $C = 0$ there is the boundary point $(0, -2, -16)$, and $g(T) = -2T(T+1)$ has the simple roots $t = 0$ and $t = -1$, which reconstruct $(-1, 1, 5)$ and $(1, -2, 8)$. So

$$F^{-1}(0, -2, 0)(\overline{\mathbb{Q}}) = \{(-1, 1, 5), (0, -2, -16), (1, -2, 8)\},$$

three *integral* points: F has an integral triple collision. The project records and formalizes only the first two (`collision0`, `collision1`). Along the project’s family, whose parameter (called t in the project README) is written τ here to keep it apart from the root $t = y + 1/x$, $F(\tau, -1/\tau, 5/\tau^2) = F(0, 2/\tau, -16/\tau^2) = (0, 2/\tau, 0)$ and the third point is $(-\tau, 2/\tau, 8/\tau^2)$. The torus action $F(x/s, sy, s^2z) = (s^2P, sQ, R/s)$ at $s = -\frac{1}{2}$ carries these three points to $(2, -\frac{1}{2}, \frac{5}{4})$, $(0, 1, -4)$ and $(-2, 1, 2)$, the fiber over $(0, 1, 0)$ displayed above; its two integral points have largest coordinate 4, against 16 for the project’s integral collision.

There are three different distributions that must not be interchanged:

Source and target	Fiber sizes	Image fraction or measure
$\mathbb{F}_2^3$	0, 1, 3	$6/8 = 3/4$
$(\mathbb{Z}/4\mathbb{Z})^3$	0, 2, 6	$28/64 = 7/16$
$\mathbb{Z}_2^3$	0, 1, 2, 3	$11/32$
$(\mathbb{Z}/2^m\mathbb{Z})^3, m \geq 3$	0, 2, 4, 6	$11/32$

The actual modular histograms for moduli 2, 4, 8 are respectively

$$(N_0, N_1, N_3) = (2, 5, 1), \quad (N_0, N_2, N_6) = (36, 26, 2), \quad (N_0, N_2, N_4, N_6) = (336, 112, 48, 16).$$

Also, passing to a finite field $\mathbb{F}_{2^r}$ is not the same operation as passing to the ring $\mathbb{Z}/2^r\mathbb{Z}$. The characteristic-two finite-field formulas in [\[4\]](#) do not by themselves imply the dyadic law.

As a consistency check, the expected integral dyadic fiber size is

$$\frac{7}{32} + 2\frac{3}{32} + 3\frac{1}{32} = \frac{1}{2}.$$

This also follows from summing the contributions of 64 source balls, each with an image of measure $1/128$. The corresponding expected fiber size at an odd prime is one. These are exactly the factors suggested by the determinant’s local absolute value, but the argument here is a finite lattice count rather than an appeal to a change-of-variables formula.

10 Sparse rational images and a measure-zero congruence closure

10.1 A closed subset of profinite measure zero

Theorem 10.1. *The compact image $F(\widehat{\mathbb{Z}}^3)$ has Haar measure zero in $\widehat{\mathbb{Z}}^3$. Nevertheless, its intersection with $\mathbb{Z}^3$ contains the Zariski-dense split-plane family of [corollary 4.4](#).*

Proof. By [theorem 5.4](#), the image is the product of the local images. For $p > 3$, [\(8.4\)](#) gives $\rho_p < 3/4$. The image is contained in the cylinder imposing local membership at any chosen N primes greater than three, and that cylinder has measure at most $(3/4)^N$. Letting N tend to infinity proves measure zero. The second assertion was already proved in [theorem 5.4](#). $\square$

This is a density statement in a compact topological space. It neither asserts emptiness nor contradicts Zariski density in a plane. Topological measure, polynomial equations, and integer counting measure detect different notions of size.

10.2 A quantitative bound for rationally soluble integral targets

Let

$$\mathcal{A}(H) = \#(F(\mathbb{Q}^3) \cap [-H, H]^3 \cap \mathbb{Z}^3), \quad H \geq 1.$$

Unlike an integer-image sieve, a sieve for $F(\mathbb{Q}^3)$ cannot simply exclude all residue classes outside $F(\mathbb{F}_p^3)$: a rational source may have a p -adic denominator. The correct necessary reduction condition is that the *binary* inverse cubic have a projective root modulo p .

Lemma 10.2. *If $(A, B, C) \in \mathbb{Z}^3 \cap F(\mathbb{Q}^3)$, then $G_{A,B,C}$ has a projective root over $\mathbb{F}_p$ for every prime $p > 2$.*

Proof. The rational source gives a rational projective root by [theorem 2.2](#). Represent it by coprime integers (T, S) . At every prime they are not both zero after reduction, and the homogeneous equation remains valid. The coefficient -2 ensures that the reduced binary polynomial is not the zero polynomial for odd p . The reduced root need not remain simple, which is why the stronger exclusion would be invalid. $\square$

Theorem 10.3 (An elementary quantitative sieve). *Let $p_1, \dots, p_N$ be distinct primes greater than three and $M = \prod_{i=1}^N p_i$. Then*

$$\mathcal{A}(H) \leq \left(\prod_{i=1}^N \alpha_{p_i} \right) (2H + 1 + M)^3, \quad \alpha_p = \frac{2}{3} + \frac{1}{3p} + \frac{1}{3p^2} - \frac{1}{3p^3} < \frac{3}{4}. \quad (10.1)$$

In particular, $F(\mathbb{Q}^3) \cap \mathbb{Z}^3$ has zero density in integer boxes. There is an absolute $c > 0$ such that

$$\mathcal{A}(H) \ll H^3 \exp(-c\sqrt{\log H}). \quad (10.2)$$

Proof. By (7.5) and [lemma 10.2](#), a fraction $I(p)/p^3$ of target residues is forbidden at p . The allowed fraction is exactly α_p . The Chinese remainder theorem makes the number of allowed classes modulo M equal to $M^3 \prod_i \alpha_{p_i}$. Each class contains at most $((2H + 1)/M + 1)^3$ points of the integer cube. This proves (10.1). For $p \geq 5$, the positive correction $1/(3p) + 1/(3p^2)$ is at most $2/25 < 1/12$, proving $\alpha_p < 3/4$.

For density zero, first fix N , divide by $(2H + 1)^3$, and let $H \rightarrow \infty$. The upper limit is at most $(3/4)^N$; then let $N \rightarrow \infty$. For the stated quantitative estimate, use the elementary Bertrand bound that between n and $2n$ there is a prime for every integer $n > 1$. Taking the primes in increasing order starting at five gives $p_i < 2^{i+2}$, and hence

$$M < 2^{N(N+5)/2}.$$

Choose the largest nonnegative integer N with $N(N+5)/2 \leq \log_2 H$. Then $M \leq H$ and $N = \sqrt{2 \log H / \log 2} + O(1)$. Substituting into (10.1) gives

$$\mathcal{A}(H) \ll H^3 \exp \left(-\log(4/3) \sqrt{\frac{2}{\log 2}} \sqrt{\log H} \right),$$

with an absolute multiplicative constant, proving the claim. Only the quantitative refinement uses Bertrand's theorem; the zero-density conclusion uses neither it nor Chebotarev. $\square$

This weak but explicit bound is not advertised as an optimal thin-set estimate. It is useful here because it is derived directly from the same inverse cubic and makes the necessary rational-versus-integral sieve correction transparent.

11 Verification artifacts and the route to formal proofs

11.1 What was actually executed

The accompanying archive contains three independent-purpose checkers and an exact rational inverse utility. They were run in the preparation of this article, with results recorded under `certificates/`. [write] In this report that directory is shipped as `data/`, with the same file names; two of the scripts still write to `../certificates/` next to their own directory ([section C.3](#)).

Symbolic identities. `verify_symbolic.py`, run with SymPy 1.14.0, verifies 27 exact identities. These include $\det JF = -2$, both inverse charts, the inverse cubic and derivative identities, the discriminant, the symbolic one-parameter factorization, all three symbolic family branches, and characteristic-three identities. Every zero test is a polynomial-numerator identity after exact rational simplification; no floating-point tolerance is involved.

Finite local certificates. `verify_finite.py` uses only the Python standard library. It checks factored and expanded polynomial evaluators, all three one-parameter branches for $2 \leq n \leq 100$ with exact fractions, the complete target-wise modulus-eight dyadic certificate in two ways, and stabilization for every target modulo 16 and 32. It also reproduces the known finite-field laws for primes from 3 through 31 and checks good-prime lifting at moduli 9, 27, 25. Finally, it constructs exact CRT preimages for four composite moduli. The finite range in these checks is not the theorem’s range; the universal conclusions are proved in the preceding sections.

The split plane. `verify_split_plane.py` uses only exact integers and fractions. It checks all 1600 distinct-root fibers with $(a, b) \in [-20, 20]^2$, including the exact denominator $\gcd g^2$. Its parameter counts through $H = 250$ provide a check on [theorem 4.3](#), not evidence substituted for the Möbius proof.

11.2 Trust boundaries

A successful Python or SymPy computation is not a Lean kernel proof. The new theorems in this article have not been compiled in Lean or Rocq, and no claim is made that they are already part of ProveIt’s trusted theorem boundary. The dyadic total is a finite, reproducible computer-assisted enumeration plus an ordinary mathematical proof of its all-precision significance. A fully kernel-checked version should verify that finite enumeration with a proved checker.

Conversely, the main integral-family and parameter-density theorems do not depend on computational discovery being trustworthy. Their proofs use explicit factorization, a complete inverse theorem, $\gcd$ identities, and elementary counting, all written out in the article. The scripts supply additional independent checks of the formulas.

11.3 A dependency-conscious formalization plan

A practical Lean extension should proceed in four layers.

Layer 1: exact algebra over a field. Reuse the repository’s definition of F . Formalize [\(2.3\)](#), [\(2.5\)](#), and [\(2.6\)](#) over a field with $2 \neq 0$. The central API should state both directions of the preimage characterization

and explicitly separate $x = 0$. Polynomial normalization proves the identities; the main logical obligation is completeness, not evaluation of examples.

Layer 2: rational denominators and local selection. Formalize [theorems 3.1](#) and [4.1](#) over $\mathbb{Q}$ and $\mathbb{Z}$. The no-integer point result needs only denominator arithmetic once completeness is available. Before introducing p -adic numbers, prove the stronger constructive congruence statement for every natural modulus by modular inverses and CRT. For each prime, the rational selected branch is already known explicitly; embedding it into $\mathbb{Q}_p$ should be a final transport step rather than a search inside a local field.

Layer 3: the finite dyadic certificate. Represent source and target residues as triples of $\mathbb{Z} \bmod 8$, prove that the evaluator is reduction of the formal polynomial, and verify the 512-entry count. The semantic bridge is crucial: a theorem about an uninterpreted list of integers is insufficient. The independent 64-ball version can instead certify image incidence using the mod-two Jacobian. No global search or oracle is needed at this layer.

Layer 4: analysis and external arithmetic. Formalize the inverse-ball lemma using completeness and contraction, then connect Layer 3 to actual $\mathbb{Z}_2$ fibers. Haar measures of cylinders finish the distribution theorem. The rational Hasse theorem is a separate branch requiring a suitable Chebotarev theorem; it is not a dependency of the explicit integral counterexamples. The parameter density and thin sieve form further analytic-number-theory branches. Keeping these branches separate prevents a deep external result from becoming an unnecessary assumption of the elementary headline family.

12 Further research questions

The following questions are proposed research targets. They are not represented as settled results, and their priority status would require additional literature review before publication.

12.1 A minimal congruence description at the prime two

Research question 12.1. Can the 512-entry dyadic certificate be compressed into a short list of explicit congruence conditions on (A, B, C) , with a hand-checkable case proof for each integral fiber size?

[Added 30 September 2026, batch 64: partly answered, for image membership only, by Part III: Proposition [32.2](#) compresses membership in $F(\mathbb{Z}_2^3)$ to the five-row table $E(A, B, C)$ of congruences modulo 8. The integral fiber sizes $j = 1, 2, 3$ are not compressed, and that proof is still the exhaustive 512-case check.]

[theorem 9.1](#) proves that modulus eight suffices and modulus four does not. Thus the remaining problem is logical and algebraic compression, not an unbounded precision search. The eight parity rows suggest a small branching structure. A compact description would improve both exposition and formal verification.

12.2 All locally soluble integral fibers, not just split integral roots

Research question 12.2. Classify the integral targets with three rational preimages, and then those with a single rational preimage and a quadratic pair, according to local integral solubility at every prime.

[Added 30 September 2026, batch 58: the completely split sector is classified in Part II (Theorem [18.2](#), with Theorem [20.1](#)), in terms of the root numerators; the rational-plus-quadratic sector remains open (Research

question 24.1.)] [Added 30 September 2026, batch 64: answered by Part III on every slice $C \neq 0$, in both sectors and in the target coefficients: Theorem 29.2 is a complete local certificate (an integer root of the monic inverse polynomial, $\gcd(3BC - 4, 27AC^2 - 4)$ a power of two, and a table modulo 8), and Corollary 33.1 the complete Hasse test for a rational point with an irreducible quadratic pair. The rational-plus-quadratic sector is no longer open.]

For arbitrary split rational fibers, lemma 3.3 already gives a criterion once denominators are known. The challenge is to express it in target coefficients without factoring every fiber individually. The quadratic case is qualitatively different: the nonrational pair can supply local integral points at primes where the rational point fails integrality. A complete classification must record both factorization and valuations of the reconstruction derivative.

12.3 Target-height asymptotics for integral Hasse failures

Research question 12.3. What is the asymptotic number of locally integral but globally nonintegral targets on $C = 2$ in boxes $|A| \leq H_A$, $|B| \leq H_B$?

[Added 30 September 2026, batch 58: for completely split fibers and square boxes $|A|, |B| \leq T$ Part II proves $\#\mathcal{H}_c^{\text{sp}}(T) \sim \kappa(c)T^{2/3}$ for every fixed $c \neq 0$, with $\kappa(2) = 27\Gamma(1/3)^2/(8\pi^2\Gamma(2/3))$ (Theorem 15.3); fibers with one rational point and a quadratic pair, unequal box sides, uniformity in c and an error term remain open.] [Added 30 September 2026, batch 64: answered by Part III for square boxes in all sectors. On $C = 2$ the locally integral targets, all integral Hasse failures and the nonsplit ones each number $\frac{27}{2\pi^2}T \log T + O(T)$ in $|A|, |B| \leq T$ (Theorem 29.3), and the integral targets $2^{5/3}T^{1/3} + O(1)$; the completely split sector ($T^{2/3}$) is not the dominant one. Unequal box sides remain open (Research question 41.4), as do other slices, a second-order term and a power-saving error.]

The density $27/(4\pi^2)$ counts ordered root parameters in a square. Passing to target heights introduces a nonlinear region, a sixfold symmetry away from the discriminant, and cusp-like ranges where one root is small. These are genuine counting issues. Establishing a main term with an explicit Euler product would sharpen the present Zariski-density result substantially.

12.4 Density in the entire three-dimensional target space

Research question 12.4. Does the set of integral Hasse failures for this fixed F form a Zariski-dense subset of all of $\mathbb{A}^3$, rather than just the plane $C = 2$?

[Added 30 September 2026, batch 58: answered affirmatively in Part II (Theorem 15.2): the failures are Zariski dense in every plane $C = c \neq 0$, hence in $\mathbb{A}^3$; the plane $C = 0$ has none, since $F(0, B, A - 4B^2) = (A, B, 0)$; the dense family persists over every $\mathbb{Z}[S^{-1}]$ (Theorem 19.3).]

The present construction does not answer this: all its targets have the same third coordinate. Rational weighted symmetries can move that coordinate, but they can introduce new prime denominators and do not preserve all local-integrality conditions automatically. A valid extension must control those exceptional primes explicitly.

12.5 Exact local laws over extensions of $\mathbb{Q}_2$

Research question 12.5. Determine integral fiber distributions over finite extensions of $\mathbb{Q}_2$, including ramified extensions.

[Added 30 September 2026, batch 64: Part III bears on this for image membership only: over the valuation ring of any finite extension, a solution modulo 8 lifts (Lemma 32.1), so membership is decided in the finite quotient (Theorem 38.1). The fiber distributions, and the least modulus deciding membership (Research question 41.6), remain open.]

Over a discretely valued extension, the same contraction proof uses the valuation of 2 in that field. It therefore gives a finite precision bound, but the number of residue classes and the rank patterns can change. It would be particularly useful to find a uniform formula in residue cardinality and ramification index, rather than a new exhaustive table for each extension.

12.6 Effective local witnesses for rational insolubility

Research question 12.6. For an integral target with irreducible inverse cubic, how small can one guarantee a prime p with no $\mathbb{Q}_p$ -preimage, in terms of the target’s height and cubic discriminant?

[theorem 6.2](#) uses qualitative Chebotarev. An effective specialized estimate could turn the abstract rational Hasse principle into a practical proof-producing negative decision procedure. Unconditional and conditional bounds should be separated, and the finite exceptional set where the inverse polynomial is not an integral separable cubic must be included in the implementation.

12.7 Finer image-density estimates

Research question 12.7. Can one obtain the correct exponent or a main-term asymptotic for $\mathcal{A}(H)$, rather than the elementary bound in [theorem 10.3](#)?

The projective root condition gives a uniform positive congruence loss, but the simple CRT argument uses the moduli inefficiently. A large-sieve approach or a direct height analysis of rational roots might yield a substantially stronger bound. Such a result would concern the arithmetic image of a concrete nonproper étale map, not merely its geometric degree.

12.8 Minimal arithmetic complexity under stable equivalence

Research question 12.8. Among the integral stable representatives in ProveIt, which realizes integral Hasse failures with the smallest target height, source denominator height, degree, or ambient dimension?

[proposition 5.3](#) guarantees existence but not small formulas after transport. Complexity should be reported as a vector of separate measures; minimizing degree alone can raise dimension and coefficient height. A search accompanied by exact certificates could produce useful representatives without claiming a global optimum from a bounded search.

12.9 A reusable formal library for finite arithmetic fibers

Research question 12.9. Can the combination of a complete inverse certificate, denominator covering, and the inverse-ball theorem become a generic proof-producing workflow for polynomial maps with small geometric fibers?

The division of labor in [section 11](#) suggests a reusable architecture: exact elimination produces a candidate inverse; algebraic lemmas certify completeness; finite local tables are reflected; and an analytic theorem

upgrades them to all precisions. This would convert many computational observations into theorems with an explicit trust boundary, rather than a collection of map-specific numerical checks.

13 Conclusion

The original polynomial witness raises an arithmetic question quite different from noninjectivity. For this particular map, rational local solubility everywhere does force a rational point, but integral local solubility everywhere does *not* force an integral point. The failure occurs in explicitly split three-point fibers, survives inversion of any fixed finite set of primes, and fills a Zariski-dense subset of a target plane. On that split-root parameter plane its exact density is $27/(4\pi^2)$.

The local side is also completely explicit at the exceptional prime: modulus eight determines the integral dyadic image and every integral fiber size, with image measure $11/32$. This is an arithmetic refinement of the existing cubic model, not a replacement for the known geometric or finite-field results. The general inverse-ball argument explains why a finite certificate can settle the infinite local question.

These results provide concrete new targets for ProveIt’s formalization program while keeping their status distinct: explicit mathematical proofs and replayed exact computations are supplied here; new kernel-checked formal proofs remain to be implemented.

Part II. Zariski-dense integral Hasse failures and split-fiber counting

Sections 1–13 and Appendices A, B and C.1–C.3 form the original report (Part I) of 24 September 2026, unchanged except for dated notes after Research questions 12.2–12.4, on the title page, and in Appendices A and C. Part II was added on 30 September 2026 in batch 58 of ProveIt’s incoming-report intake, from one manuscript of 29 September 2026 written against Part I as pinned. It answers Research question 12.4 in full, and Research questions 12.3 and 12.2 in the completely split sector only. Section 14 states exactly what is answered, fixes the notation against Part I and lists what is printed twice; Sections 15–27 are the manuscript.

[Added 30 September 2026, batch 64: Part III (Section 28) answers Research question 12.2 in the rational-plus-quadratic sector too, and Research question 12.3 for square boxes in all sectors, where the split sector counted here is not the dominant one. It added further dated notes to Parts I and II, listed in Appendix C.6.]

14 Relation to Part I

[write] This section was written when the manuscript was added to the report.

14.1 The manuscript

Part II is batch-58 manuscript 01, *Zariski-Dense Integral Hasse Failures for the ProveIt Keller Map: Complete split-fiber criteria and sharp target-height asymptotics*, dated 29 September 2026, with the author line “AI-assisted research draft prepared for Vladimir Reshetnikov” (the same in its PDF metadata). It was delivered as the archive ProveIt_Dense_Integral_Hasse and is pinned to ProveIt commit 1085b506d65e207a05b7e9c861bb1fe88432fe38; its provenance, shipped files and editorial changes are in Appendix C.4. Its title-page abstract and status note read as follows.

We study the arithmetic fibers of the three-variable polynomial map with Jacobian determinant -2 already treated in ProveIt’s Jacobian-conjecture project. We resolve the repository report’s question asking whether its integral Hasse failures are Zariski dense in the entire target space. An explicit polynomial family gives such failures densely in *every* integer plane $C = c \neq 0$; the plane $C = 0$, by contrast, has an integral preimage over every integral target. The dense construction persists over any prescribed ring of S -integers.

We also give a complete arithmetic parametrization of the distinct, completely split rational fibers. Local solubility reduces to elementary congruences at primes dividing $2c$ and a single greatest-common-divisor condition elsewhere. The dyadic calculation has three exceptional low valuation cases and a uniform higher-valuation formula. For each fixed $c \neq 0$, only finitely many completely split fibers meet $\mathbb{Z}^3$; all of these exceptions admit a finite divisor enumeration. Combining these facts with a two-dimensional lattice sieve and an explicit cusp estimate proves the sharp asymptotic

$$\#H_c^{\text{sp}}(T) \sim \kappa(c)T^{2/3},$$

where $\kappa(c) > 0$ is given by a beta integral and an explicit Euler product. In particular,

$$\kappa(2) = \frac{27 \Gamma(1/3)^2}{8\pi^2 \Gamma(2/3)}.$$

Thus the target-height question is settled in the completely split sector, [write] for square boxes $|A|, |B| \leq T$, not merely in a root-parameter square. All universal claims are proved in the text; exact computational checks and constructive congruence witnesses are supplied separately.

Status and scope. The map, the inverse cubic, and the previous split-plane construction are inherited results. The density extension, complete split-fiber local criteria, and sharp counting theorem are the contributions of this draft relative to the inspected sources. Worldwide priority is not certified. This is an unrefereed research manuscript, not a Lean or Rocq formalization, and not a new disproof of the Jacobian conjecture.

14.2 What Part II answers, and what it does not

- *Research question 12.4 is answered, affirmatively and in full.* By Theorem 15.2 the integral Hasse failures are Zariski dense in every plane $C = c \neq 0$, hence in $\mathbb{A}^3$. The plane $C = 0$ has none: every integral target $(A, B, 0)$ has the integral preimage $(0, B, A - 4B^2)$, which is Part I's boundary point (2.7). The dense family stays globally insoluble over every $\mathbb{Z}[S^{-1}]$ (Theorem 19.3); this extends the Zariski density of Part I's Corollary 5.2 from $C = 2$ to every nonzero slice (that corollary also counts the parameters on $C = 2$, which Part II does not do for other slices).
- *Research question 12.3 is answered only in the completely split sector, and only for square boxes.* It asks for the number of locally integral but globally nonintegral targets on $C = 2$ in boxes $|A| \leq H_A, |B| \leq H_B$. Theorem 15.3 proves $\#\mathcal{H}_c^{\text{sp}}(T) \sim \kappa(c)T^{2/3}$ for every fixed $c \neq 0$ in square boxes $|A|, |B| \leq T$, counting only fibers of three distinct rational points; at $c = 2$ the constant is (22.7), and the main term carries the explicit Euler product that the discussion of the question asks for. Not answered: fibers with one rational point and an irreducible quadratic pair (Research question 24.1); unequal box sides $H_A \neq H_B$; uniformity in c (Research question 24.3); an error term (Research question 24.2). The manuscript's sentence "This settles the completely split sector of the prior report's Research Question 12.3" after Theorem 15.3, and the corresponding sentence of its abstract, are printed with the restriction to square boxes added and marked [write]. [Added 30 September 2026, batch 64: this describes Part II only. The question is now answered for square boxes in all sectors by Part III (Theorem 29.3): on $C = 2$ the integral Hasse failures number $\frac{27}{2\pi^2}T \log T + O(T)$, almost all nonsplit, so the split sector above is a lower-order part of that count.]
- *Research question 12.2 is answered in the completely split sector.* Theorem 18.2 gives the complete local criterion for every completely split integral target on every slice $c \neq 0$, in terms of the root numerators, and Theorem 20.1 shows that on each slice only finitely many of them have an integral point. (On $C = 0$ every integral target has an integral point.) A criterion in the target coefficients themselves, without factoring, is Research question 24.4; the rational-plus-quadratic sector remains open (Research question 24.1). [Added 30 September 2026, batch 64: both are answered by Part III: the coefficient criterion by Theorem 29.2, and the rational-plus-quadratic sector by Corollary 33.1 (Section 28.2).]
- *Research questions 12.1 and 12.9 are related, not answered.* Research question 24.4 (coefficient-only local certificates) extends Research question 12.1 from the prime two to all primes, for split fibers; Research question 24.7 (certified finite-fiber arithmetic) is essentially Research question 12.9. Both are printed with a cross-reference. Research questions 12.5–12.8 are untouched.

14.3 Consistency with Part I

Where the two Parts meet they agree. Each item was checked at the write, by hand or by exact expansion.

- Part II's three-parameter target (19.1) at $c = 2$ is Part I's two-parameter map (4.1), written here

$\Phi_1(a, b) = (abc', 2(ab + ac' + bc'), 2)$ with $c' = 1 - a - b$ (Part I's third root c , renamed c' here):

$$\Phi(2, a, b) = (ab - ab(a + b), 2(a + b) - 2(a^2 + ab + b^2), 2) = \Phi_1(a, b).$$

- Densities. At $c = 2$ coefficient integrality forces both root numerators to be even, $a = 2s$, $b = 2t$, so every completely split fiber on $C = 2$ has integer roots, and the numerator lattice has index four in the root lattice. Accordingly $\rho(2) = \frac{9}{\pi^2} \cdot \frac{3}{16} = \frac{27}{16\pi^2} = \frac{1}{4} \cdot \frac{27}{4\pi^2}$ is one quarter of Part I's parameter density (4.6), and Part II's integer-root density $\delta(2)$ of (21.4) is $27/(4\pi^2)$ itself, as the manuscript notes.
- At $c = 2$ the two local criteria coincide. Part II's odd-prime condition (17.9) reads: $\gcd(2s - 2t, 6s - 2) = 2 \gcd(s - t, 3s - 1)$ has no odd prime factor. Its dyadic condition for $v_2(c) = 1$ (exactly one of $s, t, 1 - s - t$ odd) says that $\gcd(s - t, 3s - 1)$ is odd, since the three roots have odd sum, and are all odd exactly when $s - t$ and $3s - 1$ are both even. Together these are Part I's condition $\gcd(a - b, 3a - 1) = 1$ of Theorem 4.1.
- The constant at $c = 2$ is $\kappa(2) = I \cdot 8^{2/3} \rho(2) = 4I \cdot \frac{27}{16\pi^2} = \frac{27\Gamma(1/3)^2}{8\pi^2\Gamma(2/3)} \approx 1.81235$.
- The independent modulo-eight image of Section 23 has 176 of 512 target classes, and $176/512 = 11/32$ is Part I's dyadic image measure (Theorem 9.1).
- The plane $C = 0$: Part II's formula $F(0, B, A - 4B^2) = (A, B, 0)$ is Part I's boundary point (2.7); the collision fiber of Remark 9.2 contains its instance $(0, -2, -16)$.
- The table of integral split exceptions after Theorem 20.1 lists none on $C = 2$, as Part I's Theorem 4.1 requires: every completely split fiber there has distinct integer roots and no integral point.

The ten counts for $T = 10^3$ and $T = 10^4$ in the table of Section 22 were also recomputed at the write by an independent enumeration, not the delivered program: distinct numerators with integral coefficients and $|A|, |B| \leq T$, the three points rebuilt from the inverse chart and checked to map to the target in exact fractions, and a target counted when no point is integral and the three point denominators have greatest common divisor one (Lemma 3.3). All ten agree.

14.4 Notation: Part I against Part II

Part II keeps the manuscript's letters, with one exception, so several letters mean different things in the two Parts. Table 1 fixes them; within Part II only the Part II column applies, and the last column prints the tempting false reading.

Table 1: Symbols whose meaning differs between the Parts.

Symbol	Part I	Part II	Not to be read as
c	the third integer root, $a + b + c = 1$, in (4.1); the third target coordinate is C	the fixed third target coordinate, $\mathbf{b} = (A, B, c)$	a root
a, b, r	a, b integer roots on $C = 2$; $r \in \{a, b, c\}$ an index	numerators of the roots $a/c, b/c, r/c$, with $a + b + r = 2$ (17.1); in Section 19, s, t are integer roots and $a = cs, b = ct$. Also $k = 4r + 2$ in (19.7) and the slope $y = rx$ in (22.5)	roots
Φ	$\Phi(a, b)$ on $C = 2$ (4.1)	$\Phi(c, s, t)$ (19.1); $\Phi(2, a, b)$ is Part I's $\Phi(a, b)$ (Section 14.3)	different maps

Symbol	Part I	Part II	Not to be read as
D_r, D_j	products of differences of roots	products of differences of root numerators (17.6); D also a signed divisor of $2c$ in Theorem 20.1	the same normalization
d	$d(t) = g'(t)$ (2.5); d_i common denominators	$d = s - t$ (19.2); a signed factor $D = dv$; a divisor in (20.3)	the derivative
ρ	ρ_p , the Haar measure of $F(\mathbb{Z}_p^3)$ (Section 8)	$\rho(c)$, the density of locally integral numerator pairs (15.4)	an image measure
$27/(4\pi^2)$	density of ordered integer-root pairs on $C = 2$ (4.6)	$\delta(2)$ (21.4); the numerator density is $\rho(2) = 27/(16\pi^2)$	a density of targets
T	the indeterminate of $g_{A,B,C}(T)$	the indeterminate of g , and the height bound $ A , B \leq T$; $L = T^{1/3}$	
H	parameter heights ($M(H)$, $\mathcal{A}(H)$); box sides H_A, H_B in Research question 12.3	not used; the height bound is T	
S	a finite set of primes; the second projective variable in (2.8)	a finite set of primes (Theorem 19.3); the manuscript's sum S is printed Σ	a sum
Σ	not used	$B_0 + R_0$ in the proof of Lemma 18.1; $s + t$ in (19.4)	a summation sign
h	$h = u^2z + y^2(1 + 3u)$	the same; also the odd part of c ($c = 2h$, $c = 2^e h$) in Section 18	
I	$I(q)$ in (7.5), a finite-field count	the beta area $\Gamma(1/3)^2/(2\Gamma(2/3))$ (15.4); $\mathcal{I}_c^{\text{sp}}$ the integral split exceptions	
τ	the project's family parameter (Remark 9.2)	the number of divisors in (20.3)	
q	a finite field's cardinality	a prime in the proof of Theorem 19.3; $n = 2q + 1$ in (19.7)	a field size
R	the third coordinate of F	the same; also the cutoff $R \geq 4$ of Lemma 22.1 and the square $[-R, R]^2$ in Lemma 21.1	
x, y	source coordinates	the same; also the scaled numerators $a = Lx$, $b = Ly$ in Section 22	
e	an exponent, as in p^e	$e = v_2(c)$	
f, g	f a polynomial map (Lemma 3.3); g the inverse cubic	the same g ; also valuations f, g in Lemma 17.1 and (18.4)	
n, k	$n \geq 2$ indexes the family $\mathbf{b}_n$ (Theorem 3.1); k a field or a precision	n odd, $k \equiv 2 \pmod{4}$, $k \geq 6$ in Theorem 19.1	the index of $\mathbf{b}_n$
$\sigma, \kappa, \lambda, \delta$	not used	$\sigma(e)$ (15.3), $\kappa(c)$ (15.5), $\lambda(c)$ and $\delta(c)$ (21.4)	

The one rename: the manuscript's S for a sum of two half-numerators (in the proof of Lemma 18.1) or of two integer roots (before (19.4)) is printed Σ , because S is the finite set of primes of Theorem 19.3 and of Part I's Section 5. No normalization was changed. Part I's Theorem 3.1 (the one-parameter family $\mathbf{b}_n$ on $C = 2$) and Part II's Theorem 19.1 (the two-parameter family on every slice) are different theorems.

14.5 What is printed twice, and why

The manuscript restates parts of Part I to be self-contained. Its later sections cite these restatements, and all of its labels are kept, so they are printed where they stand, in a few lines each; they add nothing to Part I, and the

manuscript does not claim them (Section 15).

- The map and its expansion (15.1) are (1.1).
- The chart (16.1)–(16.4) is (2.1)–(2.5) with C written c . Proposition 16.1 is Part I’s Theorem 2.2 without its projective form, with the same proof; Proposition 16.2 and its proof are Part I’s Proposition 2.1. Both Parts reprove the determinant that the project kernel-checks (Section 14.6).
- The lifting fact (16.5) and the modulus-eight statement after it are special cases of Part I’s inverse-ball Lemma 8.1 (with Corollary 8.2 and Theorem 9.1 at two); they are kept as a second route, a direct contraction argument.
- New in their generality, and printed in full: the numerator inverse points (17.7), which extend Part I’s (4.2) from $C = 2$ to every slice; the gcd criterion of Proposition 17.2, which extends Part I’s Theorem 4.1 (Section 14.3); the constructive congruence argument of Section 23, which is the method of Part I’s Corollary 3.4 applied to every locally integral split target; and Theorem 19.3, proved by two assigned denominator primes on a dense two-parameter family, whereas Part I’s Theorem 5.1 uses the one-parameter family b_n .

14.6 The formal project and Part II

The manuscript cites ProveIt’s Lean file `Algebra/JacobianConjecture/Lean/JacobianConjecture/Counterexample.lean` [7] as the source of the map and names no declaration. That file defines the map as `counterexample` (namespace `LeanProofs.JacobianCounterexample`, lines 95–101) and proves `jacobianDet_counterexample`, $\det JF = -2$ over every commutative ring (line 134); Part II uses the map, reproves the determinant (Proposition 16.2), and relies on no formal statement. *None of Part II’s results is formalized.* The project has no p -adic, Hasse-principle or counting material, and the placement of this report beside its Lean and Rocq developments confers no formal status. The manuscript’s formalization order (the last subsection of Section 23) is a proposal.

15 The problem, the results, and the attribution boundary

15.1 The fixed polynomial map

Throughout, the map is literally the one used in the repository, without rescaling:

$$\begin{aligned} u &= 1 + xy, & h &= u^2z + y^2(1 + 3u), \\ F(x, y, z) &= (P, Q, R) = (uh, y + 3xh, x(5 - 3u - x^2z)). \end{aligned} \tag{15.1}$$

Equivalently,

$$\begin{aligned} P &= (1 + xy)^3z + y^2(1 + xy)(4 + 3xy), \\ Q &= y + 3x(1 + xy)^2z + 3xy^2(4 + 3xy), \\ R &= 2x - 3x^2y - x^3z. \end{aligned}$$

ProveIt provides independent Lean and Rocq/Coq developments for this map, its determinant, and its collisions [7]. The present work uses those definitions, but reproves the algebraic identities needed below. In particular, no unverified assertion about the repository is silently used as a mathematical hypothesis.

Definition 15.1. An *integral Hasse failure* for F is a target $\mathbf{b} \in \mathbb{Z}^3$ such that

$$\mathbf{b} \in F(\mathbb{Z}_p^3) \quad \text{for every prime } p, \quad \mathbf{b} \notin F(\mathbb{Z}^3),$$

and the real fiber is nonempty. All failures constructed or counted in this article have three rational preimages, so the real condition is automatic. They are failures of an *integral* local–global principle, not failures of rational solubility.

Write the target as (A, B, c) , reserving lowercase c for its fixed third coordinate. The notation a, b, r below denotes *numerators of inverse roots*, not target coordinates. This distinction is especially important because two different root parametrizations will be used.

15.2 The specific repository question resolved here

The report *Arithmetic Local–Global Dichotomies for ProveIt’s Three-Variable Keller Map* [write] (Part I of this report) proves Zariski density in the plane $C = 2$ and asks, in its Research Question 12.4, whether the integral failures are dense in all of $\mathbb{A}^3$ [8]. Its discussion correctly warns that rational weighted symmetries can introduce new denominator primes. We control those primes and prove a stronger statement.

Theorem 15.2 (Density in every nonzero slice). *For every nonzero integer c , the integral Hasse failures of F are Zariski dense in the target plane $C = c$. Consequently they are Zariski dense in $\mathbb{A}_{\mathbb{Q}}^3$. For $C = 0$, there are no integral Hasse failures:*

$$F(0, B, A - 4B^2) = (A, B, 0) \quad (A, B \in \mathbb{Z}).$$

The dense family can be chosen to remain globally insoluble over any prescribed $\mathbb{Z}[S^{-1}]$, while still being locally integral at every prime.

The proof is constructive and appears in Section 19. It does not use a density theorem for primes, Chebotarev, or a numerical search.

15.3 A sharp target-height theorem

A fiber is called *completely split* here if it consists of three distinct rational points. For $c \neq 0$, let

$$\mathcal{H}_c^{\text{sp}}(T) = \left\{ (A, B) \in \mathbb{Z}^2 : \begin{array}{l} |A|, |B| \leq T, \quad F^{-1}(A, B, c) \text{ is completely split,} \\ (A, B, c) \in F(\mathbb{Z}_p^3) \text{ for every } p, \\ (A, B, c) \notin F(\mathbb{Z}^3) \end{array} \right\}. \quad (15.2)$$

Write $e = v_2(|c|)$ and define

$$\sigma(e) = \begin{cases} 3/8, & e = 0, \\ 3/16, & e = 1 \text{ or } 2, \\ 3/2^{2e-1}, & e \geq 3. \end{cases} \quad (15.3)$$

Set

$$\rho(c) = \sigma(e) \prod_{\substack{p^v \parallel c \\ p \text{ odd}}} \frac{3}{p^{2v}} \prod_{\substack{p \geq 5 \\ p \nmid c}} \left(1 - \frac{1}{p^2}\right), \quad I = \frac{\Gamma(1/3)^2}{2\Gamma(2/3)}. \quad (15.4)$$

An empty product is one; divisibility by a negative integer is understood through its absolute value. The infinite product converges to a positive number.

Theorem 15.3 (Sharp split-fiber counting). *For every fixed nonzero integer c ,*

$$\#\mathcal{H}_c^{\text{sp}}(T) \sim \kappa(c)T^{2/3}, \quad \kappa(c) = I(2c^2)^{2/3}\rho(c) > 0. \quad (15.5)$$

The completely split fibers on $C = c$ that have an integral point form a finite, effectively enumerable set.

This settles, [write] for square boxes $|A|, |B| \leq T$, the completely split sector of the prior report's Research Question 12.3 about target-height asymptotics. [[write] That question, Research question 12.3 of Part I, asks for boxes $|A| \leq H_A$, $|B| \leq H_B$ on $C = 2$; unequal box sides are not treated here and remain open. The manuscript's sentence has no box restriction; see Section 14.2.] It does *not* count fibers having one rational point and a nonrational quadratic pair; those fibers may have a different arithmetic behavior. [Added 30 September 2026, batch 64: they do. Part III counts them on $C = \pm 2$: in square boxes they number $\frac{27}{2\pi^2}T \log T + O(T)$ (Theorem 29.3), so on these slices the split sector of this theorem is not the dominant part of the integral Hasse failures. The theorem itself is unaffected.]

15.4 What is old, what is proved here, and what remains unclaimed

The inverse cubic and the geometric degree-three model are prior work. They appear in the repository's arithmetic report, which attributes the geometric fiber stratification to Gao and the cubic model to earlier research notes [8, 3]. We give a self-contained derivation rather than claiming priority for that model. The $C = 2$ integral-root family and its root-parameter density $27/(4\pi^2)$ also belong to the prior report.

The new work here consists of the all-slice construction, the treatment of all rationally split fibers through numerator roots, the full bad-prime analysis, the finiteness of globally integral split fibers, and the target-height asymptotic including its cusp calculation. The source review was focused, not exhaustive; the results should receive independent mathematical review before being described as published breakthroughs.

The repository was inspected at commit 1085b506d65e207a05b7e9c861bb1fe88432fe38. The companion SOURCES.md records the exact files and boundaries. [write] It is shipped as 02-dense-hasse-SOURCES.md, and the manuscript's STATUS.md as 02-dense-hasse-STATUS.md. No assertion in this manuscript has been kernel-checked merely by being placed next to a formal development.

16 The complete inverse cubic, with a short determinant proof

16.1 A rational chart

On the source chart $x \neq 0$, put

$$w = \frac{1}{x}, \quad t = y + \frac{1}{x}, \quad c = R(x, y, z). \quad (16.1)$$

Then

$$x = \frac{1}{w}, \quad y = t - w, \quad z = 5w^2 - 3tw - cw^3. \quad (16.2)$$

Direct substitution into (15.1) gives

$$A = wt + t^2 - ct^3, \quad B = 2w + 4t - 3ct^2, \quad C = c. \quad (16.3)$$

For instance, $u = t/w$ and $h = w^2 + tw - ct^2w$, which immediately give the first two coordinates.

Eliminating w yields

$$g_{A,B,c}(T) = cT^3 - 2T^2 + BT - 2A, \quad g(t) = 0, \quad g'(t) = 2w = \frac{2}{x}. \quad (16.4)$$

The derivative condition is essential: only a simple root reconstructs a finite source point.

Proposition 16.1 (Complete inverse for $c \neq 0$). *Over a field of characteristic different from two, the points of $F^{-1}(A, B, c)$ with $c \neq 0$ are in bijection with the simple roots of $g_{A,B,c}$. The inverse is*

$$w = g'(t)/2, \quad (x, y, z) = (1/w, t - w, 5w^2 - 3tw - cw^3).$$

For $c = 0$, there is in addition exactly one source point on $x = 0$, namely $(0, B, A - 4B^2)$.

Proof. The forward implication follows from the chart. Conversely, the second formula in (16.3) becomes B after setting $w = g'(t)/2$. The first coordinate minus A is $g(t)/2$, hence vanishes. This gives a two-sided inverse on $x \neq 0$. If $x = 0$, evaluation of the original map gives $(z + 4y^2, y, 0)$, proving the remaining assertion. $\square$

Proposition 16.2 (Polynomial determinant identity). *In $\mathbb{Z}[x, y, z]$, one has $\det JF = -2$.*

Proof. The determinant of the coordinate change $(x, y, z) \mapsto (w, t, c)$ is $x = 1/w$. The determinant of (16.3), as a map in (w, t, c) , is

$$\det \begin{pmatrix} t & w + 2t - 3ct^2 & -t^3 \\ 2 & 4 - 6ct & -3t^2 \\ 0 & 0 & 1 \end{pmatrix} = -2w.$$

The chain rule gives -2 after localizing at x . Since the polynomial ring embeds in that localization, the identity holds globally. $\square$

Remark 16.3. The map is étale over $\mathbb{Z}[1/2]$, but not over $\mathbb{Z}$ at the prime two. None of the later arguments treats it as an everywhere étale integral map. The dyadic exception is handled explicitly.

16.2 A minimal local lifting fact

We will only need a standard elementary consequence of the determinant. For odd p , every solution modulo p lifts to a solution over $\mathbb{Z}_p$. Here is the precise argument. At a residue solution represented by $z_0 \in \mathbb{Z}_p^3$, the matrix $JF(z_0)$ is invertible over $\mathbb{Z}_p$. The map

$$u \mapsto u - JF(z_0)^{-1}(F(z_0 + u) - \mathbf{b})$$

is a contraction on $p\mathbb{Z}_p^3$: its constant term lies in that ball and the nonlinear remainder gains a factor of p in differences. Completeness produces a root. In particular,

$$p \mid c, \ p \text{ odd}, \ A, B \in \mathbb{Z} \implies (A, B, c) \in F(\mathbb{Z}_p^3), \quad (16.5)$$

because $(0, B, A - 4B^2)$ solves the equations modulo p .

For later computational checks, the corresponding safe statement at two is that a solution modulo eight lifts. Indeed, $JF(z_0)^{-1}$ has entries in $\frac{1}{2}\mathbb{Z}_2$, and the same contraction acts on $4\mathbb{Z}_2^3$ when the residual is in $8\mathbb{Z}_2^3$. Its nonlinear part contracts by at least $1/2$. Thus local membership at two is equivalent to membership in the finite image $F((\mathbb{Z}/8\mathbb{Z})^3)$. The theoretical density calculation below does not depend on enumerating that image.

17 An arithmetic parametrization of every completely split fiber

17.1 Numerator roots rather than only integer roots

Fix $c \in \mathbb{Z} \setminus \{0\}$. If the integer polynomial $g_{A,B,c}$ splits over $\mathbb{Q}$, the rational-root theorem implies that ct is an integer for every root t . Since the sum of the roots is $2/c$, we may write their numerators as

$$a, \quad b, \quad r = 2 - a - b, \quad a, b \in \mathbb{Z}. \quad (17.1)$$

Define

$$P_0 = ab(2 - a - b) = abr, \quad Q_0 = 2(a + b) - a^2 - ab - b^2 = ab + ar + br. \quad (17.2)$$

Then the target coefficients are

$$A = \frac{P_0}{2c^2}, \quad B = \frac{Q_0}{c}, \quad C = c. \quad (17.3)$$

Consequently the exact integrality conditions are

$$2c^2 \mid P_0, \quad c \mid Q_0. \quad (17.4)$$

Conversely, these divisibilities give an integral target whose inverse cubic is

$$g(T) = c(T - a/c)(T - b/c)(T - r/c). \quad (17.5)$$

We always exclude coincidences among a, b, r when speaking of a completely split fiber.

Put

$$D_a = (a - b)(a - r), \quad D_b = (b - a)(b - r), \quad D_r = (r - a)(r - b). \quad (17.6)$$

The three source points are explicitly

$$X_j = \left(\frac{2c}{D_j}, \quad \frac{2j - D_j}{2c}, \quad \frac{D_j(10D_j - 12j - D_j^2)}{8c^2} \right), \quad j \in \{a, b, r\}. \quad (17.7)$$

These follow from $g'(j/c) = D_j/c$ and Proposition 16.1. Thus the parametrization is exhaustive, not merely a supply of examples.

17.2 Odd primes in the leading coefficient

The coefficient divisibilities have a particularly simple local form.

Lemma 17.1 (Odd-prime integrality). *Let p be odd and $p^\nu \parallel c$. The conditions*

$$v_p(P_0) \geq 2\nu, \quad v_p(Q_0) \geq \nu$$

are equivalent to

$$(a, b) \equiv (0, 0), (0, 2), \text{ or } (2, 0) \pmod{p^\nu}. \quad (17.8)$$

The three residue classes are disjoint and have total density $3/p^{2\nu}$.

Proof. Modulo p , the monic polynomial with roots a, b, r becomes $U^3 - 2U^2 = U^2(U - 2)$. Therefore two roots are divisible by p and the third is a unit congruent to two. Call the two divisible roots u, v , and put $f = v_p(u)$, $g = v_p(v)$, allowing infinite valuations for zero. If $f \neq g$, then $v_p(Q_0) = \min(f, g)$ because the unit root multiplies $u + v$, and uv has strictly higher valuation. Thus both valuations are at least ν . If $f = g < \nu$, then $v_p(P_0) = 2f < 2\nu$, a contradiction. The remaining case again has $f, g \geq \nu$. Conversely, two roots divisible by p^ν plainly give the two required divisibilities. Their sum with the third root is exactly two, proving (17.8). $\square$

No extra local-solubility condition is needed at these odd primes, by (16.5). This is the precise control of the new denominator primes that a rational rescaling alone would not provide.

17.3 Odd primes away from the leading coefficient

Let $p \nmid 2c$. The last two coordinates in (17.7) are automatically p -integral. The first is p -integral exactly when D_j is a unit. Among three roots, all three products D_j are divisible by p exactly when all three roots have the same residue modulo p . Since $a + b + r = 2$, this gives the following.

Proposition 17.2 (The single gcd obstruction). *For an integral, completely split target, local solubility at every odd prime not dividing c is equivalent to*

$$\gcd(a - b, 3a - 2) \text{ has no odd prime divisor outside } c. \quad (17.9)$$

The prime three can never be an obstruction in (17.9). For each $p \geq 5$ not dividing c , the unique excluded residue class is

$$a \equiv b \equiv 2 \cdot 3^{-1} \pmod{p},$$

of density $1/p^2$ in the numerator-parameter plane.

Proof. The simultaneous congruences $a = b = r$ are equivalent to $a - b \equiv 0$ and $3a - 2 \equiv 0$. If $p = 3$, the second congruence is impossible. At any other odd prime away from c , it is exactly one class of pairs. Completeness of the inverse description ensures that there is no further p -adic branch to consider. $\square$

18 The dyadic classification and its exact density

The prime two cannot be absorbed into the preceding gcd criterion. This section proves the factor $\sigma(e)$ in (15.3), including all exceptional low valuations.

18.1 Odd c

If c is odd, the three inverse roots lie in $\mathbb{Z}_2$. Write them as α, β, γ , with sum $2/c$. At a root α ,

$$w = \frac{c}{2}(\alpha - \beta)(\alpha - \gamma).$$

For the corresponding source point to be integral, $x = 1/w$ must be integral and $y = \alpha - w$ must be integral. Together these force w to be a unit. Conversely, a unit w makes all three source coordinates integral. Hence one difference must be odd and the other must have valuation exactly one.

The roots modulo two are either all even or have two odd roots and one even root, because their sum is even. The first case fails. In the second, the two odd roots must have residues one and three modulo four. Their sum is zero modulo four, while $2/c \equiv 2 \pmod{4}$, so the even root must be two modulo four. Multiplication by the odd number c preserves the set $\{1, 2, 3\}$ modulo four. We have proved:

$$c \text{ odd : } F^{-1}(A, B, c)(\mathbb{Z}_2) \neq \emptyset \iff \{a, b, r\} \pmod{4} = \{1, 2, 3\}. \quad (18.1)$$

The product abr is always even, so coefficient integrality introduces no separate dyadic condition when c is odd. There are six allowed ordered pairs modulo four, giving $\sigma(0) = 6/16 = 3/8$.

18.2 One factor of two in c

Let $c = 2h$ with h odd. Coefficient integrality forces a, b even, because Q_0 is even only when a, b are both even. Write

$$a = 2A_0, \quad b = 2B_0, \quad r = 2R_0, \quad A_0 + B_0 + R_0 = 1. \quad (18.2)$$

For $e = 1$, these evenness conditions already suffice for dyadic coefficient integrality. The inverse roots are $A_0/h, B_0/h, R_0/h$, and w at a root is h times the product of the two root differences. It is integral automatically, and its inverse is integral precisely when both differences are odd. Since the root sum is odd, this happens exactly when one of A_0, B_0, R_0 is odd and the other two are even. The density is

$$\frac{1}{4} \cdot \frac{3}{4} = \frac{3}{16} = \sigma(1).$$

18.3 At least two factors of two

Write $c = 2^e h$, h odd, and $e \geq 2$. Again use (18.2). Since

$$P_0 = 8A_0B_0R_0, \quad Q_0 = 4(A_0B_0 + A_0R_0 + B_0R_0),$$

the dyadic coefficient conditions become

$$v_2(A_0B_0R_0) \geq 2e - 2, \quad v_2(A_0B_0 + A_0R_0 + B_0R_0) \geq e - 2. \quad (18.3)$$

For $e = 2$, these are equivalent to exactly one of the three half-numerators being odd. The other two are even, making their product divisible by four; the second inequality is automatic. This has density $3/16$ in (a, b) .

For $e \geq 3$, there is again a unique odd half-numerator. Denote it by A_0 , and write the other two as B_0, R_0 . Let their valuations be $f, g \geq 1$. If $f \neq g$, the valuation of the pair sum in (18.3) is $\min(f, g)$. If $f = g$, the product condition requires $f \geq e - 1$. It follows that the allowed possibilities are exactly

$$\begin{aligned} f, g \geq e - 1; \quad \text{or} \\ \{f, g\} = \{e - 2, g'\} \quad \text{with } g' \geq e. \end{aligned} \quad (18.4)$$

The convention that zero has infinite valuation is harmless here.

Lemma 18.1 (No further dyadic obstruction when $4 \mid c$). *For a completely split integral target with $v_2(c) \geq 2$, at least one of its rational inverse points is in $\mathbb{Z}_2^3$.*

Proof. Use the odd half-numerator A_0 just identified, and put

$$\Sigma = B_0 + R_0, \quad J = B_0R_0, \quad A_0 = 1 - \Sigma, \quad K = (A_0 - B_0)(A_0 - R_0) = 1 - 3\Sigma + 2\Sigma^2 + J.$$

Here K is odd. Its inverse point has coordinates

$$x = \frac{2^{e-1}h}{K}, \quad y = \frac{A_0 - K}{2^{e-1}h}, \quad z = \frac{K(5K - 3A_0 - 2K^2)}{2^{2e-2}h^2}. \quad (18.5)$$

The first is integral. The identity

$$A_0 - K = 2\Sigma - 2\Sigma^2 - J$$

shows the same for the second, since in the first pattern of (18.4) one has $v_2(\Sigma) \geq e - 1$, and in the second one has $v_2(\Sigma) = e - 2$, while always $v_2(J) \geq 2e - 2$. For $e = 2$ the two even half-numerators give $v_2(\Sigma) \geq 1$ and $v_2(J) \geq 2$, which are also sufficient.

For the third coordinate, the exact cancellation is

$$5K - 3A_0 - 2K^2 = J - 16\Sigma^2 + 24\Sigma^3 + 12\Sigma J - 8\Sigma^4 - 8\Sigma^2 J - 2J^2. \quad (18.6)$$

Every term has valuation at least $2e - 2$ in either pattern. In particular, when $v_2(\Sigma) = e - 2$ and $e \geq 3$, the potentially lowest quadratic term $16\Sigma^2$ has valuation $2e$, not merely $2e - 4$. This factor of sixteen is the useful cancellation. The case $e = 2$ follows from $v_2(\Sigma) \geq 1$, $v_2(J) \geq 2$. Thus z is integral as well. $\square$

To compute the density for $e \geq 3$, fix which of the three half-numerators is odd. The first pattern in (18.4) has measure 2^{-2e+2} in the other two. The second has measure

$$2 \cdot 2^{-(e-1)} \cdot 2^{-e} = 2^{-2e+2}.$$

The three choices of the odd root are disjoint. Finally the substitution $(a, b) = (2A_0, 2B_0)$ contributes $1/4$. The resulting density is

$$\frac{1}{4} \cdot 3 \cdot 2^{-2e+3} = \frac{3}{2^{2e-1}} = \sigma(e).$$

18.4 The complete local criterion

Theorem 18.2 (All local conditions for split fibers). *Let $c \neq 0$, let $a, b, r = 2 - a - b$ be distinct integers, and suppose (17.4) holds. The associated target is in $F(\mathbb{Z}_p^3)$ for every prime p if and only if (17.9) holds and the following dyadic condition holds:*

$v_2(c)$	Additional condition after coefficient integrality
0	The numerator residues a, b, r modulo four are 1, 2, 3.
1	Among $a/2, b/2, r/2$, exactly one is odd.
≥ 2	No additional condition.

The density of the combined coefficient-integrality and dyadic conditions at two is exactly $\sigma(v_2(c))$.

Proof. At odd primes dividing c , use the lifting argument (16.5). At the other odd primes, use Proposition 17.2. The three dyadic cases were proved above. The inverse model is complete over $\mathbb{Q}_p$, so these conditions are both necessary and sufficient, not just sufficient tests. $\square$

19 A dense polynomial family in every nonzero plane

The full numerator classification is useful for counting. For a simple explicit construction, a different parametrization is shorter: require two inverse roots to be integers s, t . The third is $2/c - s - t$. The target is

$$\Phi(c, s, t) = \left(st - \frac{c}{2}st(s+t), \quad 2(s+t) - c(s^2 + st + t^2), \quad c \right). \quad (19.1)$$

This is integral for all integer parameters because $st(s+t)$ is even. The parameters in Section 17 are $a = cs$, $b = ct$.

Put

$$d = s - t, \quad U = c(2s + t) - 2, \quad V = c(s + 2t) - 2. \quad (19.2)$$

The first coordinates of the three inverse points are

$$x_s = \frac{2}{dU}, \quad x_t = -\frac{2}{dV}, \quad x_0 = \frac{2c}{UV}. \quad (19.3)$$

The other coordinates follow from (16.2). If desired, the third branch can be written without denominators involving c : with $\Sigma = s + t$ and $J = (2s + t)(s + 2t)$,

$$y_0 = 2\Sigma - \frac{c}{2}J, \quad z_0 = \frac{UV}{8}(2J - 36\Sigma^2 + 12c\Sigma J - c^2J^2). \quad (19.4)$$

Thus primes dividing c do not introduce hidden odd denominators in these coordinates.

Theorem 19.1 (A universal arithmetic family). *Let $c \neq 0$, let n be a positive odd integer, and let $k \geq 6$ satisfy $k \equiv 2 \pmod{4}$. Set*

$$s = nk, \quad t = n(k - 1). \quad (19.5)$$

Then $\Phi(c, s, t)$ is an integral Hasse failure with exactly three rational preimages. None of those points has an integral first coordinate.

Proof. Here $d = n$, and

$$\gcd(d, 3cs - 2) = \gcd(n, 3cnk - 2) = \gcd(n, 2) = 1.$$

Therefore every odd prime is locally soluble: this is the root-difference criterion away from c , and (16.5) at primes dividing c . At two, if c is odd, then $s \equiv 2 \pmod{4}$ and t is odd, so the three roots have residues 1, 2, 3 modulo four. If $v_2(c) = 1$, the two integer roots s, t have opposite parity and the third has the parity needed for a unique odd root. If $v_2(c) \geq 2$, apply Lemma 18.1. These statements can also be read directly from Theorem 18.2 after passing to the numerator roots $a = cs$, $b = ct$, $r = 2 - cs - ct$.

The differences are nonzero. More quantitatively,

$$|U| \geq |c|n(3k - 3) \geq 15|c|n, \quad |V| \geq |c|n(3k - 4) \geq 14|c|n.$$

All three numbers in (19.3) are nonzero and have absolute value strictly less than one. Consequently no branch is integral. Completeness follows from Proposition 16.1, and rationality supplies a real point. $\square$

Example 19.2 (A small odd-slice failure). Take $c = 1, s = 1, t = 3$, which also satisfies the local conditions although it is not the particular parametrization (19.5). Then

$$\Phi(1, 1, 3) = (-3, -5, 1),$$

and its complete fiber is

$$\left(-\frac{1}{3}, 4, 81\right), \quad \left(\frac{1}{5}, -2, -45\right), \quad \left(\frac{2}{15}, -\frac{19}{2}, -\frac{765}{8}\right).$$

No point is integral. The first point is integral at every prime except three, and the second at every prime except five. Hence these two branches alone cover all local requirements. The obstruction is not a single bad prime; the permissible branch changes with the prime.

19.1 Dominance, not just infinitely many examples

A direct determinant calculation gives

$$\det \frac{\partial \Phi}{\partial(c, s, t)} = -\frac{1}{2}(s-t)(c(2s+t)-2)(c(s+2t)-2) = -\frac{1}{2}dUV. \quad (19.6)$$

For fixed c , the same expression is the two-by-two determinant of the first two target coordinates with respect to s, t . The change $(n, k) \mapsto (nk, n(k-1))$ has determinant n . Thus the parametrization in Theorem 19.1 has a nonzero Jacobian polynomial.

For clarity, the elementary dominance principle being used is this: a polynomial map in characteristic zero with nonzero square Jacobian has algebraically independent coordinate polynomials. Indeed, a nonzero polynomial relation of least degree would, after differentiation and inversion of the Jacobian over the rational-function field, yield a nonzero partial derivative relation of smaller degree, a contradiction. A nonconstant relation has a nonzero partial derivative in characteristic zero.

An infinite product of arithmetic progressions is Zariski dense: prove this by induction on the number of variables, using that a nonzero univariate polynomial has only finitely many zeros. Therefore restricting n to positive odd integers and k to positive integers congruent to two modulo four does not destroy density. This proves density in every plane $C = c \neq 0$. Taking infinitely many integer values of c proves density in $\mathbb{A}^3$. The formula at $c = 0$ was already established. This proves the first part of Theorem 15.2.

One can make the entire family an ordinary integer-coefficient polynomial map by writing $n = 2q + 1$, $k = 4r + 2$, where $q \geq 0$ and $r \geq 1$. Its target is

$$\begin{aligned} A &= n^2(2r+1)(4r+1)(2-cn(8r+3)), \\ B &= 2n(8r+3) - cn^2(48r^2+36r+7), \\ C &= c. \end{aligned} \quad (19.7)$$

The determinant in variables (c, q, r) is $-4n^2UV$, which is nonzero.

19.2 Persistence over every prescribed ring of S -integers

Theorem 19.3. *Fix $c \neq 0$ and a finite set S of primes. A Zariski-dense subset of $C = c$ consists of targets from Theorem 19.1 with no preimage in $\mathbb{Z}[S^{-1}]^3$.*

Proof. Choose distinct odd primes $\ell, q \notin S$ not dividing c , with $q \neq 3$. Impose

$$n \equiv 0 \pmod{\ell}, \quad n \equiv 1 \pmod{q}, \quad n \equiv 1 \pmod{2},$$

and

$$k \equiv 2 \pmod{4}, \quad k \equiv 3^{-1}(1+2c^{-1}) \pmod{q}.$$

The Chinese remainder theorem supplies independent infinite arithmetic progressions for n and k , and we take positive values with $k \geq 6$. The prime ℓ divides $d = n$ but not the numerators two in the first two fractions of (19.3); it excludes those branches from S -integrality. At q , one has $U \equiv 0$, $V = U - cn \not\equiv 0$, while $2c$ is a unit, so the third branch is excluded by q . The original theorem still supplies local points at every prime. The two arithmetic progressions remain Zariski dense in parameter space, so the same dominance argument proves the asserted density. $\square$

20 Only finitely many split fibers can have an integral point

This finiteness is stronger than a density-zero assertion, and is the reason that the later asymptotic need not subtract a second main term.

Theorem 20.1 (Finite integral split exceptions). *For fixed $c \neq 0$, the set*

$$\mathcal{I}_c^{\text{sp}} = \{(A, B) \in \mathbb{Z}^2 : F^{-1}(A, B, c) \text{ consists of three distinct rational points and meets } \mathbb{Z}^3\}$$

is finite. It has a complete divisor enumeration using the positive divisors of $2|c|$.

Proof. Choose an integral source point in such a fiber and label its root numerator a . By (17.7), its first coordinate is $2c/D_a$, so

$$D_a = (a - b)(2a + b - 2) \mid 2c. \quad (20.1)$$

Choose a nonzero signed divisor D of $2c$ and a signed factorization $D = dv$. The equations

$$a - b = d, \quad 2a + b - 2 = v$$

have the unique possible solution

$$a = (d + v + 2)/3, \quad b = (v + 2 - 2d)/3. \quad (20.2)$$

There are finitely many choices. Retain integral a, b , distinct roots, integral coefficients (17.3), and an integral point (17.7). This gives an exhaustive finite procedure. No height cutoff is used. $\square$

A simple, nonoptimal upper bound for the number of candidates is

$$4 \sum_{d \mid 2|c|} \tau(d) = 4 \prod_{p^v \parallel 2c} \frac{(\nu + 1)(\nu + 2)}{2}. \quad (20.3)$$

The actual number of targets can be much smaller because of integrality conditions, repeated roots, and repetitions of the same target. The exact program in the archive gives, for example,

c	Completely split targets on $C = c$ with an integral point
1	$(-1, -1, 1)$
2	none
3	$(-1, -3, 3)$
4	$(0, -2, 4)$
8	none
12	$(0, -2, 12)$
16	none

These rows are finite exact enumerations justified by the theorem, not claims inferred from a bounded brute-force search.

Corollary 20.2. *On each nonzero integer slice, all but finitely many locally integral, completely split targets are integral Hasse failures.*

21 The exact numerator-parameter density

The finite-prime conditions of Sections 17 and 18, together with the one-class exclusions at all remaining primes, yield the density $\rho(c)$ in (15.4). We need the density not just in squares, but in bounded regions suitable for a height calculation.

Lemma 21.1 (A lattice sieve in bounded regions). *Fix $c \neq 0$. Let $\Omega \subset \mathbb{R}^2$ be bounded with boundary of Lebesgue measure zero. As $L \rightarrow \infty$, the number of pairs $(a, b) \in L\Omega \cap \mathbb{Z}^2$ satisfying coefficient integrality and local solubility at every prime, with $a, b, 2 - a - b$ distinct, is*

$$\rho(c) \operatorname{vol}(\Omega) L^2 + o(L^2). \quad (21.1)$$

Proof. First impose the complete conditions at primes dividing $2c$, and the one-class exclusions for primes $5 \leq p \leq Y$ not dividing c . All of these are residue conditions modulo a fixed integer. The Chinese remainder theorem makes their density the product of the local densities:

$$\sigma(v_2(c)) \prod_{\substack{p^v \parallel c \\ p \text{ odd}}} \frac{3}{p^{2v}} \prod_{\substack{5 \leq p \leq Y \\ p \nmid c}} (1 - p^{-2}).$$

A Riemann-sum argument for each residue class gives this constant times $\operatorname{vol}(\Omega) L^2 + o(L^2)$. The distinctness exclusions are three affine lines, contributing only $O_\Omega(L)$ points.

It remains to justify the infinite-prime limit. Enclose Ω in a square $[-R, R]^2$. Apart from $a = b$, a prime causing a failure must divide $a - b$, so it is at most $2RL$. For each $p \geq 5$, the bad pairs are one residue class and number at most $(2RL/p + 1)^2$. Summing for $Y < p \leq 2RL$ and bounding prime sums by integer sums gives

$$O_R \left(\frac{L^2}{Y} + L \log L + L \right). \quad (21.2)$$

Thus, after division by L^2 , the omitted tail has upper limit $O_R(1/Y)$. Let L tend to infinity and then Y tend to infinity. The Euler product converges since $\sum_p p^{-2} < \infty$. $\square$

Using the classical identity $\zeta(2) = \pi^2/6$, the product away from c can be written without an infinite numerical computation:

$$\rho(c) = \frac{9}{\pi^2} \sigma(v_2(c)) \prod_{\substack{p^v \parallel c \\ p \text{ odd}}} \frac{3}{p^{2v}} \prod_{\substack{p \mid c \\ p \geq 5}} (1 - p^{-2})^{-1}. \quad (21.3)$$

The prime three is absent from the sieve factor because three coincident numerator residues would have sum zero rather than two.

21.1 Density in the simpler two-integer-root family

For comparison with the earlier $C = 2$ report, suppose s, t are the integer roots used in (19.1). Their everywhere-local density in large parameter squares is

$$\delta(c) = \frac{9}{\pi^2} \lambda(c) \prod_{\substack{p \mid c \\ p \geq 5}} (1 - p^{-2})^{-1}, \quad \lambda(c) = \begin{cases} 3/8, & c \text{ odd,} \\ 3/4, & c \equiv 2 \pmod{4}, \\ 1, & 4 \mid c. \end{cases} \quad (21.4)$$

At odd primes away from $3c$, the bad condition is $s = t$, $3cs = 2$ modulo p , again one class. For odd c , the dyadic condition is that the three roots have residues 1, 2, 3 modulo four. For $c \equiv 2 \pmod{4}$, the only bad parity class is s, t both odd. For $4 \mid c$, all integer-root parameters pass the dyadic test. The same sieve proof gives (21.4).

In particular, $\delta(2) = 27/(4\pi^2)$ recovers the prior split-plane parameter density. This is a consistency check and an extension to other slices, not a claim that the old $C = 2$ density is new. The distinction between $\delta(c)$ and $\rho(c)$ matters: the latter counts *all* rationally split fibers through their numerator lattice.

22 Sharp target-height asymptotics

We now prove Theorem 15.3. The main issues are the unbounded cusp in root-parameter space, the finite-to-one passage to targets, and the removal of integral exceptions.

22.1 The exact height domain

Let $N_c(T)$ be the number of ordered numerator pairs satisfying all local and integrality conditions, with distinct roots and

$$|ab(2 - a - b)| \leq 2c^2T, \quad |2(a + b) - a^2 - ab - b^2| \leq |c|T. \quad (22.1)$$

Every locally integral, completely split target contributes exactly six such ordered pairs: choose the first two of its three distinct numerator roots in order. Therefore

$$\#\mathcal{H}_c^{\text{sp}}(T) = \frac{1}{6}N_c(T) - \#\{(A, B) \in \mathcal{I}_c^{\text{sp}} : |A|, |B| \leq T\}. \quad (22.2)$$

The second term is $O_c(1)$ by Theorem 20.1.

Set $L = T^{1/3}$. On any fixed bounded region in scaled coordinates $a = Lx$, $b = Ly$, the first inequality in (22.1) tends to

$$|xy(x + y)| \leq 2c^2. \quad (22.3)$$

The second inequality is eventually automatic there, because its left side is $O(L^2)$ and its right side is $|c|L^3$. The limiting domain is unbounded, so a compact-region argument alone would not prove the theorem.

22.2 An explicit cusp bound

Lemma 22.1 (Uniform tail control). *For fixed $c \neq 0$, the number of integer pairs satisfying (22.1) and $\max(|a|, |b|, |2 - a - b|) > RL$ is*

$$O_c(L^2/R + L^{3/2}), \quad R \geq 4. \quad (22.4)$$

The estimate holds before imposing any congruence conditions.

Proof. The quadratic form gives a global bound. Completing the square,

$$Q_0 = \frac{4}{3} - \left((a - \frac{2}{3})^2 + (a - \frac{2}{3})(b - \frac{2}{3}) + (b - \frac{2}{3})^2 \right).$$

Hence every numerator is $O_c(T^{1/2}) = O_c(L^{3/2})$. Let M be the largest absolute numerator. Since the three numerators sum to two, when $M \geq 4$ the second largest has absolute value at least $M/4$. The product bound then forces the smallest absolute numerator to be at most $8c^2T/M^2$.

Choose which numerator is largest and which is smallest, and choose the sign of the largest. This costs only an absolute factor. For a fixed integer largest numerator of absolute value m , there are $O_c(T/m^2 + 1)$ possibilities for the smallest; the third is determined by the sum. Summing from RL to $O_c(L^{3/2})$ gives

$$O_c \left(T \sum_{m > RL} m^{-2} + L^{3/2} \right) = O_c(L^2/R + L^{3/2}).$$

The $+1$ term is important: it includes lines on which a numerator is zero, and shows directly that they are lower order. $\square$

This argument also explains why the quadratic restriction changes no leading constant: it is eventually inactive on each fixed scaled compact set, and the part of the actual height domain outside such sets is controlled by the lemma. One must not simply drop it from the exact integer count: the cubic inequality alone contains infinite zero-product lines. The $O_c(T^{1/2})$ term explicitly controls those lines after the quadratic bound has been imposed.

22.3 The beta integral

The region

$$\mathcal{B} = \{(x, y) \in \mathbb{R}^2 : |xy(x + y)| \leq 1\}$$

is divided by $x = 0$, $y = 0$, and $x + y = 0$ into six cones. Permuting the three linear forms x , y , $-x - y$, together with simultaneous negation, identifies these cones by linear maps of determinant ± 1 that preserve the absolute cubic. Each cone therefore has the same area as the positive quadrant piece. On that piece put $y = rx$:

$$\begin{aligned} \text{vol}\{x, y > 0 : xy(x + y) \leq 1\} &= \int_0^\infty \int_0^{[r(1+r)]^{-1/3}} x \, dx \, dr \\ &= \frac{1}{2} \int_0^\infty r^{-2/3} (1 + r)^{-2/3} \, dr \\ &= \frac{1}{2} B(1/3, 1/3) = \frac{\Gamma(1/3)^2}{2\Gamma(2/3)} = I. \end{aligned} \tag{22.5}$$

The integral converges both at zero and infinity. The beta–gamma identity used here is standard [9], and the preceding integral is already an independent exact description of the constant. Thus

$$\text{vol}\{(x, y) : |xy(x + y)| \leq 2c^2\} = 6I(2c^2)^{2/3}. \tag{22.6}$$

22.4 Completion of the asymptotic proof

Fix R . On a bounded scaled region, the moving height domain converges to (22.3) except on its algebraic boundary, which has area zero. Sandwiching it between fixed inner and outer regions and applying Lemma 21.1 gives the expected local density times its limiting area. Equivalently, one can use the two fixed conditions $|xy(x + y)| \leq 2c^2 \pm \varepsilon$ inside a fixed square, then let ε tend to zero.

Lemma 22.1 bounds the remaining lattice points by $O_c(L^2/R + L^{3/2})$. The corresponding limiting continuous tail also tends to zero, by the finite area just computed. Let $L \rightarrow \infty$ first and then $R \rightarrow \infty$. We obtain

$$N_c(T) \sim 6\rho(c)I(2c^2)^{2/3}T^{2/3}.$$

Divide by six in (22.2) and subtract the bounded set of integral exceptions. This proves (15.5).

22.5 Constants and reproducible counts

The constant depends only on $|c|$. Some values are

c	$\rho(c)$	$\kappa(c)$
1	0.3419589948	1.4384663498
2	0.1709794974	1.8123540336
3	0.1139863316	2.0746274748
4	0.1709794974	4.5668459937
8	0.0854897487	5.7538653990

For $c = 2$ the theorem specializes to

$$\#\mathcal{H}_2^{\text{sp}}(T) \sim \frac{27\Gamma(1/3)^2}{8\pi^2\Gamma(2/3)}T^{2/3}. \quad (22.7)$$

There are no integral exceptions on this slice. The companion enumeration gives the following *actual Hasse-failure* counts, after subtracting the finite exceptions:

c	$T = 10^3$	$T = 10^4$	$T = 10^5$
1	103	540	2692
2	121	653	3304
3	134	727	3730
4	282	1576	8118
8	330	1903	9975

These finite data are not used to infer the asymptotic. Convergence of the normalized count can be visibly slow because the cutoff in the cusp is of order $T^{1/2}$, only a factor $T^{1/6}$ below the main scale $T^{2/3}$. The theorem asserts an equivalent for fixed c , not a uniform error estimate as c grows.

23 Constructive certificates and formalization boundaries

23.1 Every congruence has an explicit solution

For any locally integral split target, the three rational points (17.7) suffice to construct a solution modulo every positive integer M . Factor $M = \prod p^{e_p}$. At each p , choose a branch whose coordinate denominators are coprime to p , reduce each fraction using a modular inverse modulo p^{e_p} , and combine the three coordinates by the Chinese remainder theorem. Because the original polynomial has integer coefficients, the resulting point solves $F(x, y, z) \equiv (A, B, c) \pmod{M}$.

This does not produce a single rational branch that is integral at all primes. In Example 19.2, one branch is needed at three and a different branch at five. Confusing a coherent choice of residues for one modulus with a globally integral branch is precisely the invalid local–global inference.

23.2 The companion programs

The archive contains three main programs.

code/split_fibres.py

Uses Python integers and exact fractions to reconstruct all three points, decide the local criterion, enumerate all integral split exceptions for fixed c , generate the universal family, and produce modular witnesses. It has no third-party dependencies.

code/verify_results.py

Runs the exact symbolic identities, an independent image calculation modulo eight, exhaustive finite checks of the bad-prime density factors, odd-prime residue checks, and exact rational tests of the dense family. It writes a structured verification report. Symbolic checks use SymPy; vectorized finite residue checks use NumPy integers in ranges safely within signed 64-bit arithmetic.

code/count_heights.py

Reproduces the target-height count table by exhaustive numerator enumeration, exact divisibility and gcd tests, and subtraction of the complete finite exception list. Floating-point arithmetic is used only for displaying the theoretical constants and normalized counts, not for membership decisions.

[write] In this report the three programs are shipped as `code/02-dense-hasse-split_fibres.py`, `code/02-dense-hasse-verify_results.py` and `code/02-dense-hasse-count_heights.py`, and their recorded outputs under `data/` with the same prefix; run them only on a copy with the delivered names (Appendix C.5).

The executed verification contains twelve passing groups. In particular, the modulo-eight image has 176 elements, and the complete bad-prime numerator densities were independently checked for $c = 1, 2, 3, 4, 5, 6, 8, 12, 16$. Exact reconstruction and local tests were also compared on a fixed-seed sample, and the universal family was tested on 108 signed-slice parameter choices. Finite tests supplement the proofs; they do not establish universal statements on their own.

23.3 An economical Lean/Rocq route

A useful formalization order is: first the chart identities and the complete simple-root inverse; then the numerator parametrization and odd-prime valuation lemmas; then the explicit dense family and its nonintegrality; then the dyadic cancellation. The constructive congruence theorem can be formalized before introducing p -adic fields, using reduced denominators, modular inverses, and CRT.

The counting theorem is a separate analytic branch. It requires lattice point asymptotics for fixed congruence classes, the elementary large-prime tail estimate, the cusp bound, and the beta integral. The finite integral exception theorem needs only divisibility and the explicit inverse. No Chebotarev theorem is needed anywhere in this paper.

The original repository declarations are not recompiled here, and the new results have not been encoded as Lean or Rocq proofs. The package is an English proof manuscript with reproducible exact checks.

24 Further research questions of Part II

The results above close the full-dimensional density question and provide a sharp counting theorem for the split sector. They also distinguish several genuinely different remaining problems.

Research question 24.1 (The rational-plus-quadratic sector). For fixed $c \neq 0$, classify integral Hasse failures whose inverse cubic has one rational root and an irreducible quadratic factor over $\mathbb{Q}$. What is their target-height exponent and leading constant?

The quadratic pair can supply local points at primes where the rational branch is nonintegral. The split denominator-cover argument alone does not classify this phenomenon. A useful first step is to express the quadratic discriminant and the reconstruction valuations in terms of the rational root and the remaining coefficients. [write] This is the second half of Part I's Research question 12.2, with a counting question added. [Added 30 September 2026, batch 64: the classification is answered by Part III for every $c \neq 0$ (Theorem 29.2 and Corollary 33.1); the count for $c = \pm 2$ in square boxes is $\frac{27}{2\pi^2}T \log T + O(T)$ (Theorem 29.3). The exponent and constant for other c remain open (Research question 41.3).]

Research question 24.2 (A quantitative remainder). Can one prove an explicit error term $\#\mathcal{H}_c^{\text{sp}}(T) = \kappa(c)T^{2/3} + O_c(T^{2/3-\eta})$ with a specified $\eta > 0$, or identify a secondary term of order $T^{1/2}$?

The proof here deliberately separates the cutoff limits and gives an $o(T^{2/3})$ remainder. A secondary term must account for the quadratic height cutoff, affine shifts of the three cusp lines, lattice effects, and congruence conditions simultaneously. The finite count table suggests that these effects are numerically significant, but it does not identify a secondary constant.

Research question 24.3 (Uniformity in the third coordinate). How far can the asymptotic be made uniform for $0 < |c| \leq T^\theta$? What is the asymptotic when all three target coordinates are bounded by a common height?

The factors $\sigma(v_2(c))$ and $3/p^{2\nu_p(c)}$ are explicit, but their moduli grow with c . The fixed- c sieve and cusp bounds cannot simply be summed without uniform error control. The dyadic maximum near $v_2(c) = 3$ is an additional feature worth understanding in a global count.

Research question 24.4 (Coefficient-only local certificates). Can the complete split local criterion be expressed directly in (A, B, c) by a short collection of valuations and congruences, without first factoring the inverse cubic?

This paper supplies an exhaustive root-numerator parametrization, not a factorization-free coefficient criterion. The modulo-eight target image is finite, but the odd-prime root-collision condition also interacts with the splitting assumption. A compact certificate format could be more useful computationally than a single large formula. [write] This extends Part I's Research question 12.1, which asks the same for the dyadic certificate alone. [Added 30 September 2026, batch 64: answered by Part III. The integral roots of a completely split target's monic inverse polynomial are its root numerators, so by Theorem 29.2 such a target is locally integral everywhere exactly when $\gcd(3Bc - 4, 27Ac^2 - 4)$ is a power of two and the modulo-eight table $E(A, B, c)$ holds; no factorization is needed (Section 28.2). Applied to completely split targets, this criterion reproduces all fifteen counts of the table in Section 22 (Section 28.3).]

Research question 24.5 (Number fields and S -integer counting). What replaces the constant $\kappa(c)$ over the integers of a number field, and what is the exact counting constant after inverting a finite set of primes?

The numerator-root method and denominator covering have natural ideal versions. Archimedean regions, ideal classes, residue characteristic two, and the unit group introduce new issues. The existence theorem for

S -integers proved here does not by itself give their height asymptotics. [Added 30 September 2026, batch 64: Part III’s Theorem 38.1 gives a complete local membership certificate over every ring $\mathcal{O}_{K,S}$, for all fibers; the counting constants asked for here remain open (Research question 41.8 asks the same for nonsplit fibers).]

Research question 24.6 (The finite integral exceptional set). Can the divisor enumeration of $\mathcal{I}_c^{\text{sp}}$ be simplified to a closed arithmetic classification of those c for which it is empty, together with an exact formula for its cardinality?

The proof reduces the problem to finitely many signed factorizations of divisors of $2c$, congruences modulo three, and explicit integrality conditions. This is a bounded arithmetic problem with no remaining geometric elimination. A classification would refine, rather than replace, the current effective procedure.

Research question 24.7 (Certified finite-fiber arithmetic). Can the inverse-certificate, local-selection, and finite-exception arguments be packaged as a reusable Lean/Rocq library for polynomial maps with small rational fibers?

The natural abstraction is a proved bijection between a polynomial fiber and simple roots, followed by exact denominator data. This could support both positive congruence witnesses and globally negative integrality certificates without invoking an untrusted solver. [write] Cf. Part I’s Research question 12.9, which asks essentially the same question (complete inverse certificates, denominator covering and the inverse-ball theorem as a generic workflow).

25 Conclusion of Part II

The obstruction exhibited here is not geometric nonexistence: every counted fiber has three rational points. It is the impossibility of choosing one branch that is integral at all primes, even though an integral branch exists separately at each prime.

For the fixed ProveIt map, this obstruction is Zariski dense in every nonzero integer slice and hence in the full target space. In the split sector it admits a complete arithmetic description, an exact dyadic factor, a finite list of globally integral exceptions, and a sharp $T^{2/3}$ target-height law. The resulting theorem resolves the concrete full-dimensional question left in the repository report while keeping the unsplit arithmetic sector, uniform counting, and formal verification as explicitly separate next problems. [Added 30 September 2026, batch 64: the unsplit sector is treated in Part III, which classifies it on every slice and shows that on $C = \pm 2$ it carries the dominant $T \log T$ term of the integral Hasse failures.]

26 A compact proof and certificate ledger

[write] This is the manuscript’s Appendix A.

Claim	Main proof ingredient	Computational check
Complete inverse and determinant	Rational chart, chain rule, localization	Symbolic identities
Odd-prime integrality	Two divisible roots, valuation comparison	Finite residue checks
Odd-prime local obstruction	All three residues coincide	Exact rational comparison

Claim	Main proof ingredient	Computational check
Dyadic factor	Root parities and cancellation (18.6)	Independent mod-eight image and bad-prime densities
Dense family	Explicit inverse first coordinates, gcd one, nonzero Jacobian	108 exact signed-slice tests
S -integer density	Two assigned denominator primes, CRT	Proof, not claimed as a finite exhaustive test
Finite integral exceptions	$D_a \mid 2c$ and signed factorizations	Complete divisor enumeration
Numerator density	CRT and (21.2)	No numerical inference used
Sharp target asymptotic	Cusp bound, beta area, six-to-one map	Exact finite counts for comparison

27 Reproduction commands of Part II

[write] This is the manuscript’s Appendix B.

From the extracted archive directory, install the optional verification packages and run:

```
python -m pip install -r requirements.txt
python code/verify_results.py
python code/split_fibres.py 1 1 3 --modulus 18144000
python code/count_heights.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The exact-fraction utility `split_fibres.py` needs only the Python standard library. The other packages are needed for symbolic verification, fast residue enumeration, and numerical display of constants. The retained outputs are under `data/`. This archive does not modify the ProveIt repository.

[write] In this report the files carry the prefix `02-dense-hasse-`: the programs are under `code/`, the `requirements.txt` is `data/02-dense-hasse-requirements.txt`, and the manuscript’s `Makefile` is `code/02-dense-hasse-Makefile`, which still names the delivered paths. The manuscript is printed here as Part II; its `article.pdf` is not shipped. The commands above are the delivery’s: `count_heights.py` imports `split_fibres` by its delivered name, and the scripts write under delivered names into `../data/`, so they must be run on a copy (Appendix C.5). On Windows use `py` or `uv run` for python.

Part III. Nonsplit fibers: a complete local certificate and a $T \log T$ law

Part III was added on 30 September 2026 in batch 64 of ProveIt’s incoming-report intake, from one manuscript of 30 September 2026 written against Parts I and II as pinned. Parts I and II are unchanged except for dated notes and the title-page spacing (listed in Appendix C.6). Part III answers Research question 12.2 on every slice $C \neq 0$, and Research question 12.3 for square boxes in all sectors: on $C = 2$ the integral Hasse failures number $\frac{27}{2\pi^2}T \log T + O(T)$, almost all of them nonsplit, so the completely split sector that Part II counts ($T^{2/3}$) is not the dominant one. Section 28 states exactly what is answered, fixes the notation against Parts I and II and lists what is printed twice; Sections 29–46 are the manuscript.

28 Relation to Parts I and II

[write] This section was written when the manuscript was added to the report.

28.1 The manuscript

Part III is batch-64 manuscript 02, *Beyond Split Fibers: Exact Local Certificates and a $T \log T$ Law for Integral Hasse Failures*, subtitled *Arithmetic of ProveIt’s three-variable Keller map* and dated 30 September 2026. It was delivered as the archive `ProveIt_Beyond_Split_Fibers` and is pinned to ProveIt commit `6d04e1e385f2fe7d1cfb8bd8fd8d4e45bd6b6c72`; its provenance, shipped files and editorial changes are in Appendix C.6. Its title-page abstract and evidence note read as follows.

We resolve the rational-plus-quadratic local-solubility question left open in ProveIt’s arithmetic-fiber report. For its explicit three-variable polynomial map F with Jacobian determinant -2 , an integral target (A, B, C) is locally integral at every prime precisely when a monic inverse cubic has an integer root, the integer $\gcd(3BC - 4, 27AC^2 - 4)$ has no odd prime factor, and an explicit table of congruences modulo 8 holds. Thus no separate analysis of the quadratic factor’s splitting fields is needed once a rational root is known.

On the slice $C = 2$, this criterion becomes a primitive parametrization: $A = r^3 - r^2 + rb$, $B = 2b$, and $\gcd(3r - 1, 3b - 1) = 1$. We prove that nonsplit integral Hasse failures in $|A|, |B| \leq T$ number $\frac{27}{2\pi^2}T \log T + O(T)$; the same formula counts all integral Hasse failures on the slice. This differs fundamentally from the $T^{2/3}$ scale previously obtained for completely split fibers. We also determine the entire integral image on $C = 2$: it has two polynomial parametrizations with disjoint integer images, containing $2^{5/3}T^{1/3} + O(1)$ targets in the same box. A number-field and S -integer extension, exact algorithms, finite certificates, and further research questions complete the study.

Evidence and novelty boundary. All principal claims have proofs below. The computations are reproducible checks and finite certificates, not Lean or Rocq formalizations. The classification and counting theorems are candidate new contributions relative to the pinned repository and primary sources reviewed here; worldwide priority has not been established. The map, inverse-cubic geometry, and earlier split-fiber results are explicitly attributed. This article does not claim a new disproof of the Jacobian conjecture.

28.2 What Part III answers, and what it does not

- *Research question 12.2 is answered, on every slice $C \neq 0$ and in both sectors.* Theorem 29.2 is a complete local certificate for every integral target: an integer root of the monic inverse polynomial H , the gcd G of $U = 3BC - 4$ and $V = 27AC^2 - 4$ a power of two, and the modulus-eight table E . Corollary 33.1 turns it into a complete Hasse test for fibers with one rational point and an irreducible quadratic pair, the second sector of the question, which Part II left open. (Part II's Theorem 18.2 had classified the completely split sector in terms of root numerators.) On $C = 0$ every integral target has an integral point.
- *Research question 12.3 is answered for square boxes, in all sectors.* On $C = 2$ (and, by reflection, $C = -2$), in $|A|, |B| \leq T$, the locally integral targets, all integral Hasse failures and the nonsplit ones each number $\frac{27}{2\pi^2}T \log T + O(T)$ (Theorem 29.3), and the integral targets $2^{5/3}T^{1/3} + O(1)$ (Corollary 35.4). The completely split sector counted by Part II, $\#\mathcal{H}_2^{\text{sp}}(T) \sim \kappa(2)T^{2/3}$ (Theorem 15.3), is therefore *not* the dominant term: almost every integral Hasse failure on $C = 2$ is nonsplit (Corollary 37.2). The constant $27/(2\pi^2)$ is twice the Euler product $\prod_{p \neq 3}(1 - p^{-2}) = 27/(4\pi^2)$, the main term with an explicit Euler product that the discussion of the question asks for. Not answered: unequal box sides $H_A \neq H_B$ (Research question 41.4); a second-order term or a power saving (Research questions 41.1 and 41.2); other slices (Research question 41.3).
- *Part II's Research question 24.4 is answered.* For a completely split integral target the numerators of the roots are integer roots of H (Section 28.3), so the root gate holds automatically and local solubility everywhere is “ G is a power of two and $E(A, B, C)$ holds”: a criterion in (A, B, c) that needs no factorization of the inverse cubic.
- *Part II's Research question 24.1 is answered in its classification half on every slice, and in its counting half on $C = \pm 2$ (order $T \log T$, constant $27/(2\pi^2)$).* The exponent and constant on other slices remain open (Research question 41.3).
- *Partly answered or related.* Research question 12.1: membership in the dyadic image is compressed to the five-row table E in the target coefficients (Proposition 32.2); the integral fiber sizes $j = 1, 2, 3$ are not compressed, and the proof is still the exhaustive 512-case check, not a hand case proof. Research question 12.5 and Part II's Research question 24.5: Theorem 38.1 gives a membership certificate over every ring $\mathcal{O}_{K,S}$, with the uniform dyadic modulus $8R$; fiber distributions, minimal moduli and height counts over number fields remain open (Research questions 41.6 and 41.8). Research question 12.9 and Part II's Research question 24.7 are related to Research question 41.9. Research questions 12.4 (answered by Part II), 12.6–12.8 and Part II's Research questions 24.2, 24.3 and 24.6 are untouched.

Part II's results stand as proved: its split-sector theorems are not superseded, and its $T^{2/3}$ law is the exact order of the split sector, which Part III bounds only by $O(T)$ (Lemma 37.1). What changes is the scope of the sentence “Research question 12.3 is answered only in the completely split sector”: dated notes in Parts I and II now point here.

28.3 Consistency with Parts I and II

Where the Parts meet they agree. Each item was checked at the write, by hand or by an exact computation of its own (not the delivered programs).

- *Part II's numerators are the roots of H .* Since $H(Ct) = C^2g(t)$, the roots of the monic H are C times the roots of Part I's $g_{A,B,C}$; on $C = c$ these are exactly Part II's root numerators a, b, r , with

$$a + b + r = 2 \quad (17.1).$$

- *Odd primes.* At an odd prime $p \nmid c$, Part II's obstruction (17.9) is $a \equiv b \equiv r \equiv 2 \cdot 3^{-1}$, that is, $\bar{H} = (S - 2/3)^3$, which is exactly $p \mid U$ and $p \mid V$ in the proof of Theorem 31.3.
- *The slice $C = 2$.* Part III's polynomial f with $H(2T) = 8f(T)$ is $g_{A,B,2}/2$, so its integer roots are Part I's roots a, b, c of (4.1); $d = f'(r)$ is Part I's D_r , Part I's $d(t) = g'(t)$ is $2d$ there, and (34.6) is Part I's (4.2). For a completely split target, Theorem 34.1 reduces to Part I's criterion (4.4): by (34.7) its condition is $\gcd(3r - 1, d) = 1$, and with $u = r - s$, $v = r - t$ one has $3r - 1 = u + v$ and $d = uv$; a prime divides both $u + v$ and uv exactly when it divides u and v , so the condition is $\gcd(u, v) = 1$, which is Part I's $g = 1$.
- *The dyadic table.* The program of Section 46 was rerun at the write: the image of F on $(\mathbb{Z}/8\mathbb{Z})^3$ has 176 classes and equals E pointwise, and the residue fiber histogram is 336, 112, 48, 16 for sizes 0, 2, 4, 6, which is Part I's (9.5) at $m = 3$; $176/512 = 11/32$ is (9.4).
- *Constants.* Part III's $\alpha = 27/(4\pi^2)$ is Part I's parameter density (4.6), from the same Möbius sum over d prime to 3; $27/(\pi^2) = 1.3678359792\dots$ and $2^{5/3}(10^5)^{1/3} = 147.36\dots$ (against $I_2(10^5) = 148$).
- *The split counts.* By Lemma 35.2 every locally integral completely split target on $C = 2$ is a Hasse failure, so the manuscript's $\Sigma(T)$ is Part II's $\#\mathcal{H}_2^{\text{sp}}(T)$. Its recorded values 121, 653, 3 304 at $T = 10^3, 10^4, 10^5$ (table in Section 39) are Part II's recorded exact counts for $c = 2$ (data/02-dense-hasse-height_counts.txt). Moreover, the coefficient criterion of Theorem 29.2 with the integrality test of Corollary 30.2, applied by an independent enumeration of completely split targets on $C = c$, reproduces all fifteen counts of Part II's table in Section 22 ($c = 1, 2, 3, 4, 8$ and $T = 10^3, 10^4, 10^5$).
- *Integral points.* Part II's table of integral split exceptions after Theorem 20.1 lists none on $C = 2$, as Lemma 35.2 requires; the integral image on $C = 2$ (Theorem 35.3) consists of nonsplit fibers and the repeated-root target $(0, 0, 2)$. The formulas $F(1/2, -2, 4) = (0, 4, 2)$, (34.9) for $2 \leq b < 40$, (35.2) for $|r| \leq 30$ and $F(1, 0, 0) = (0, 0, 2)$ were evaluated exactly.
- *The plane $C = 0$:* (30.1) is Part I's boundary point (2.7) and the display in Part II's Theorem 15.2.

28.4 Notation: Parts I and II against Part III

Part III keeps the manuscript's letters, with one exception, so several letters mean different things in the three Parts. Table 3 fixes them; within Part III only the Part III column applies, and the last column prints the tempting false reading. Table 1 compares Parts I and II.

Table 3: Symbols of Part III whose meaning differs in Parts I and II.

Symbol	Parts I and II	Part III	Not to be read as
H	I: parameter heights $(M(H), \mathcal{A}(H))$, box sides H_A, H_B in Research question 12.3; II: $\mathcal{H}_c^{\text{sp}}(T)$ is a set	the monic inverse polynomial $H(S)$ (29.2); the counts $H_2(T), H_2^{\text{ns}}(T), H_2^{\text{sp}}(T)$ of Hasse failures on $C = 2$	a height
G	I: the binary cubic $G_{A,B,C}(T, S)$ (2.8)	the positive integer $\gcd(U , V)$ (29.2)	a cubic
U, V	I: $u = 1 + xy$ (lower case); II: an indeterminate U in the proof of Lemma 17.1, and auxiliary integers U, V in the proof of Theorem 19.3	$U = 3BC - 4, V = 27AC^2 - 4$	
E	not used	the dyadic condition $E(A, B, C)$ (table before Theorem 29.2)	

Symbol	Parts I and II	Part III	Not to be read as
S	a finite set of primes; I: also the second projective variable in (2.8); II: a sum, printed Σ	the indeterminate of $H(S)$; a finite set of finite places in Section 38; the manuscript's split count $S(T)$ is printed $\Sigma(T)$	a count
Σ	II: a sum of two half-numerators or roots	$\Sigma(T)$, the locally integral completely split targets on $C = 2$ in the box ($= \#\mathcal{H}_2^{\text{sp}}(T)$)	a sum
a, b, r	I: integer roots a, b, c on $C = 2$, r an index; II: root numerators with $a + b + r = 2$	a : an integer root of H (a root numerator, $a = Ct$); $b = B/2$ and r an integer root of f on $C = 2$ (34.1); b is the target	roots of g
C, c	I: C the third coordinate, c a root; II: c the third coordinate	C the third coordinate (the questions write c for a fixed slice); c_0 an implied constant in Lemma 36.1	a root
d, D	I: $d(t) = g'(t)$, D_r products of root differences; II: $d = s - t$, $D = dv$, D_r for numerators	$D = H'(a)$ (30.2); $d = f'(r)$ (34.4), so $D = 4d$ and d is Part I's D_r on $C = 2$; d also a divisor in Lemmas 36.2 and 36.3	Part I's $d(t)$
Δ	I: the discriminant of the inverse cubic	Δ_a (33.2) and Δ (34.4), discriminants of the <i>quadratic</i> factor, $\Delta_{2r} = 4\Delta$ on $C = 2$	the cubic's
f, g	I: g the inverse cubic; II: also valuations	g the same (proof of Proposition 30.1); $f(T) = H(2T)/8 = g(T)/2$ on $C = 2$ (34.2)	
T	I: the indeterminate of g ; II: also the height bound	the height bound; also the indeterminate of f and g	
t	I: $t = y + 1/x$; II: an integer root	$t = y + 1/x$; a root t in Lemmas 35.2 and 37.1; the parameter of the cusp curve	
P, Q, R	the coordinates of F ; II: also the cutoff R	the coordinates; $P(T)$ the marked-root count (Section 36); $R = \lfloor \sqrt{T/8} \rfloor$ in Lemma 36.1; $R = O_{K,S}$ in Section 38	
L, I, N	I: $L = 2H + 1$, $I(q)$ (7.5), N a number of primes; II: $L = T^{1/3}$, the beta area I , $\mathcal{I}_c^{\text{sp}}$	$L_2(T)$, $I_2(T)$, $N(T)$ target counts on $C = 2$; L an interval length in Lemma 36.2; $\mathcal{I}_r(T)$ an interval	
α	I: α_p in Theorem 10.3, automorphisms α, β ; II: roots α, β, γ	$\alpha = 27/(4\pi^2)$ (36.3), Part I's density (4.6)	
e	I: $e = v_p(\det J)$ in Lemma 8.1; II: $e = v_2(c)$	$e = v(2)$, the valuation of $\det J$ up to a unit, as in Part I	Part II's e
q, τ, δ	I: q a field size, τ the project's family parameter; II: q a prime, τ a divisor count, $\delta(c)$ a density	$q = r $; τ the divisor function, as in Part II; δ a Newton correction in Lemma 32.1	

The one rename: the manuscript's split count $S(T)$ (Sections 37, 39 and 45) is printed $\Sigma(T)$, as Part II printed its sum S as Σ , because S is also the indeterminate of H and the finite set of places. The manuscript's implied constant C in the proof of Lemma 36.1 is printed c_0 , because C is the third coordinate. No normalization was changed.

28.5 What is printed twice, and why

The manuscript restates parts of Parts I and II to be self-contained. Its later sections cite these restatements, and all of its labels are kept, so they are printed where they stand; the manuscript attributes them (its attribution table in Section 29).

- The map (29.1) is (1.1), and (30.1) is (2.7).

- Proposition 30.1 is Part I’s Theorem 2.2 (and Part II’s Proposition 16.1) in the monic coordinate $a = Ct$: Part I’s $w = g'(t)/2 = 1/x$ is $D/(2C)$. The proof here is a direct substitution.
- Lemma 30.3 is the Chebotarev mechanism of Part I’s Lemma 6.1 and Theorem 6.2, applied to integral local points; the manuscript lists it as inherited.
- Lemma 31.1 at odd primes and Lemma 32.1 at $p = 2$ are cases of Part I’s inverse-ball Lemma 8.1 (with Corollary 8.2), and of Part II’s lifting fact (16.5). Lemma 32.1 is stated over any complete discrete valuation ring of characteristic zero, which Theorem 38.1 uses; it is kept as a second route, a Newton iteration.
- Modulus-eight sufficiency and the measure $11/32$ are Part I’s Theorem 9.1; the residue histogram in the Remark after Proposition 32.2 is (9.5) (pointer added there). The table E itself is new.
- Lemma 35.2 is Part I’s Theorem 4.1 (“It has no integral point”), with the same proof; the manuscript does not cite it, and a pointer was added there. (34.6) is Part I’s (4.2) for split fibers.
- (29.8) is Part II’s Theorem 15.3 at $c = 2$; the manuscript does not use it, and Lemma 37.1 is a weaker bound proved independently.
- New in their generality: Theorems 29.2, 31.3, the table E of Proposition 32.2, Corollary 33.1, Theorem 34.1 (which extends Part I’s criterion (4.4) from split targets to every locally integral target on $C = 2$), Corollary 34.2, Proposition 35.1, Theorem 35.3, the counting results of Sections 36 and 37, and Theorem 38.1, which extends the method of Part I’s Theorem 6.2 from rational to S -integral points.

28.6 The formal project and Part III

The manuscript cites ProveIt’s Lean file `Algebra/JacobianConjecture/Lean/JacobianConjecture/Counterexample.lean` at its pin `6d04e1e385f2fe7d1cfbdbd8fd8d4e45bd6b6c72`; the file is identical at Part II’s pin and at the placement commit, so it is cited here as [7]. The file defines the map as `counterexample` (namespace `LeanProofs.JacobianCounterexample`, lines 95–101) and proves `jacobianDet_counterexample`, $\det JF = -2$ over every commutative ring (line 134), and the collision `collision` (line 165). Part III uses the map, checks the determinant symbolically, and relies on no formal statement. *None of Part III’s results is formalized.* The project has no p -adic, Hasse-principle or counting material, and the placement of this report beside its Lean and Rocq developments confers no formal status. The manuscript’s formalization plan (Section 40) is a proposal; it keeps Chebotarev as an explicitly imported theorem.

29 The question, the answer, and its scope

29.1 A specific unresolved question in ProveIt

The repository’s report *Arithmetic Local–Global Dichotomies for ProveIt’s Three-Variable Keller Map* develops split integral Hasse failures and exact local fiber laws [write] (Parts I and II of this report). Its Part II classifies completely split fibers and counts them in fixed slices. It explicitly leaves open the case of one rational point accompanied by an irreducible quadratic pair, and the corresponding target-height asymptotics. Part I, Section 12.2, asks for local integral classification beyond split fibers; Section 12.3 asks for target-height asymptotics. The appended Part II records the nonsplit case as still open and asks for coefficient-level certificates [write] (Research questions 24.1 and 24.4).

The difficulty is genuine: a rational point can fail integrality at a prime where the two nonrational geometric points become locally rational and provide integral alternatives. Testing only the denominator of the rational point therefore gives the wrong local answer. Nevertheless, for this particular map the entire odd-prime phenomenon collapses to two coefficient congruences.

We use the snapshot 6d04e1e385f2fe7d1cfbdb8fd8d4e45bd6b6c72. The report is located at

SetTheory/Cardinals/docs/reports/jacobian-conjecture/
arithmetic-local-global-fibers/article.tex.

[write] That is the present report, whose Parts I and II are unchanged from this pin to the placement commit 3025c15df; the pinned commit is abbreviated 6d04e1e38 elsewhere in this report. Its mathematical assertions are treated as research claims rather than as formal consequences of residing in a repository. The proofs of our main theorems are independent of its unformalized counting arguments.

29.2 The map and the terminology

Throughout, let

$$\begin{aligned} u &= 1 + xy, & h &= u^2 z + y^2(1 + 3u), \\ F(x, y, z) &= (uh, y + 3xh, x(5 - 3u - x^2 z)). \end{aligned} \tag{29.1}$$

Thus its coordinate polynomials are

$$\begin{aligned} P &= (1 + xy)^3 z + y^2(1 + xy)(4 + 3xy), \\ Q &= y + 3x(1 + xy)^2 z + 3xy^2(4 + 3xy), \\ R &= 2x - 3x^2 y - x^3 z. \end{aligned}$$

The identity $\det JF = -2$ and explicit collisions are formalized in the repository [7] [write] (declarations `jacobianDet_counterexample` and `collision`; Section 28.6). The map is attributed there and by Gao to Alpöge; Gao establishes its complex fiber geometry [3].

Definition 29.1. An integral target $\mathbf{b} = (A, B, C) \in \mathbb{Z}^3$ is *everywhere locally integral* if $\mathbf{b} \in F(\mathbb{Z}_p^3)$ for every rational prime p . It is an *integral Hasse failure* if it is everywhere locally integral but does not belong to $F(\mathbb{Z}^3)$. A fiber is *nonsplit* in this article if its inverse cubic factors over $\mathbb{Q}$ as a linear factor times an irreducible quadratic. A *completely split* fiber has three distinct rational points unless repeated roots are explicitly mentioned.

Every everywhere locally integral target studied here has a rational preimage, and hence a real preimage. Adding an archimedean-solubility condition would not change the definition. An integral Hasse failure is not a rational Hasse failure.

29.3 Main classification

Put

$$\begin{aligned} H(S) &= S^3 - 2S^2 + BCS - 2AC^2, \\ U &= 3BC - 4, & V &= 27AC^2 - 4, & G &= \gcd(|U|, |V|). \end{aligned} \tag{29.2}$$

For integer coefficients $U \neq 0$, so G is positive. Let $E(A, B, C)$ be the following condition. All congruences in the first column are modulo 8.

C	Condition $E(A, B, C)$
0	no condition on A, B
4	B is even
2 or 6	A and B are even
odd	$[A \text{ even and } B \equiv 2 \pmod{4}]$ or $[A \text{ odd and } BC \equiv 2A + 1 \pmod{8}]$

Theorem 29.2 (Complete local certificate). *For every $(A, B, C) \in \mathbb{Z}^3$, the following are equivalent:*

- (i) $(A, B, C) \in F(\mathbb{Z}_p^3)$ for every prime p ;
- (ii) H has an integer root, G is a power of 2 (including 1), and $E(A, B, C)$ holds.

If $C = 0$, these conditions always hold and the target already has an integral preimage. If $C \neq 0$, global integral solubility is decided by the finitely many simple integer roots of H and the reconstruction formulas in Section 30.

The rational-root condition cannot be dropped. The theorem is a finite coefficient certificate *together with* an ordinary rational-root test; it is not a bounded congruence characterization of rational solubility.

29.4 Main counting and integral-image results

Let $L_2(T)$ count everywhere locally integral targets $(A, B, 2)$ with $|A|, |B| \leq T$. Let $I_2(T)$ count their globally integral subset, and let $H_2(T)$ and $H_2^{\text{ns}}(T)$ count all and nonsplit integral Hasse failures, respectively. Targets, not roots or ordered parameter tuples, are counted.

Theorem 29.3 (The dominant nonsplit law). *As $T \rightarrow \infty$,*

$$L_2(T) = \frac{27}{2\pi^2} T \log T + O(T), \quad (29.3)$$

$$H_2(T) = \frac{27}{2\pi^2} T \log T + O(T), \quad (29.4)$$

$$H_2^{\text{ns}}(T) = \frac{27}{2\pi^2} T \log T + O(T). \quad (29.5)$$

The entire integral image on $C = 2$ is parametrized without duplication by

$$(A, B) = (-2r^3 + r^2 + \varepsilon r, -6r^2 + 4r + 2\varepsilon), \quad r \in \mathbb{Z}, \quad \varepsilon \in \{-1, 1\}. \quad (29.6)$$

Consequently

$$I_2(T) = 2^{5/3} T^{1/3} + O(1). \quad (29.7)$$

The same conclusions, after reflecting the second coordinate, hold on $C = -2$.

The previous split theorem [write] (Part II, Theorem 15.3 at $c = 2$) gives

$$H_2^{\text{sp}}(T) \sim \frac{27\Gamma(1/3)^2}{8\pi^2\Gamma(2/3)} T^{2/3}. \quad (29.8)$$

[[write] In Part II's notation $H_2^{\text{sp}}(T) = \#\mathcal{H}_2^{\text{sp}}(T)$, and the constant is (22.7); the manuscript's $\Sigma(T)$ below counts the same targets (Section 28.3).] We do not claim that result anew and do not need its proof to establish (29.3)–(29.5): an elementary $O(T)$ bound for split fibers suffices. The new main term comes from the rational-plus- quadratic sector, not from refining the split-family constant.

29.5 Attribution and what is not claimed

Ingredient or result	Status here	Scope
Map, determinant, collisions	Prior; independently checked	ProveIt formal source; Gao
Inverse cubic and missing cusp	Prior geometry; rederived	All relevant odd characteristics
Rational Hasse argument	Inherited mechanism	Chebotarev plus cubic Galois groups
Modulus-8 sufficiency and image size	Prior dyadic phenomenon	Explicit coefficient table reverified
G -criterion for all rooted cubics	Main refinement	Includes the previously open nonsplit case
Primitive parametrization on $C = 2$	Main refinement	All locally integral targets
$T \log T$ and two integral curves	Main counting results	Square target boxes on $C = \pm 2$
General number-field certificate	Extension	Finite local tests outside a finite set

We make no claim of a sharp second term in (29.4), a uniform asymptotic for every varying nonzero C , or a new solution of the plane Jacobian problem. The limited literature search did not locate these specific arithmetic and counting statements in the inspected primary sources; that finding is not a global priority proof.

30 Inverse cubics and exact reconstruction

30.1 The boundary $C = 0$

Direct substitution gives

$$F(0, B, A - 4B^2) = (A, B, 0). \quad (30.1)$$

Thus no integral target on $C = 0$ is an integral Hasse failure. Formula (30.1) is also the reason primes dividing C will cause no odd-prime obstruction.

30.2 The affine inverse chart

Proposition 30.1 (Inverse chart). *Let k be a field of characteristic different from 2, and let $A, B, C \in k$ with $C \neq 0$. The points of $F^{-1}(A, B, C)(k)$ are in bijection with the simple roots $a \in k$ of H . If*

$$D = H'(a) = 3a^2 - 4a + BC \neq 0, \quad (30.2)$$

then the corresponding point is

$$\boxed{x = \frac{2C}{D}, \quad y = \frac{2a - D}{2C}, \quad z = \frac{D(10D - 12a - D^2)}{8C^2}.} \quad (30.3)$$

For a point in the fiber, the inverse coordinate is $a = C(y + 1/x)$.

Proof. Since $C = x(2 - 3xy - x^2z) \neq 0$, we have $x \neq 0$. Put $t = y + 1/x$ and $g(T) = CT^3 - 2T^2 + BT - 2A$. Substitution of (29.1), followed by collecting terms, gives

$$g(t) = 0, \quad g'(t) = \frac{2}{x}. \quad (30.4)$$

These identities can also be derived by eliminating z from the third coordinate, using $z = 2/x^2 - 3y/x - C/x^3$, and then substituting in the first two coordinates. As $H(Ct) = C^2g(t)$, differentiation yields $H'(Ct) = Cg'(t) = 2C/x \neq 0$.

Conversely, if $H(a) = 0$ and $D \neq 0$, set $t = a/C$ and $x = 2C/D$. Then $y = t - 1/x = (2a - D)/(2C)$. Solving the third coordinate for z gives exactly (30.3). Substitution in the first two coordinates reduces their residuals to multiples of $H(a)$, so they equal A, B . All operations divide only by 2, C , and D , which are nonzero. The recovered value $C(y + 1/x) = a$ proves injectivity of the construction. $\square$

The same root chart appears in the prior arithmetic report, expressed with t rather than the monic coordinate a [write] (Part I, Theorem 2.2; Part II, Proposition 16.1). Gao's degree- three and missing-fiber geometry is consistent with this description [3]. We use the monic version because an integral coefficient polynomial has integral rational roots.

Corollary 30.2 (Exact integral test). *For $C \neq 0$ and a simple integer root a of H , the reconstructed point is integral if and only if*

$$D \mid 2C, \quad 2C \mid 2a - D, \quad 8C^2 \mid D(10D - 12a - D^2). \quad (30.5)$$

An integral target belongs to $F(\mathbb{Z}^3)$ if and only if either $C = 0$ or at least one simple integer root satisfies (30.5).

Proof. Each rational coordinate in (30.3) is integral precisely under the corresponding divisibility condition. Every rational root of H is an integer by the rational-root theorem, and Proposition 30.1 accounts for the entire rational fiber. $\square$

30.3 Why local solubility supplies a rational root

Lemma 30.3 (The cubic rational-root gate). *If $(A, B, C) \in \mathbb{Z}^3$ is everywhere locally integral, then H has an integer root. For $C \neq 0$, a rational root of H guarantees that H has a simple rational root.*

Proof. When $C = 0$, $H = S^2(S - 2)$. Suppose $C \neq 0$. If H were irreducible over $\mathbb{Q}$, its Galois group would be a transitive subgroup of S_3 , hence C_3 or S_3 . Both contain a 3-cycle. By the Chebotarev density theorem [6, Theorem 8.31 and Example 8.36], there are infinitely many good primes, excluding $2C$ and the discriminant, for which H has irreducible reduction. At such a prime it has no root in $\mathbb{Q}_p$: any $\mathbb{Q}_p$ -root of the monic integral polynomial would be p -integral and reduce to a root. Proposition 30.1 rules out a local preimage, a contradiction.

A reducible cubic has a rational root, which is integral here. If a rational root is repeated but not triple, the remaining linear factor has a simple rational root. The triple case would be $H = (S - 2/3)^3$, impossible for a monic integer polynomial. This proves the last assertion. $\square$

The argument uses qualitative Chebotarev, not an unproved hypothesis. Its purpose is only to justify the rational-root gate. The subsequent local tests and the entire counting proof are elementary.

31 Every odd prime is controlled by one gcd

31.1 From integral points to simple roots modulo a prime

Lemma 31.1. *For every odd prime p ,*

$$\mathbf{b} \in F(\mathbb{Z}_p^3) \iff \overline{\mathbf{b}} \in F(\mathbb{F}_p^3).$$

If $p \nmid C$, the right side holds precisely when $\overline{H}$ has a simple root in $\mathbb{F}_p$. If $p \mid C$, it always holds.

Proof. Reduction gives necessity. At any residue preimage, $\det JF = -2$ is a unit modulo p , so the multivariable Hensel argument lifts it to the prescribed target over $\mathbb{Z}_p$. When $p \nmid C$, apply Proposition 30.1 over $\mathbb{F}_p$. When $p \mid C$, use (30.1) modulo p and then lift. $\square$

Lemma 31.2 (The only obstruction for a rooted cubic). *Let $q(S)$ be a monic cubic over a field and suppose it has a root in that field. It has no simple root in the field if and only if $q(S) = (S - a)^3$ for some field element a .*

Proof. If a root is simple there is nothing to prove. Otherwise divide by its square. The remaining factor is linear. Its root is simple unless it coincides with the repeated root, which is exactly the triple case. The converse is immediate in every characteristic. $\square$

31.2 The coefficient criterion

Theorem 31.3 (Odd-prime classification). *Assume H has an integer root. For each odd prime p ,*

$$(A, B, C) \in F(\mathbb{Z}_p^3) \iff p \nmid \gcd(3BC - 4, 27AC^2 - 4). \quad (31.1)$$

Proof. The integer root reduces to a root modulo every prime. If $p \mid C$, Lemma 31.1 gives local solubility, and $3BC - 4 \equiv -4 \not\equiv 0 \pmod{p}$ gives the right side.

Suppose $p \nmid C$. By Lemmas 31.1 and 31.2, failure occurs exactly when $\overline{H}$ is a cube of a linear factor. In characteristic 3, a linear cube has no S^2 term, whereas H has coefficient $-2 \not\equiv 0$; moreover $3BC - 4 \not\equiv 0 \pmod{3}$.

For $p \neq 2, 3$, the S^2 coefficient forces the triple root to be $2/3$. Comparing the other coefficients with $(S - 2/3)^3$ gives

$$BC = \frac{4}{3}, \quad AC^2 = \frac{4}{27} \quad \text{in } \mathbb{F}_p.$$

These are exactly $p \mid U$ and $p \mid V$. $\square$

Remark 31.4 (The geometric source of the gcd). The locus $U = V = 0$ is the missing cusp in the inverse-cubic geometry; in characteristic zero it is parametrized by $(A, B, C) = (4/(27t^2), 4/(3t), t)$, $t \neq 0$ [3]. The new arithmetic point is that, *after a rational root is known*, every odd local failure is reduction into this one locus. Merely avoiding the cusp is not sufficient for an arbitrary irreducible cubic.

31.3 Why the quadratic pair need not be analyzed separately

Suppose a rational root a is known. At a good odd prime, its reduction may be repeated. If it is double rather than triple, the remaining linear factor has a simple residue root and lifts to a local integral point. That point need not arise from a globally rational geometric point. Thus quadratic splitting does enter the mechanism, but does not survive as an independent condition in (31.1).

This explains why tracking discriminants, Legendre symbols, and the valuation of each reconstructed nonrational point would overcomplicate this problem. Such data are useful for finer fiber distributions, but not for the yes/no local-solubility classification under the root gate.

32 The prime two and a finite coefficient table

32.1 A lifting lemma with a nonunit determinant

Lemma 32.1 (Uniform inverse correction). *Let O be a complete discrete valuation ring of characteristic zero, with valuation v , and put $e = v(2)$. For the map F , suppose $a_0 \in O^3$ and $b \in O^3$ satisfy*

$$\min_i v(F_i(a_0) - b_i) > 2e.$$

Then there is $a \in O^3$ with $F(a) = b$. In particular, over $\mathbb{Z}_2$ a solution modulo 8 lifts to a solution over $\mathbb{Z}_2$.

Proof. At each integral point, $J^{-1} = -(1/2) \operatorname{adj}(J)$ has entries of valuation at least $-e$. If the current error has minimum valuation $k > 2e$, take the Newton correction $\delta = -J^{-1}(F(a) - b)$. Then $v(\delta_i) \geq k - e > e \geq 0$. The integral polynomial Taylor expansion has linear term canceled and all remaining terms of degree at least two in δ . Hence the new error has minimum valuation at least $2k - 2e$.

Iterating, the excess $k - 2e$ at least doubles, and corrections tend to zero in valuation. Completeness gives an integral limit, and polynomial continuity gives $F(a) = b$. For $\mathbb{Z}_2$, an error divisible by 8 has $k \geq 3 > 2$. $\square$

32.2 The complete residue certificate

Proposition 32.2 (Dyadic coefficient criterion). *For every integer target,*

$$(A, B, C) \in F(\mathbb{Z}_2^3) \iff E(A, B, C). \quad (32.1)$$

The set of admissible targets modulo 8 has cardinality 176.

Proof. Lemma 32.1 reduces the statement to the finite image of F on $(\mathbb{Z}/8\mathbb{Z})^3$. Substituting the 512 input triples and collecting outputs gives the table defining E in Section 29. Section 46 [write] (the manuscript's Appendix D) gives a full exhaustive certificate. It uses only integer arithmetic and finite loops, and establishes both inclusions, not just the number of admissible targets.

The certificate has been executed in the supplied verification package. The finite identity it checks, together with the lifting lemma, proves the statement at every 2-adic precision. $\square$

The size 176 also follows immediately from the coefficient table: 64 targets for $C = 0$, 32 for $C = 4$, 16 for each of $C = 2, 6$, and 12 for each of the four odd values of C . The corresponding normalized image measure is $176/512 = 11/32$, agreeing with the prior dyadic law [write] (Part I, (9.4) in Theorem 9.1). This measure is not claimed as a new result.

Remark 32.3 (Finite fibers are not 2-adic fibers). The counts of residue targets with respectively 0, 2, 4, 6 preimages modulo 8 are 336, 112, 48, 16. These are finite-ring fiber cardinalities. They must not be confused with cardinalities of fibers over $\mathbb{Z}_2$. Our argument needs only membership in the image. [write] These counts are Part I's (9.5) at $m = 3$, $2^9 \cdot (21, 7, 3, 1)/32$, and its table in the proof of Theorem 9.1 (total row); Part I also gives the 2-adic fiber law (9.3) that this remark distinguishes them from.

33 Complete classification, including nonsplit fibers

33.1 Proof of the main certificate

Proof of Theorem 29.2. If a target is locally integral everywhere, Lemma 30.3 gives the integer root. Theorem 31.3 then says no odd prime divides G , while Proposition 32.2 gives E .

Conversely, an integer root and a power-of-two gcd imply local integral solubility at every odd prime by Theorem 31.3. Condition E supplies the prime 2. If $C = 0$, formula (30.1) gives a global integral point; the conditions are indeed automatic, since $H = S^2(S - 2)$ and $G = 4$. $\square$

33.2 An explicit rational-plus-quadratic formulation

For an integer root a , factor

$$H(S) = (S - a)(S^2 + (a - 2)S + a^2 - 2a + BC). \quad (33.1)$$

The discriminant of the quadratic is

$$\Delta_a = -3a^2 + 4a + 4 - 4BC = (3a - 2)^2 - 4D. \quad (33.2)$$

Here and below, “a square in $\mathbb{Z}$ ” means the square of an integer; negative integers are not squares.

Corollary 33.1 (Complete nonsplit Hasse test). *Suppose $C \neq 0$, $H(a) = 0$ for an integer a , and Δ_a is not a square in $\mathbb{Z}$. Then the fiber has exactly one rational point, given by (30.3). It is an integral Hasse failure if and only if G is a power of two, $E(A, B, C)$ holds, and at least one of the three divisibilities in (30.5) fails.*

Proof. The discriminant condition says the quadratic factor is irreducible over $\mathbb{Q}$. In particular $D \neq 0$, since otherwise a would also be a root of that factor. Proposition 30.1 gives the unique rational point. Apply Theorem 29.2 and Corollary 30.2. $\square$

This resolves the rational-plus-quadratic local-solubility question for *all* integral target slices, not only $C = 2$. It also supplies the requested coefficient certificate at every odd prime. What remains restricted to $C = 2$ is the sharp counting law.

33.3 The three tests are genuinely different

The following examples isolate the conditions. In the first column, all targets have $C = 2$.

Target	Rational-root gate	Odd gcd test	Dyadic test
(2, 0, 2)	fails	passes	passes
(8, 4, 2)	passes	fails at 5	passes
(1, 2, 2)	passes	passes	fails
(0, 4, 2)	passes	passes	passes

For (2, 0, 2) the relevant cubic is $T^3 - T^2 - 2$, which has no rational root and already has no root modulo 3. The gcd equals 4; avoiding the cusp alone does not help. For (8, 4, 2) the root is 2, and the cubic becomes $(T - 2)^3$ modulo 5. For (1, 2, 2) the inverse cubic is $(T - 1)(T^2 + 1)$, but the target fails the parity condition at 2. Finally, (0, 4, 2) will be our smallest displayed nonsplit integral Hasse failure, with unique rational point $(1/2, -2, 4)$.

34 A primitive parametrization on the slice $C = 2$

34.1 All locally integral targets

Theorem 34.1 (Primitive root coordinates). *An integer target $(A, B, 2)$ is everywhere locally integral if and only if there exist $r, b \in \mathbb{Z}$ such that*

$$A = r^3 - r^2 + rb, \quad B = 2b, \quad \gcd(3r - 1, 3b - 1) = 1. \quad (34.1)$$

Every integer root of the corresponding cubic satisfies the gcd condition, not just one chosen root.

Proof. The dyadic table forces B and A to be even. Write $B = 2b$ and define

$$f(T) = T^3 - T^2 + bT - A. \quad (34.2)$$

The relation with the monic inverse polynomial is $H(2T) = 8f(T)$. The rational-root gate is therefore equivalent to an integer root r of the monic polynomial f , or equivalently the first equation of (34.1).

Now

$$G = 4 \gcd(3b - 1, 27A - 1),$$

and the exact congruence

$$27A - 1 \equiv (3r - 1)^3 \pmod{3b - 1} \quad (34.3)$$

shows that its odd prime support is the common odd prime support of $3r - 1$ and $3b - 1$. In fact the difference in (34.3) is $9r(3b - 1)$. Modulo 2, $A = r^3 - r^2 + rb \equiv rb$. The gcd of $3r - 1$ and $3b - 1$ is even exactly when both r and b are odd, exactly when A is odd. Thus the odd gcd test and the dyadic test combine into $\gcd(3r - 1, 3b - 1) = 1$. The argument applies to any integer root of f . The converse follows from the same identities and Theorem 29.2. $\square$

34.2 Reconstruction and the quadratic discriminant

Put

$$d = 3r^2 - 2r + b = f'(r), \quad \Delta = (3r - 1)^2 - 4d = -3r^2 + 2r + 1 - 4b. \quad (34.4)$$

Then

$$f(T) = (T - r)(T^2 + (r - 1)T + r^2 - r + b). \quad (34.5)$$

Whenever $d \neq 0$, the point belonging to r is particularly simple:

$$(x, y, z) = \left(\frac{1}{d}, r - d, 5d^2 - 3rd - 2d^3 \right). \quad (34.6)$$

Indeed, in (30.3) take $a = 2r$, $C = 2$, and $D = 4d$. The second and third coordinates are integers automatically. Thus the entire global integrality obstruction is $d \neq \pm 1$.

For reference, the primitive gcd may also be tested with the derivative:

$$\gcd(3r - 1, d) = \gcd(3r - 1, 3b - 1), \quad (34.7)$$

because $3d \equiv 3b - 1 \pmod{3r - 1}$ and 3 is coprime to $3r - 1$. This is an equality of positive gcDs, not merely an equality of their prime supports.

34.3 An infinite nonsplit Hasse family

Corollary 34.2. *For every integer $b \geq 2$, the target*

$$(0, 2b, 2) \tag{34.8}$$

is an integral Hasse failure with exactly one rational preimage,

$$\left(\frac{1}{b}, -b, 5b^2 - 2b^3\right). \tag{34.9}$$

Proof. Take $r = 0$. The primitive gcd is $\gcd(-1, 3b - 1) = 1$, so all local conditions hold. The remaining quadratic is $T^2 - T + b$, whose discriminant $1 - 4b$ is negative. Formula (34.6) gives (34.9), and its first coordinate is not integral. $\square$

This family makes branch switching visible. At an odd prime dividing b , the rational point is not integral, but the reduction of $T(T^2 - T + b)$ is $T^2(T - 1)$: its root 1 is simple. The quadratic branch therefore supplies an integral local point. For $b = 2$, the nonrational pair has discriminant -7 , and the rational point is $(1/2, -2, 4)$.

35 Repeated roots and the entire integral image

35.1 The unique repeated-root exception

Proposition 35.1. *The only everywhere locally integral target on $C = 2$ whose inverse cubic has a repeated root is $(A, B, 2) = (0, 0, 2)$. Its inverse polynomial is $T^2(T - 1)$; its unique rational point in the fiber is $(1, 0, 0)$.*

Proof. A repeated root of a monic cubic over $\mathbb{Q}$ is rational, hence integral. Write

$$f(T) = (T - s)^2(T - r), \quad r = 1 - 2s, \quad b = 2s - 3s^2.$$

Then

$$3b - 1 = -(3s - 1)^2,$$

so Theorem 34.1, applied to the root s , requires $|3s - 1| = 1$. The only integer solution is $s = 0$; the apparent alternative $s = 2/3$ is not integral. Therefore $r = 1$, $b = A = 0$. The repeated root 0 does not correspond to a point of the inverse chart; the simple root 1 gives $(1, 0, 0)$. A triple root would require $3s = 1$ and is impossible. $\square$

The repeated root is counted once when marking distinct integer roots, not twice according to multiplicity. This distinction will produce the constant corrections in the exact counting identities.

35.2 Split distinct fibers have no integral point

Lemma 35.2. *On $C = 2$, a completely split fiber with three distinct rational points has no integral point.*

Proof. The three roots r, s, t of the monic polynomial f are integers and satisfy $r + s + t = 1$. If the point belonging to r were integral, then

$$f'(r) = (r - s)(r - t) = \pm 1$$

by (34.6). Both integer differences must be 1 or -1 . Since $s \neq t$, the other roots must be $r - 1, r + 1$. Their sum would be $3r = 1$, impossible for integer r . $\square$

This proof is specific to the sum of roots on $C = 2$; it should not be exported without modification to an arbitrary slice. [write] This is Part I's Theorem 4.1 ("It has no integral point"), reproved here by the same argument; the manuscript does not cite it. The three roots r, s, t of f are Part I's integer roots a, b, c of (4.1), and $f'(r)$ is Part I's D_r (Section 28.5). Part II's Theorem 20.1 gives the finitely many exceptions on other slices.

35.3 Two polynomial curves

Theorem 35.3 (Exact integral-image curves). *The targets in $F(\mathbb{Z}^3)$ with third coordinate 2 are precisely*

$$\begin{aligned} A &= -2r^3 + r^2 + \varepsilon r, \\ B &= -6r^2 + 4r + 2\varepsilon, \quad r \in \mathbb{Z}, \quad \varepsilon \in \{-1, 1\}. \end{aligned} \tag{35.1}$$

The corresponding integral preimage is

$$(x, y, z) = (\varepsilon, r - \varepsilon, 5 - \varepsilon(3r + 2)). \tag{35.2}$$

Distinct pairs (r, ε) give distinct targets.

Proof. Any integral target is locally integral, so $B = 2b$ and the inverse cubic f is available. Its point corresponds to a simple integer root r . By (34.6), integrality is equivalent to $d = \varepsilon \in \{-1, 1\}$. Solving for b gives $b = \varepsilon - 3r^2 + 2r$. Substitution into $A = r^3 - r^2 + rb$ and $B = 2b$ gives (35.1), and (34.6) becomes (35.2). Conversely, these formulas substitute directly into F and give integral points.

If two different parameter pairs gave the same target, they could not have the same r , since B determines ε once r is fixed. There would therefore be two different simple rational roots. A cubic with two distinct simple rational roots must have three distinct rational roots. Lemma 35.2 contradicts the existence of the integral point. This also handles possible intersections of the two curves. The repeated-root target has only one simple root, so is not an exception to injectivity. $\square$

Corollary 35.4 (Exact integral-image scale). *In the square target box,*

$$I_2(T) = 2^{5/3}T^{1/3} + O(1).$$

Except for $(0, 0, 2)$, every integral target on this slice has a nonsplit fiber.

Proof. For each fixed ε , the first coordinate in (35.1) is $-2r^3 + O(r^2)$. The inequality $|A| \leq T$ therefore restricts r to endpoints $\pm(T/2)^{1/3} + O(1)$. More explicitly, for all sufficiently large positive or negative r the cubic is strictly monotone on that tail, and its real threshold roots have these estimates. The number of integer r is $2(T/2)^{1/3} + O(1)$. For these r , $|B| = O(T^{2/3}) < T$ for sufficiently large T , so the second-coordinate bound removes no additional asymptotic range. There are two disjoint integer-parametrized images, proving the formula. A rational cubic factorization other than nonsplit would be completely split or repeated; Lemma 35.2 and Proposition 35.1 leave only the stated exception. $\square$

36 Marked-root counting and the $T \log T$ term

36.1 Why a logarithm should appear

The parametrization is nearly quadratic in its center but linear in b . For a nonzero root r , keeping A bounded forces b into an interval of length about $2T/|r|$ centered at $-r^2 + r$. The bound on $B = 2b$ permits $|r|$ up to a constant multiple of $\sqrt{T}$. Summing $T/|r|$ through this range produces $T \log T$. The constant is not an arbitrary density: it is the mean primitive density in the arithmetic progressions $3r \pm 1$.

36.2 The exact parameter region

Let $P(T)$ denote the number of pairs $(r, b) \in \mathbb{Z}^2$ satisfying (34.1) and $|A|, |B| \leq T$. These pairs mark integer roots, so they are not yet target counts. We may first assume T is an integer; replacing a real T by $\lfloor T \rfloor$ does not change the conclusions.

For $r \neq 0$, the permitted real interval for b is

$$\mathcal{I}_r(T) = [-T/2, T/2] \cap [-r^2 + r - T/|r|, -r^2 + r + T/|r|]. \quad (36.1)$$

For $r = 0$, every $|b| \leq T/2$ is primitive, yielding $2\lfloor T/2 \rfloor + 1 = O(T)$ pairs.

Lemma 36.1 (Range and boundary reduction). *Only $|r| = O(\sqrt{T})$ contribute. If $R = \lfloor \sqrt{T/8} \rfloor$, then, for sufficiently large T and $3 \leq |r| \leq R$, the second interval in (36.1) is contained in the first. The total contribution from $|r| > R$, together with $|r| < 3$, is $O(T)$.*

Proof. Nonemptiness implies

$$r^2 - r \leq T/2 + T/|r|.$$

In terms of $q = |r|$, the left side is at least $q^2 - q$, so $q = O(\sqrt{T})$. For example the explicit bound $q \leq \lfloor \sqrt{2T} \rfloor + 3$ is safe for every integer $T \geq 1$: beyond it, $q^2 - q - T/2 > T/q$.

If $3 \leq q \leq R$, the absolute center is at most $T/8 + \sqrt{T/8}$, and the radius is at most $T/3$. Their sum is smaller than $T/2$ once T is sufficiently large. Thus the interval is uncut there and has length $2T/q$.

For $R < q \leq c_0\sqrt{T}$ [write] (the manuscript writes $C\sqrt{T}$; c_0 is the implied constant of $q = O(\sqrt{T})$ above), forget the gcd condition and bound the number of possible integers by $2T/q + 1$. The harmonic sum over this fixed-ratio range is $O(1)$, so its contribution is $O(T)$. Each of the finitely many roots $|r| < 3$ also contributes $O(T)$. $\square$

36.3 Primitive points in each interval

Lemma 36.2 (A uniform interval count). *For fixed $r \in \mathbb{Z}$ and any interval of length L , the number of integers b in the interval with $\gcd(3r - 1, 3b - 1) = 1$ is*

$$L \frac{\varphi(|3r - 1|)}{|3r - 1|} + O(\tau(|3r - 1|)), \quad (36.2)$$

with an absolute implied constant. Here τ is the divisor function.

Proof. Write $n = |3r - 1|$, so $3 \nmid n$. Inclusion–exclusion gives

$$\sum_{b \in I \cap \mathbb{Z}} \sum_{d | \gcd(n, 3b-1)} \mu(d) = \sum_{d | n} \mu(d) \#\{b \in I \cap \mathbb{Z} : 3b \equiv 1 \pmod{d}\}.$$

Each congruence is one residue class, hence contributes $L/d + O(1)$. The main term is $L \sum_{d|n} \mu(d)/d = L\varphi(n)/n$, and summing the errors gives the assertion. $\square$

Lemma 36.3 (The arithmetic mean). *For each sign,*

$$\sum_{1 \leq r \leq X} \frac{\varphi(3r \pm 1)}{3r \pm 1} = \alpha X + O(\log(2X)), \quad \alpha = \prod_{p \neq 3} (1 - p^{-2}) = \frac{27}{4\pi^2}. \quad (36.3)$$

Consequently

$$\sum_{1 \leq r \leq X} \frac{\varphi(3r \pm 1)}{r(3r \pm 1)} = \alpha \log X + O(1). \quad (36.4)$$

Proof. Expand $\varphi(n)/n = \sum_{d|n} \mu(d)/d$. Since 3 never divides $3r \pm 1$, only $3 \nmid d$ occur. For each such d , the condition $3r \pm 1 \equiv 0 \pmod{d}$ specifies one residue class, with $X/d + O(1)$ representatives up to X . Thus

$$\sum_{r \leq X} \frac{\varphi(3r \pm 1)}{3r \pm 1} = X \sum_{\substack{d \leq 3X+1 \\ 3 \nmid d}} \frac{\mu(d)}{d^2} + O\left(\sum_{d \leq 3X+1} \frac{1}{d}\right).$$

The tail of the absolutely convergent main sum is $O(1/X)$. Its Euler product is

$$\alpha = \frac{1}{\zeta(2)(1 - 3^{-2})} = \frac{27}{4\pi^2}.$$

This proves (36.3). Partial summation gives (36.4); the error integral $\int_1^X O(\log(2t))t^{-2} dt$ is bounded. $\square$

[write] The constant α is the density $27/(4\pi^2)$ of Part I's (4.6), obtained there from the same Möbius sum $\sum_{3 \nmid d} \mu(d)/d^2$; the counted quantities differ (here a mean of $\varphi(3r \pm 1)/(3r \pm 1)$ over roots, there a density of ordered root pairs). Part I's α_p in Theorem 10.3 is unrelated (Table 3).

Proposition 36.4 (Marked-root asymptotic).

$$P(T) = \frac{27}{2\pi^2} T \log T + O(T). \quad (36.5)$$

Proof. Apply Lemmas 36.1 and 36.2 on the full intervals:

$$P(T) = 2T \sum_{3 \leq |r| \leq R} \frac{\varphi(|3r - 1|)}{|r| |3r - 1|} + O(T).$$

The accumulated divisor errors are harmless: since the integers $|3r - 1|$ lie below $O(\sqrt{T})$ and each is attained at most a bounded number of times,

$$\sum_{|r| \leq R} \tau(|3r - 1|) \ll \sum_{n \ll \sqrt{T}} \tau(n) \ll \sqrt{T} \log(2T) = O(T).$$

The last inequality follows directly from $\sum_{n \leq Y} \tau(n) = \sum_{d \leq Y} \lfloor Y/d \rfloor \ll Y \log(2Y)$.

Positive roots produce $3r - 1$, negative roots produce $3|r| + 1$. Lemma 36.3 therefore makes the displayed sum $2\alpha \log R + O(1)$. Finally $\log R = \frac{1}{2} \log T + O(1)$, giving $P(T) = 2\alpha T \log T + O(T)$ and the stated constant. $\square$

There is no hidden equidistribution hypothesis in this computation. The primitive density is obtained by a uniformly controlled divisor sum.

37 From roots to targets: exact multiplicity corrections

37.1 An elementary bound for split targets

Let $\Sigma(T)$ be the number of locally integral targets in the box with three distinct rational roots, and let $N(T)$ count locally integral nonsplit targets. By Lemma 35.2, every target counted by $\Sigma(T)$ is already an integral Hasse failure.

Lemma 37.1. $\Sigma(T) = O(T)$.

Proof. For a split cubic write the integer roots as r, s, t with sum 1. The second elementary symmetric function is b , so

$$r^2 + s^2 + t^2 = (r + s + t)^2 - 2b = 1 - 2b \leq T + 1.$$

Thus r, s range in a disk inside a square of side $O(\sqrt{T})$, and $t = 1 - r - s$ is fixed by them. There are $O(T)$ ordered pairs, hence no more than $O(T)$ targets. Imposing the A bound and the primitive gcd can only reduce the count. $\square$

This deliberately elementary estimate is enough for our main theorem. The stronger prior formula (29.8) is compatible with, but not required by, the argument.

37.2 Exact identities

For every integer $T \geq 1$, Proposition 35.1 yields

$$P(T) = 3\Sigma(T) + N(T) + 2, \tag{37.1}$$

$$L_2(T) = \Sigma(T) + N(T) + 1 = P(T) - 2\Sigma(T) - 1. \tag{37.2}$$

The 2 in the first identity is the number of distinct integer roots 0, 1 of $T^2(T - 1)$; the 1 in the second counts its single target. Every other locally integral cubic is either nonsplit, with one integer root, or split distinct, with three.

All integral targets except this repeated-root target are nonsplit. Therefore the exact Hasse counts are

$$H_2^{\text{ns}}(T) = P(T) - 3\Sigma(T) - I_2(T) - 1, \tag{37.3}$$

$$H_2(T) = P(T) - 2\Sigma(T) - I_2(T) - 1. \tag{37.4}$$

These formulas are also used by the supplied counting program. They prevent the common error of dividing every marked-root count by three.

Proof of Theorem 29.3. Insert Proposition 36.4, Lemma 37.1, and Corollary 35.4 into (37.2), (37.3), and (37.4). The curve statement is Theorem 35.3. Finally direct substitution in (29.1) gives

$$F(-x, -y, z) = (P(x, y, z), -Q(x, y, z), -R(x, y, z)).$$

This bijection preserves square target heights and all integral and local-solubility conditions, transferring the results to $C = -2$. $\square$

37.3 Density consequences and the distinction of sample spaces

Corollary 37.2. *Among everywhere locally integral targets on $C = 2$ in growing square boxes, the proportion that are integral Hasse failures tends to 1. Among those Hasse failures, the proportion that are nonsplit also tends to 1. More precisely,*

$$\frac{I_2(T)}{L_2(T)} \sim \frac{2^{5/3} 2\pi^2}{27} \frac{T^{-2/3}}{\log T}.$$

In contrast, the proportion of everywhere locally integral targets among all integer targets in the square tends to 0.

Proof. Use $L_2(T) \sim (27/(2\pi^2))T \log T$, $I_2(T) \sim 2^{5/3}T^{1/3}$, and $\Sigma(T) = O(T)$. The ambient box has $(2\lfloor T \rfloor + 1)^2$ targets, which is of order T^2 . $\square$

Thus “almost every” here has a specified conditioning set. No positive ambient density is asserted. The arithmetic image can be thin in the ambient plane while local-to-global failure is overwhelmingly typical inside its locally soluble part.

38 Number fields and rings of S -integers

The collapse of odd local conditions is not special to rational primes. It uses a rooted cubic over a finite residue field and the invertibility of 2. We record a general extension, including the effect of dyadic ramification.

Let K be a number field and S a finite set of finite places. Put $R = \mathcal{O}_{K,S}$. For a finite place $v \notin S$, write $\mathcal{O}_v$ for the valuation ring of the completion. All target coordinates now lie in R . The ring $R/8R$ is finite; if 2 is a unit in R , it is the zero ring and its image condition is vacuous.

Theorem 38.1 (Finite certificate over S -integers). *Let $C \neq 0$. A target $\mathbf{b} \in R^3$ belongs to $F(\mathcal{O}_v^3)$ for every finite $v \notin S$ if and only if all three conditions hold:*

- (a) H has a simple root in K ;
- (b) for every $v \notin S$ of odd residue characteristic, at least one of U, V is a v -adic unit;
- (c) the residue of $\mathbf{b}$ in $(R/8R)^3$ lies in $F((R/8R)^3)$.

Condition (b) is equivalently that the ideal $(U, V)R$ has no prime support outside S and the dyadic places. For $C = 0$, every target has the global R -point (30.1).

Proof. Suppose local solubility holds outside S . If H is irreducible over K , its transitive cubic Galois group has a 3-cycle. Chebotarev gives infinitely many residue fields of irreducible reduction, avoiding the finite sets S , the dyadic places, the primes of C , and the bad reduction primes. No local root is possible there. Hence H is reducible. A reducible cubic has a simple K -root unless it is a triple linear factor. The triple case has no point over any extension field by Proposition 30.1, contradicting local solubility. This proves (a).

A root in K of the monic polynomial $H \in R[T]$ belongs to the integrally closed ring R . Reducing it at any $v \notin S$ gives a residue root. At odd residue characteristic the proof of Theorem 31.3 works verbatim over the residue field: places dividing C are automatically soluble, residue characteristic 3 permits no triple root, and in every other odd characteristic the only obstruction is $U = V = 0$. This proves the equivalence of the odd local conditions with (b), in both directions.

It remains to handle dyadic places not inverted in R . Necessity of (c) follows by reducing the local solutions modulo $8O_v$ and combining their residues by the Chinese remainder theorem. Indeed, $R/8R$ is the product of the corresponding finite local quotients. Conversely, (c) supplies an integral approximate solution at each such place. If $e = v(2)$, its error has valuation at least $3e > 2e$, so Lemma 32.1 gives a genuine local point. Thus (a)–(c) are sufficient as well. $\square$

Remark 38.2 (Why “simple” is explicit in the extension). Over $\mathbb{Z}$, the triple polynomial $(S - 2/3)^3$ cannot have integral coefficients. Over S -integers with 3 inverted it can. The word “simple” in condition (a) prevents importing that integer-specific shortcut into the more general theorem. Equivalently, one could allow any K -root and separately exclude the triple case.

The dyadic modulus $8R$ is a uniform sufficient modulus, including ramified dyadic fields. We do not claim that it is minimal. The compact rational coefficient table E is special to $\mathbb{Z}/8\mathbb{Z}$; for a general number field the finite image in (c) must be computed in its actual quotient ring, not by applying that table componentwise to coordinates of an arbitrary integral basis.

Once a simple root is known, global R -membership is likewise finite: reconstruct all simple K -roots by (30.3) and test membership of their coordinates in R . The theorem therefore supplies a complete local/global decision procedure, although no optimal complexity bound is claimed for number-field factorization.

39 Algorithms and reproducible checks

39.1 An exact classification algorithm

For integer input (A, B, C) , the following procedure has no prime-search loop and needs no factorization of G .

1. If $C = 0$, return the point $(0, B, A - 4B^2)$.
2. Compute $G = \gcd(|3BC - 4|, |27AC^2 - 4|)$ and repeatedly divide out 2. Reject local solubility if the remaining odd part exceeds 1, or if $E(A, B, C)$ fails.
3. Compute the rational roots of the monic cubic H . They are integers. If none exist, reject local solubility.
4. Reconstruct all simple roots exactly. Return “integral image” if a reconstructed point is integral and “integral Hasse failure” otherwise. The quadratic discriminant distinguishes the nonsplit case.

The order is chosen for speed: the inexpensive gcd and dyadic tests can reject many inputs before polynomial factorization. Correctness follows from Theorem 29.2; the root gate is never silently omitted. The supplied code/classify.py implements this procedure with exact rational arithmetic and SymPy’s rational-root computation.

39.2 Exact counting without target-box scanning

For each root r , compute the integer endpoints of (36.1). For each squarefree divisor d of $|3r - 1|$, the permitted residue class is $b \equiv 3^{-1} \pmod{d}$. Inclusion–exclusion counts all primitive b in the interval exactly by floor differences. This evaluates $P(T)$ without scanning $O(T^2)$ target pairs.

To count $\Sigma(T)$, enumerate ordered roots $r < s < t$ with $r + s + t = 1$ and $r^2 + s^2 + t^2 \leq T + 1$. Compute $A = rst$, $B = 2(rs + rt + st)$, check the box and gcd, and count each unordered triple once. The integral curves give $I_2(T)$ directly. Equations (37.3) and (37.4) then give the exact Hasse counts.

This is a practical exact algorithm, not a claimed optimal complexity result. The delivered split enumerator uses $O(T)$ candidate pairs; trial division of $|3r - 1|$ is used in the marked-root counter. Both can be improved by sieving, but no asymptotic theorem here depends on such an optimization.

39.3 Recorded exact counts

The following outputs were generated by the included programs. The last column is illustrative floating-point arithmetic; all counts are exact.

T	$P(T)$	$\Sigma(T)$	$I_2(T)$	$H_2^{\text{ns}}(T)$	$H_2^{\text{ns}}/(T \log T)$
10^2	633	21	14	555	1.205167
10^3	9 462	121	32	9 066	1.312438
10^4	126 107	653	68	124 079	1.347171
10^5	1 575 664	3 304	148	1 565 603	1.359865

The predicted leading constant is $27/(2\pi^2) \approx 1.3678359792$. For comparison, the total Hasse counts in these rows are 576, 9 187, 124 732, and 1 568 907. The proximity of the ratios is evidence of consistency, not a proof of the asymptotic or its error term.

39.4 Independent verification layers

The package records the following completed checks.

- (i) Exact symbolic checks of the determinant, inverse identities, reconstruction, factorization, discriminant, and the coefficient-gcd identity, using SymPy.
- (ii) Exhaustive comparison of the dyadic coefficient table with all 512 targets modulo 8.
- (iii) Exhaustive image and root-cusp comparison over $p = 3, 5, 7, 11, 13, 17, 19, 23, 29, 31$, totaling 82 142 target triples.
- (iv) Direct substitution and inverse recovery for 1 331 integer input points in $[-5, 5]^3$.
- (v) Independent target-box enumeration at $T = 5, 10, 25, 50$, checking the exact multiplicity corrections, integral curves, and nonsplit counts; and 100 exact instances of Corollary 34.2.

The finite dyadic enumeration is a complete finite certificate for the residue-table claim. The other finite tests do not establish statements at untested primes or arbitrarily large heights; those statements have separate proofs in the article.

40 Formalization plan and dependency audit

40.1 What is and is not kernel checked

The source repository contains a Lean declaration for the displayed map and a theorem computing its formal multivariate-polynomial Jacobian determinant [7]. This article did not run that Lean build. Its own classification, counting, and number-field results are not formalized in Lean or Rocq, and no such status is inferred from the existing project.

The arithmetic proofs naturally decompose into a mostly elementary formalization target. First, formalize the inverse chart as identities in a localization, with the conditions $2CD \neq 0$ explicit. Second, formalize the finite-field cubic lemma and coefficient comparison. Third, implement the dyadic table as a decidable finite statement and prove the Newton lifting lemma. Fourth, formalize the primitive parametrization and the two integral curves. These four stages contain the main classification innovation. Chebotarev is needed only for the necessity of the global rational-root gate, and should remain an explicit imported theorem or an explicitly stated dependency until an appropriate formal library result is available.

For the counting theorem, the critical analytic inputs are modest: Möbius inversion, the Euler product for $\zeta(2)$, an elementary divisor-sum bound, and partial summation. A clean formal proof should keep P, S, N, L, I, H distinct rather than hiding multiplicity corrections inside a single counting function.

40.2 Dependency graph

Claim	Essential dependencies
Odd gcd criterion	Inverse chart; unit-Jacobian lifting; rooted cubic lemma
Dyadic table	512-case finite identity; nonunit Newton lifting
Complete local classification	Odd criterion; dyadic table; cubic Chebotarev gate
Primitive parametrization	Classification; exact coefficient congruence
Integral curves	Slice reconstruction; distinct-root derivative product
$T \log T$ count	Primitive parametrization; divisor inversion; harmonic mean; multiplicity correction
Number-field extension	Same rooted cubic lemma; Chebotarev over K ; integrally closed rings; dyadic lifting and CRT

No proof of the main counting constant uses numerical extrapolation, the previously claimed split asymptotic, or a conjectural equidistribution statement.

41 Further research questions of Part III

The results above settle the nonsplit local-solubility question and the leading square-box asymptotic on $C = 2$. They also suggest several substantial extensions. The questions below specify what remains open in this article; they are not all claimed to be previously published open problems.

Research question 41.1 (A genuine second-order term). Does the limit

$$\lim_{T \rightarrow \infty} \frac{H_2^{\text{ns}}(T) - \frac{27}{2\pi^2} T \log T}{T}$$

exist, and can it be expressed by an explicit convergent arithmetic series and a boundary integral?

The present proof absorbs the clipped intervals and finitely many small roots into $O(T)$, precisely the scale of the desired term. Improving it requires retaining their contributions rather than merely sharpening the divisor error. The summatory mean in Lemma 36.3 is already strong enough to make such a refinement plausible.

Research question 41.2 (Power-saving remainder). After a second main term is identified, can the remainder be bounded by $O(T^{1/2+\epsilon})$, or is a larger fluctuation forced by the changing integer boundaries and split-root subtraction?

The known split contribution is of order $T^{2/3}$, so the answer for all Hasse failures and for nonsplit failures may require explicitly subtracting different $T^{2/3}$ terms. A proposed error smaller than that scale must account for this distinction. [write] Part II's Research question 24.2 asks the analogous question for the split count $\#\mathcal{H}_c^{\text{sp}}(T)$ itself.

Research question 41.3 (Other fixed slices). For each fixed nonzero integer c , determine the order and leading constant for nonsplit integral Hasse failures on $C = c$ in square boxes. Which c share the $T \log T$ law, and what local factors replace $27/(2\pi^2)$?

The complete local certificate already applies. The difficulty is the height geometry and the divisibility conditions in the monic-root parametrization. The special slice $c = 2$ removes denominator congruences and makes two reconstructed coordinates automatically integral; that simplification cannot simply be assumed for other c . [write] This is the part of Part II's Research question 24.1 (exponent and leading constant of the rational-plus-quadratic sector) that Part III leaves open: its classification half is answered on every slice by Corollary 33.1, its counting half on $C = \pm 2$ by Theorem 29.3. The split sector of every slice is Part II's Theorem 15.3.

Research question 41.4 (Unequal boxes and phase changes). Determine the asymptotic in $|A| \leq H_A$, $|B| \leq H_B$ uniformly in natural ranges of the ratio $H_A/H_B^{3/2}$.

The root interval has radius $H_A/|r|$ but the admissible centers extend to $|r| \asymp \sqrt{H_B}$. The logarithmic range can shrink or disappear as these two scales cross. A uniform result would explain the mechanism behind the square-box logarithm rather than only one specialization. [write] This is what remains of Part I's Research question 12.3, which asks for boxes $|A| \leq H_A$, $|B| \leq H_B$; Part III answers it for square boxes in all sectors.

Research question 41.5 (Quadratic fields and branch-switching statistics). Among nonsplit Hasse failures, what is the distribution of the quadratic field $\mathbb{Q}(\sqrt{\Delta})$ and of the primes at which the unique rational point is nonintegral but a quadratic point is locally integral?

The gcd criterion removes these data from the membership test, but not from finer arithmetic statistics. A field-conditioned count would involve square classes of $-3r^2 + 2r + 1 - 4b$ in the primitive region.

Research question 41.6 (Minimal dyadic precision over number fields). For a finite extension of $\mathbb{Q}_2$, what is the least ideal modulus that determines membership in $F(\mathcal{O}_v^3)$? How does it depend on ramification and residue degree?

Theorem 38.1 supplies the uniform modulus $8\mathcal{O}_v$, not an optimal one. A classification of the finite image could generalize the small rational table without an exponential enumeration of the entire quotient cube. [write] Cf. Part I's Research question 12.5, which asks for the integral fiber distributions over finite extensions of $\mathbb{Q}_2$; this question asks only for the membership modulus.

Research question 41.7 (Beyond this cubic map). Which polynomial maps with an inverse polynomial of degree $d > 3$ admit an analogous collapse from local factorization data to a small set of coefficient obstructions after one rational branch is known?

Lemma 31.2 is unusually strong in degree three. In degree four a repeated rational factor can leave an irreducible quadratic with no residue root, so the same proof fails. Identifying the correct replacement would separate genuinely cubic behavior from a reusable arithmetic method.

Research question 41.8 (Counting over number fields and S -integers). For a fixed number field and a specified archimedean height, determine the asymptotic number of locally integral but globally nonintegral nonsplit targets on a fixed slice. How do units and the ideal class group alter the logarithmic factor?

The certificate in Section 38 settles the local membership problem but does not settle height counting. The meaning of a box and the treatment of units must be fixed before formulating a leading constant. [write] Part II’s Research question 24.5 asks the same for split fibers (the constant $\kappa(c)$ over number fields and after inverting finitely many primes).

Research question 41.9 (Proof-producing arithmetic software). Can the rational-root, gcd, dyadic-residue, and reconstruction certificates be emitted by a small program and checked inside ProveIt’s Lean project, leaving only the general rational-root necessity theorem dependent on Chebotarev?

Such a checker would certify individual global and integral-Hasse examples without trusting a computer algebra system. For positive local certificates, the root and gcd witnesses are especially compact. [write] Cf. Part I’s Research question 12.9 and Part II’s Research question 24.7, which ask for a reusable library; this question asks for per-target certificates.

42 Conclusion of Part III

The unresolved nonsplit sector is not a minor correction to the split case. It changes the dominant arithmetic scale. Once a rational root is available, every odd local obstruction is detected by reduction to the missing cubic cusp; the entire infinite family of odd-prime tests reduces to the gcd of two explicit target polynomials. The dyadic test is finite and uniform.

On $C = 2$, these facts give a primitive integer parametrization. Its harmonic-width geometry yields the coefficient $27/(2\pi^2)$ in a $T \log T$ law. Global integrality, meanwhile, forces a derivative to be a unit and collapses the image to two curves with only $T^{1/3}$ targets. This establishes a sharp qualitative separation: locally integral targets are sparse in the ambient plane, but almost all of them are nonsplit integral Hasse failures.

43 Package contents and replay instructions of Part III

[write] This is the manuscript’s Appendix A. The archive contains this article as `article.tex` and `article.pdf`, together with:

File	Purpose
<code>code/arithmetic.py</code>	Exact map, residue test, reconstruction, primitive interval counts, and target counts
<code>code/classify.py</code>	Exact integer-target classifier with JSON output
<code>code/verify.py</code>	Dyadic certificate, finite-field checks, inverse checks, independent small-box enumeration
<code>code/symbolic.py</code>	Exact symbolic identity checks
<code>code/count.py</code>	Reproduce the counting table
<code>data/*.json</code>	Recorded outputs of the verification and counting runs
<code>README.md</code>	Build and execution commands
<code>STATUS.md</code>	Proof, novelty, formalization, and computation boundaries
<code>SOURCES.md</code>	Pinned source and attribution ledger
<code>SHA256SUMS</code>	Integrity checks for the delivered files

From the package root, run

```
python code/verify.py
python code/symbolic.py
python code/count.py
python code/classify.py 0 4 2
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The exact arithmetic and finite verification programs need only Python’s standard library. Symbolic verification and the general classifier use SymPy. Counts and proofs do not depend on floating-point calculations. The package uses Python 3.10 or later; the recorded run used Python 3.13 and SymPy 1.14.0.

[write] In this report the files carry the prefix `03-nonsplit-hasse-`: the programs and the Makefile are under `code/`, the recorded outputs and `requirements.txt` under `data/`, and `STATUS.md` and `SOURCES.md` at the report root. The manuscript is printed here as Part III; its `article.pdf`, `README.md` and `SHA256SUMS` are not shipped. The commands above are the delivery’s: `classify.py`, `count.py`, `symbolic.py` and `verify.py` import `arithmetic` by its delivered name, and `verify.py`, `symbolic.py` and `count.py` write `verification.json`, `symbolic.json` and `counts.json` into `../data/` relative to their own directory, so they must be run on a copy (Appendix C.7). On Windows use `py` or `uv run` for python.

44 Explicit finite bounds used by the programs of Part III

[write] This is the manuscript’s Appendix B. The marked-root enumerator loops through $|r| \leq \lfloor \sqrt{2T} \rfloor + 3$. To verify that this is safe, put $q = |r|$. A feasible pair would satisfy

$$q(q^2 - q - T/2) \leq T.$$

For $q \geq \sqrt{2T} + 2$, the expression $q^2 - q - T/2$ is at least $3T/2 + 3\sqrt{2T} + 2 > T$, so feasibility is impossible. The chosen integer bound is slightly looser, not tighter, than needed.

For split counting, $r^2 + s^2 + t^2 \leq T + 1$ bounds each root. Enumerating $r < s < t$ with $t = 1 - r - s$ is unique; it does not divide an ordered count by six near repeated roots. For integral curves, the inequality $|-6r^2 + 4r + 2\epsilon| \leq T$ implies a quadratic bound on $|r|$; $\lfloor \sqrt{T+1} \rfloor + 3$ is again deliberately safe. The sharp $T^{1/3}$ scale is proved from the first coordinate, not imposed as an unverified truncation in the program.

45 Small-box cross-checks of Part III

[write] This is the manuscript’s Appendix C. These values were obtained both from the root-parametrized algorithm and by independently scanning all integer target pairs in the indicated boxes and computing their roots.

T	P	Σ	I_2	L_2	H_2^{ns}	H_2
5	14	1	4	11	6	7
10	33	2	5	28	21	23
25	104	5	8	93	80	85
50	261	10	12	240	218	228

For example, $P = 14$ at $T = 5$ is not a target count: its three-root fiber contributes three markings, and the repeated-root target contributes two. The corrections give $L_2 = 14 - 2 \cdot 1 - 1 = 11$.

46 The complete dyadic finite certificate of Part III

[write] This is the manuscript's Appendix D. This short standalone program verifies the exact equality between the residue image and the coefficient table. No symbolic algebra package is used. There are 512 source triples and 512 target triples; the set comprehension computes the former image and the final loop checks every possible target. Polynomial evaluation is performed over the integers and then reduced modulo 8, which agrees with evaluation in the quotient ring.

```

from itertools import product

def F(x, y, z):
    u = 1 + x*y
    h = u*u*z + y*y*(1 + 3*u)
    return u*h, y + 3*x*h, x*(5 - 3*u - x*x*z)

def E(A, B, C):
    c = C % 8
    if c == 0: return True
    if c == 4: return B % 2 == 0
    if c in (2, 6): return A % 2 == B % 2 == 0
    return ((A % 2 == 0 and B % 4 == 2)
            or (A % 2 == 1 and (B*C - 2*A - 1) % 8 == 0))

image = {tuple(v % 8 for v in F(*a))
          for a in product(range(8), repeat=3)}
assert len(image) == 176
for b in product(range(8), repeat=3):
    assert (b in image) == E(*b)

```

The recorded run passed every assertion. The same comparison appears in `code/verify.py`, which also records the finite fiber histogram. For readers replacing executable enumeration by a proof-assistant certificate, the final assertion is a decidable statement on a finite type. Its formalization is separate from the valuation-theoretic argument lifting the residue image to $\mathbb{Z}_2$.

The cardinality check alone would not prove the coefficient table: two different subsets can both have 176 elements. The final loop certifies the stronger pointwise equivalence used in the theorem.

A Replay guide and artifact inventory

The archive is self-contained except for standard TeX packages and the optional SymPy dependency. From its root directory:

```
python3 code/verify_finite.py --output certificates/finite_checks.json
python3 code/verify_split_plane.py
python3 code/verify_symbolic.py
python3 code/rational_inverse.py 0 -4 2
pdflatex -interaction=nonstopmode -halt-on-error arithmetic_fibers.tex
pdflatex -interaction=nonstopmode -halt-on-error arithmetic_fibers.tex
```

The first two commands use only Python’s standard library. The last two Python utilities need SymPy; the executed symbolic version was 1.14.0. The PDF build uses ordinary packages supplied in a full TeX Live installation, and does not require shell escape or external figures.

Path	Purpose
arithmetic_fibers.tex	Complete article source, bibliography included
arithmetic_fibers.pdf	Compiled article
code/verify_finite.py	Exact local tables, CRT construction, finite checks
code/verify_split_plane.py	Split-plane gcd and parameter counts
code/verify_symbolic.py	Universal algebraic identity checks
code/rational_inverse.py	Complete rational inverse utility
certificates/*.json	Machine-readable executed results
certificates/*.txt	Human-readable execution transcripts
README.md	Build instructions and statement-status summary
SOURCE_REVIEW.md	Pinned source paths and novelty boundaries

No external article PDFs, font files, or copies of third-party research repositories are redistributed in this archive.

[write] In ProveIt the article is `article.tex` (this text, rewritten as described in [section C](#)), `article.pdf` is a build of it, `certificates/` is shipped as `data/`, the delivered README is replaced by the report README, and the delivered PDF `arithmetic_fibers.pdf` is not shipped. The commands above are the delivery’s; [section C.3](#) gives commands that work with the shipped names. [Added 30 September 2026, batch 58: Part II’s delivered replay commands are printed in [Section 27](#); commands that work with its shipped names are in [Appendix C.5](#).] [Added 30 September 2026, batch 64: Part III’s delivered replay commands are printed in [Section 43](#); commands that work with its shipped names are in [Appendix C.7](#).]

B A characteristic-three simplification

The separate characteristic-three finite-field count has a direct algebraic explanation. Reducing [\(1.1\)](#) modulo 3 gives

$$Q = y, \quad P = z + y^2 - y^3 R, \quad R = -x - x^3 z.$$

For a target (A, B, C) , set $\zeta = A - B^2 + B^3 C$. The first two equations force $y = B$, $z = \zeta$; the remaining equation is

$$\zeta x^3 + x + C = 0.$$

Its derivative with respect to x is one. If $\zeta = 0$, it has exactly one geometric root; if $\zeta \neq 0$, it has three distinct geometric roots. Over a finite field its solution set is either empty or a translate of the kernel of the

additive map $x \mapsto \zeta x^3 + x$, whose kernel has one or three elements. This confirms the possible rational fiber sizes 0, 1, 3, without carrying characteristic-zero cusp formulas through an invalid division by three. It is an explanatory calculation, not a new finite-field histogram claim.

C Provenance of this report

C.1 One manuscript

[write] This report is built from *one* manuscript: manuscript 03 of batch 36 of ProveIt’s incoming reports, the archive ProveIt_Arithmetic_Local_Global (inner directory proveit_arithmetic_fibers, main file arithmetic_fibers.tex, a 26-page PDF dated 24 September 2026, author line “Research draft prepared for Vladimir Reshetnikov”, PDF metadata “Research draft prepared with ChatGPT”), pinned to ProveIt commit e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9. The archive arrived in commit e13affd32 and was placed in 1a1396d4d as a new report. Nothing was merged and nothing was selected away: every result, proof, example, table, question and limitation of the manuscript is printed. The manuscript contributed everything except the text marked [write]. [Added 30 September 2026, batch 58: this subsection describes Part I. The report is now built from *two* manuscripts; Part II is batch-58 manuscript 01, pinned to 1085b506d65e207a05b7e9c861bb1fe88432fe38, and its provenance, shipped files and editorial changes are in Appendix C.4.] [Added 30 September 2026, batch 64: the report is now built from *three* manuscripts; Part III is batch-64 manuscript 02, pinned to 6d04e1e385f2fe7d1cfdbbd8fd8d4e45bd6b6c72, and its provenance is in Appendix C.6.]

The editorial changes are these.

- (1) Every label carries the prefix alg:. The source’s 77 labels are kept, unchanged after the prefix. Six were added: alg:sub:attribution, alg:rem:collisionfiber, alg:app:report, alg:app:onesource, alg:app:pinned and alg:app:rerun.
- (2) Attribution was made precise: a bracketed [write] clause in the abstract, the status note on the title page, and [section 1.3](#), which names the formalized statements of the project and says where the complex stratification and the finite-field laws come from.
- (3) [Remark 9.2](#) records an observation made at the write, not a claim of the manuscript. It is the only new numbered item and the last one of its section, so no source number moved.
- (4) Notes marked [write] were added in [section 11](#) and in Appendix A (shipped names), and this appendix is new.

No mathematical statement of the manuscript was changed and no symbol was renamed. The manuscript’s $t = y + 1/x$ is not the project’s t (which is xy in Algebra/JacobianConjecture/Research/README.md and the family parameter in the project README, which [remark 9.2](#) writes τ), and its p, q are a prime and a field size, not the project’s $p = xy^2, q = x^2yz$. The PDF bookmark of the title page was fixed (no duplicate page anchor), and its top space was reduced from 0.65 to 0.3 inch so that the added status note fits.

C.2 Repository statements at the pin

[write] The source read the root README, the project README and Algebra/JacobianConjecture/Research/README.md at its pin (its delivered SOURCE_REVIEW.md). Between the pin and the placement commit the project did not change, so its statements about ProveIt are current. They are accurate: the formal

developments, the integral stable shears of [proposition 5.3](#) (unitriangular source and target maps over $\mathbb{Z}$), and the refusal to upgrade the project’s SymPy frontier certificates to kernel proofs. The manuscript claims no gap in the project; the question (1.2) is its own. The two external sources were checked at the write ([section 1.3](#)); Milne was not.

C.3 Rerunning the checks

[write] Rerun on a copy with the delivered layout, never in the report: `verify_split_plane.py` and `verify_symbolic.py` always write `split_plane_checks.json` and `symbolic_checks.json` into `../certificates/` relative to their own directory, and fail after passing their checks if it does not exist.

```
mkdir <scratch>           # outside the repository
cp -r code <scratch>/code
cp -r data <scratch>/certificates
cd <scratch>
python code/verify_finite.py --output certificates/finite_checks.json
python code/verify_split_plane.py
python code/verify_symbolic.py
python code/rational_inverse.py 0 -4 2
```

The last two need SymPy (the recorded runs used 1.14.0 and Python 3.13.5; no requirements file was delivered). At the write the four commands were run so with Python 3.13.5 and SymPy 1.14.0: every regenerated file equals the shipped one apart from Windows line endings. `data/build_review.json` is a static build record, not regenerated.

C.4 Part II: the second manuscript

[write] [Added 30 September 2026, batch 58.] Part II is built from *one* manuscript: manuscript 01 of batch 58 of ProveIt’s incoming reports, the archive `ProveIt_Dense_Integral_Hasse` (inner directory of the same name, main file `article.tex`, a 20-page US-Letter PDF dated 29 September 2026, author line and PDF metadata “AI-assisted research draft prepared for Vladimir Reshetnikov”), pinned to ProveIt commit `1085b506d65e207a05b7e9c861bb1fe88432fe38`. The archive arrived in commit `b2b626d85` and was placed in `b30441a8c` as an addition to this report. The report is therefore built from two manuscripts: Part I (batch 36, manuscript 03; [Appendix C.1](#)) and Part II. Nothing was merged across the Parts and nothing was selected away: every result, proof, example, table, question and limitation of the manuscript is printed in [Sections 15–27](#), and it contributed everything there except the text marked [write]. [Section 14](#) is new.

Part I is unchanged from Part II’s pin to the placement commit (the only change to this directory in between is the placed files), so the manuscript’s references to Sections 2, 4 and 12 of Part I and to Research questions 12.2–12.4 are current. The manuscript also read the project’s `Algebra/JacobianConjecture/Research/README.md` and the README of the neighbouring report `weighted-keller-rigidity`; it uses no result of either (`02-dense-hasse-SOURCES.md`).

The editorial changes are these.

- (1) *Labels*. The manuscript’s 70 labels carry the prefix `alg:dh:`, unchanged after it. Twenty were added for Part II: the section, its six subsections and the notation table (`alg:dh:sec:relation`, `alg:dh:sec:source`, `alg:dh:sec:answers`, `alg:dh:sec:consistency`, `alg:dh:sec:notation`, `alg:dh:sec:duplicates`, `alg:dh:sec:formal`, `alg:dh:tab:notation`); the manuscript’s seven research questions (`alg:dh:q:quadratic`, `alg:dh:q:remainder`, `alg:dh:q:uniform`,

- `alg:dh:q:coefficient`, `alg:dh:q:numberfield`, `alg:dh:q:exceptions`, `alg:dh:q:library`); its Conclusion and appendices (`alg:dh:sec:conclusion`, `alg:dh:sec:ledger`, `alg:dh:sec:repro`); and the two subsections of this appendix (`alg:dh:app:provenance`, `alg:dh:app:rerun`). In Part I five labels were added to existing subsections of Section 12 (`alg:sub:q-dyadic`, `alg:sub:q-split`, `alg:sub:q-height`, `alg:sub:q-density`, `alg:sub:q-library`); no existing label was renamed or removed, and no Part I number moved.
- (2) *Numbering.* Manuscript Section k ($k = 1, \dots, 9$) is printed as Section $k + 14$, so its Theorem $k.m$ and equation $(k.m)$ are Theorem $(k + 14).m$ and equation $((k + 14).m)$. Its seven research questions, numbered 1–7 by a counter of their own, are Research questions 24.1–24.7 under the report’s shared counter; its Conclusion is Section 25 and its Appendices A and B are Sections 26 and 27. The titles of Sections 24, 25 and 27 gained “of Part II”.
 - (3) *Scope of the counting claim.* The sentence after Theorem 15.3, “This settles the completely split sector of the prior report’s Research Question 12.3”, and the matching sentence of the abstract (Section 14.1) are printed with the restriction to square boxes $|A|, |B| \leq T$ added and marked [write]. The theorem is unchanged; the sentence overstated it, because Research question 12.3 asks for boxes $|A| \leq H_A$, $|B| \leq H_B$. The delivered 02-dense-hasse-STATUS.md keeps the unrestricted wording.
 - (4) *Notation.* The manuscript’s sum S is printed Σ (in the proof of Lemma 18.1 and before (19.4)); nothing else was renamed (Section 14.4). Macros: the manuscript’s `\pin` is `\dhsha` here and its `\pathtext` is the report’s `\rpath`; its definitions of `\Z`, `\Q`, `\R`, `\C`, `\A`, `\Zp`, `\Qp`, `\F`, `\Ztwo`, `\im`, `\vol`, `\lcm` and `\target` duplicate the report’s and were dropped; its unused `\disc`, `\ord`, `\rad` and `\sgn` were not carried over; `\Hsplit`, `\Isplit` and `\calB` were added, and the package `needspace` is loaded for its `\Needspace`. The manuscript’s separate question counter was replaced by the report’s.
 - (5) *Citations.* The manuscript’s `proveitformal`, `priorreport` and `dlnf` are the entries [7], [8] and [9]; its `priorreport` is Part I itself. Its entry for Gao’s paper was merged into Part I’s entry [3], with its note kept.
 - (6) *Notes marked* [write] were added in Section 15 (Part I and the shipped name of SOURCES.md), at the two square-box sentences, after the program list of Section 23, after three research questions (cross-references to Part I), and at the end of Section 27 (shipped names). The manuscript’s title page is printed as the abstract and status note of Section 14.1.
 - (7) *Part I* gained dated notes after Research questions 12.2, 12.3 and 12.4, on the title page, in Appendix A and in Appendix C.1; the Part II heading, its running head and two contents entries are new.

No mathematical statement of the manuscript was changed.

Where the addition had to choose: (a) the manuscript became Part II of this report, not a new report, because it studies the same map through the same chart and cubic and extends Part I’s $C = 2$ family; (b) its restatements of Part I are kept in place and marked (Section 14.5), since all of its labels are kept; (c) of the two colliding letters, the sum S was renamed rather than the height T , which occurs throughout and whose other use (the indeterminate of g) is Part I’s own; (d) the counting claim was restricted to square boxes in the printed text.

Delivered file	In this report
article.tex	printed as Part II (Sections 15–27)
README.md, article.pdf	not shipped (the report README and article.pdf replace them)
SOURCES.md, STATUS.md	02-dense-hasse-SOURCES.md, 02-dense-hasse-STATUS.md
Makefile	code/02-dense-hasse-Makefile
requirements.txt	data/02-dense-hasse-requirements.txt
code/split_fibres.py,	code/02-dense-hasse-split_fibres.py,
verify_results.py, count_heights.py	code/02-dense-hasse-verify_results.py, code/02-dense-hasse-count_heights.py
data/build_review.json, example.json,	the same names under data/ with the prefix
height_counts.json,	02-dense-hasse-
height_counts.txt, verification.json,	
verification.txt	

Every shipped file is byte-identical to the delivery; no checksum ledger was delivered. The shipped 02-dense-hasse-SOURCES.md, 02-dense-hasse-STATUS.md and code/02-dense-hasse-Makefile still name the delivered paths. Two of the recorded outputs are not what the delivered programs write: data/02-dense-hasse-example.json is the standard output of `split_fibres.py 1 1 3 --modulus 18144000` (identical at the write), whereas `verify_results.py` writes an `example.json` without its "modular_witness" block; and data/02-dense-hasse-verification.txt is a captured console log of a different run (0.524 s) from the one that wrote data/02-dense-hasse-verification.json (0.511 s).

C.5 Rerunning the checks of Part II

[write] [Added 30 September 2026, batch 58.] Rerun on a copy with the delivered names, never in the report: `count_heights.py` imports `split_fibres` by its delivered name, and `verify_results.py` and `count_heights.py` write `verification.json`, `example.json`, `height_counts.json` and `height_counts.txt` into `../data/` relative to their own directory.

```
W=<scratch>                # outside the repository
mkdir -p "$W/code" "$W/data"
for f in count_heights split_fibres verify_results; do
  cp "code/02-dense-hasse-$f.py" "$W/code/$f.py"; done
cd "$W"
uv run --no-project --with sympy==1.14.0 --with numpy==2.3.5 \
  --with mpmath==1.3.0 python code/verify_results.py
uv run --no-project --with numpy==2.3.5 --with mpmath==1.3.0 \
  python code/count_heights.py
py code/split_fibres.py 1 1 3 --modulus 18144000
```

At the write these commands were run so, with Python 3.13.5 under uv, SymPy 1.14.0, NumPy 2.3.5 and mpmath 1.3.0: all twelve groups passed (1.5 s); the regenerated `verification.json` and `height_counts.json` equal the shipped ones apart from their `elapsed_seconds` fields and Windows line endings, `height_counts.txt` is identical apart from line endings, and the regenerated `example.json` is the shipped one without its "modular_witness" block, which the third command prints. `split_fibres.py` needs only the standard library. `data/02-dense-hasse-build_review.json` is a static record of the delivered 20-page PDF, not regenerated.

C.6 Part III: the third manuscript

[write] [Added 30 September 2026, batch 64.] Part III is built from *one* manuscript: manuscript 02 of batch 64 of ProveIt’s incoming reports, the archive ProveIt_Beyond_Split_Fibers (inner directory ProveIt_Nonsplit_Hasse, main file `article.tex`, a 24-page US-Letter PDF dated 30 September 2026, author line and PDF metadata “AI-assisted research draft prepared for Vladimir Reshetnikov”), pinned to ProveIt commit `6d04e1e385f2fe7d1cfb8dbd8fd8d4e45bd6b6c72`. The archive arrived in commit `7747fcfdd` and was placed in `3025c15df` as an addition to this report, with the file prefix `03-nonsplit-hasse-`. The report is therefore built from three manuscripts: Part I (batch 36, manuscript 03; Appendix C.1), Part II (batch 58, manuscript 01; Appendix C.4) and Part III. Nothing was merged across the Parts and nothing was selected away: every result, proof, example, table, question and limitation of the manuscript is printed in Sections 29–46, and it contributed everything there except the text marked [write]. Section 28 is new.

Parts I and II are unchanged from Part III’s pin to the placement commit (the only change to this directory in between is the placed files), so the manuscript’s references to Part I’s Research questions 12.2 and 12.3 and to Part II’s open questions and split constant are current. The manuscript also read this directory’s README and the collection’s SetTheory/Cardinals/docs/reports/README.md and SetTheory/Cardinals/README.md at its pin, for scope and placement only (`03-nonsplit-hasse-SOURCES.md`).

The editorial changes are these.

- (1) *Labels.* The manuscript’s 78 labels carry the prefix `alg:ns:`, unchanged after it. Twenty were added for Part III: the relation section, its six subsections and the notation table (`alg:ns:sec:relation`, `alg:ns:sec:source`, `alg:ns:sec:answers`, `alg:ns:sec:consistency`, `alg:ns:sec:notation`, `alg:ns:sec:duplicates`, `alg:ns:sec:formalproject`, `alg:ns:tab:notation`); the manuscript’s nine research questions (`alg:ns:q:second`, `alg:ns:q:power`, `alg:ns:q:slices`, `alg:ns:q:unequal`, `alg:ns:q:quadfields`, `alg:ns:q:dyadicprec`, `alg:ns:q:beyond`, `alg:ns:q:numberfield`, `alg:ns:q:software`); its Conclusion (`alg:ns:sec:conclusion`); and the two subsections of this appendix (`alg:ns:app:provenance`, `alg:ns:app:rerun`). In Part I the label `alg:sub:q-extensions` was added to the existing subsection 12.5. No existing label was renamed or removed, and no number of Part I or Part II moved.
- (2) *Numbering.* Manuscript Section k ($k = 1, \dots, 14$) is printed as Section $k + 28$, so its Theorem $k.m$ and equation $(k.m)$ are Theorem $(k + 28).m$ and equation $((k + 28).m)$; its Appendices A–D are Sections 43–46, with the manuscript’s reference to “Appendix D” printed as a section reference. Its research questions already shared the theorem counter. The titles of its research-question, Conclusion and appendix sections gained “of Part III”.
- (3) *Citations.* The manuscript’s entry `prior` reads: “*Arithmetic Local–Global Dichotomies for ProveIt’s Three-Variable Keller Map*, with Part II, *Zariski-dense integral Hasse failures: complete split-fiber criteria and sharp target-height asymptotics*, AI-assisted research reports dated September 24 and 29, 2026, merged in ProveIt, snapshot [the pin]. Unrefereed and not formalized. See Part I, Sections 12.2–12.3, and the Part II discussion of remaining nonsplit and coefficient-certificate questions.” It is this report, so its four citations became cross-references to Parts I and II, marked [write], and Part I’s section numbers 12.2 and 12.3 became cross-references (the printed numbers are unchanged). Its entry `formal` names the same Lean file at its own pin, identical to Part II’s entry, and is cited as [7]. Its entries for Gao and Milne were merged into Part I’s entries [3] and [6], with their notes kept.
- (4) *Notation.* The split count $S(T)$ is printed $\Sigma(T)$, and the implied constant C in the proof of Lemma 36.1 is printed c_0 (Section 28.4); nothing else was renamed. Macros: the manuscript’s `\pin` is `\nssha` here; `\NS` and `\spc` were added; its definitions of $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{F}$, $\mathbb{A}$, $\mathbb{Z}_p$, $\mathbb{Q}_p$, $\mathbb{O}_h$, $\mathbb{G}_a$, $\mathbb{D}_{\text{isc}}$,

`\im`, `\lcm`, `\target` and `\rpath` duplicate the report’s and were dropped; its unused `\odd` was not carried over.

- (5) *Notes marked* [write] were added in Section 29 (the report and the pin, the formal declarations, Part II’s split constant), at the four former citations, after the Remark following Proposition 32.2 (Part I’s residue counts), after Lemma 35.2 (Part I’s Theorem 4.1, which the manuscript does not cite), after Lemma 36.3 (Part I’s density), at the constant c_0 , after six research questions (cross-references to Parts I and II), at the heads of Sections 43–46, and at the end of Section 43 (shipped names). The manuscript’s title page is printed as the abstract and evidence note of Section 28.1; its author line is recorded only here.
- (6) *Parts I and II* gained dated notes on the title page, after Research questions 12.1, 12.2, 12.3 and 12.5, in the introductory paragraph of Part II, in Section 14.2, after Theorem 15.3, after Part II’s Research questions 24.1, 24.4 and 24.5, in Section 25, in Appendices A and C.1, and in the bibliography entries [3], [6] and [7]. The title page’s vertical spaces were reduced so that its note fits on one page. The Part III heading, its running head and its contents entry are new.

No mathematical statement of the manuscript was changed.

Where the addition had to choose: (a) the manuscript became Part III of this report, not a new report, because it studies the same map through the same chart and cubic and answers questions of Parts I and II; (b) its restatements of Parts I and II are kept in place and marked (Section 28.5), since all of its labels are kept; in particular its proof that split fibers on $C = 2$ have no integral point is kept beside a pointer to Part I’s Theorem 4.1; (c) of the colliding letters, the count $S(T)$ was renamed rather than the indeterminate S or the set of places, following Part II’s rename of its sum, and the constant C was renamed rather than the coordinate; (d) Part II’s split-sector theorems are kept as proved, and the sentences that restrict the answer to Research question 12.3 to the split sector are re-scoped by dated notes, not edited; (e) the bibliography entry for this report itself became cross-references.

Delivered file	In this report
<code>article.tex</code>	printed as Part III (Sections 29–46)
<code>README.md</code> , <code>article.pdf</code> , <code>SHA256SUMS</code>	not shipped (the report <code>README</code> and <code>article.pdf</code> replace the first two)
<code>SOURCES.md</code> , <code>STATUS.md</code>	<code>03-nonsplit-hasse-SOURCES.md</code> , <code>03-nonsplit-hasse-STATUS.md</code>
<code>Makefile</code>	<code>code/03-nonsplit-hasse-Makefile</code>
<code>requirements.txt</code>	<code>data/03-nonsplit-hasse-requirements.txt</code>
<code>code/arithmic.py</code> , <code>classify.py</code> , <code>count.py</code> , <code>symbolic.py</code> , <code>verify.py</code>	the same names under <code>code/</code> with the prefix <code>03-nonsplit-hasse-</code>
<code>data/build_review.json</code> , <code>classification_example.json</code> , <code>counts.json</code> , <code>symbolic.json</code> , <code>verification.json</code>	the same names under <code>data/</code> with the prefix <code>03-nonsplit-hasse-</code>

Every shipped file is byte-identical to the delivery, and matches the delivered `SHA256SUMS`. The shipped `03-nonsplit-hasse-SOURCES.md`, `03-nonsplit-hasse-STATUS.md` and `code/03-nonsplit-hasse-Makefile`, and the manuscript’s own text (Sections 39 and 43), still name the delivered paths. `data/03-nonsplit-hasse-build_review.json` records the delivered 24-page PDF, which is not shipped, and `data/03-nonsplit-hasse-classification_example.json` is the standard output of `classify.py 0 4 2`; no program writes that file.

C.7 Rerunning the checks of Part III

[write] [Added 30 September 2026, batch 64.] Rerun on a copy with the delivered names, never in the report: `classify.py`, `count.py`, `symbolic.py` and `verify.py` import arithmetic by its delivered name, and `verify.py`, `symbolic.py` and `count.py` (without `--output`) write `verification.json`, `symbolic.json` and `counts.json` into `../data/` relative to their own directory.

```
W=<scratch>                # outside the repository
mkdir -p "$W/code" "$W/data"
for f in arithmetic classify count symbolic verify; do
  cp "code/03-nonsplit-hasse-$f.py" "$W/code/$f.py"; done
cd "$W"
py code/verify.py
uv run --no-project --with sympy==1.14.0 python code/symbolic.py
py code/count.py
uv run --no-project --with sympy==1.14.0 python code/classify.py 0 4 2
```

At the write these commands were run so, with Python 3.14.4 for the standard-library programs and Python 3.13.5 under uv with SymPy 1.14.0: all exit with status zero, `verify.py` and `symbolic.py` report "status": "PASS", and the regenerated `verification.json`, `symbolic.json` and `counts.json` and the printed classification equal the shipped files apart from Windows line endings. `verify.py`, `count.py` and `arithmetic.py` need only the standard library.

References

- [1] Vladimir Reshetnikov, *ProveIt: Jacobian conjecture project*, Algebra/JacobianConjecture/README.md and associated Lean/Rocq sources. Repository snapshot e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9, inspected September 24, 2026. [Pinned project documentation](#).
- [2] Vladimir Reshetnikov, *Searching for a simpler counterexample*, Algebra/JacobianConjecture/Research/README.md, in ProveIt, same pinned snapshot. [Pinned research documentation](#).
- [3] Shuhong Gao, *Counterexamples to the Jacobian conjecture in dimensions greater than two*, arXiv:2608.00222v1 (2026), especially Theorems 3.3–3.4 and Appendix A.1. <https://arxiv.org/html/2608.00222v1>. [write] Part II cites the abstract page <https://arxiv.org/abs/2608.00222> (arXiv:2608.00222, 2026); its entry reads: “Used for attribution of the pre-existing geometric setting; none of its generalizations is an assumption of the proofs here.” [write] Part III cites arXiv:2608.00222v1, July 31, 2026, with the note: “Theorems 3.3–3.4 provide the relevant degree-three map and fiber geometry; the map is attributed to Alpöge’s announcement.”
- [4] royvanrijn/jacobian-research, *Exact finite-field value distribution*, archived research note, research/archive/legacy-notes/FINITE_FIELD_VALUE_DISTRIBUTION.md. Repository snapshot caf5685db1099cb90e608556d4b8d825734f29cb, inspected September 24, 2026. [Pinned finite-field note](#). The repository note is cited as prior public work, not as an independently peer-reviewed source.
- [5] royvanrijn/jacobian-research, *An integral Jacobian-one Hasse failure*, archived research note, research/archive/non-elliptic/verified/INTEGRAL_HASSE_JACOBIAN_ONE.md, same pinned snapshot. [Pinned degree-five note](#). Its distinct map and rational obstruction are used only for comparison, not as assumptions in any theorem of this article.
- [6] James S. Milne, *Algebraic Number Theory*, version 3.08, July 19, 2020, Theorem 8.31 and Example 8.36. <https://www.jmilne.org/math/CourseNotes/ANT.pdf>. [write] Part III cites the same version, adding “printed pages 147–148”.
- [7] V. Reshetnikov, *ProveIt: Jacobian-conjecture formal development*, repository at commit 1085b506d65e207a05b7e9c861bb1fe88432fe38, inspected September 29, 2026. <https://github.com/VladimirReshetnikov/ProveIt/blob/1085b506d65e207a05b7e9c861bb1fe88432fe38/Algebra/JacobianConjecture/Lean/JacobianConjecture/Counterexample.lean>. [write] Cited by Part II, and by Part III at its pin 6d04e1e385f2fe7d1cfbdbd8fd8d4e45bd6b6c72, where the file is identical.
- [8] *Arithmetic Local–Global Dichotomies for ProveIt’s Three-Variable Keller Map: Split integral Hasse failures, a dense parameter family, and exact p -adic fiber laws*, research draft prepared for Vladimir Reshetnikov, September 24, 2026; ProveIt research-report collection, with later repository additions. Sections 2, 4, and 12, especially Research Questions 12.2–12.4. <https://github.com/VladimirReshetnikov/ProveIt/blob/1085b506d65e207a05b7e9c861bb1fe88432fe38/SetTheory/Cardinals/docs/reports/jacobian-conjecture/arithmetic-local-global-fibers/article.tex>. [write] Cited by Part II; this is Part I of the present report, at Part II’s pin.
- [9] NIST Digital Library of Mathematical Functions, Section 5.12, *Beta Function*, especially the beta–gamma identity and Euler’s integral. <https://dlmf.nist.gov/5.12>. [write] Cited by Part II.